SAMSUNG

REFRIGERATOR

SIDE BY SIDE TYPE

Basic Model : RS22R5561**
Project Name : RS5300TC
RS5300T
Model Code : RS22T5561**/**
RS22T5201**/**
RS22T5200**/**
RS27T5561**/**
RS27T5200**/**
RS28T5B00**/**

SERVICE Manual

REFRIGERATOR

RS22T5561**
RS27T5561**

RS22T5200**
RS22T5201**
RS27T5200**

RS28T5B00**

CONTENTS

WARNING

IMPORTANT SAFETY NOTICE

The service guide is for service technicians with sufficient background in electrical, electronic and mechanical engineering.
Any attempt to repair the appliance yourself may result in personal injury and property damage.
The manufacturer or dealer will not be held responsible for the interpretation of this information.

Contents

Contents

1. PRECAUTIONS (SAFETY WARNINGS)

- Unplug the appliance before the changing or repairing the electric parts.
- Use rated electronic Control equipment.
  → Make sure to check out Model name, Rated voltage, Rated current, Operation Temp, etc.
- Upon repair, make sure that harnesses are not to be water-penetrated and are bundled tight.
  → Should not be detached by a certain amount of external force.
- Upon repair, completely remove dust or other foreign substances from housing, harness, connector, etc.
  → To prevent fire by tracking, short, etc.
- Check out whether water has penetrated into the electronic Control system.
  → If there is any kind of trace, take necessary measures such as related component change, insulation tapping, etc.
- After repair, check out the assembled state of parts.
  → It should be the same as the previous state.
- Check out the surrounding conditions.
  → Change the location, if the fridge is located at humid, wet places or the installed state is unstable.
- In order to reduce the risk of electric shock the appliance must be properly grounded.
- Do not allow consumers to overload a certain outlet.
- Check out whether the power cord or the outlet is broken, squeezed, chopped off or heatdeformed.
  → Repair or replace the defective power cord/outlet immediately.
  → Make sure the power cord is not punctuated or stomped down.
- Do not allow consumers to keep food frayed or place bottles in the Freezer Room.
- Do not allow consumers to repair the fridge by themselves.
- Do not allow consumers to keep things except for food.
  → Pharmaceutical, Chemical substances : These are not possible to be fine-Controlled with a consumer fridge.
  → Flammable material (alcohol, benzene, ether, LPG, etc) : possibility of explosion.

PRECAUTIONS(SAFETY WARNINGS)

Read all instructions before repairing the product and follow the instructions in order to prevent danger or property damage.

CAUTION/WARNING SYMBOLS DISPLAYED

Warning	Indicates that a danger of death or serious injury exists.
Caution	Indicates that a risk of personal injury or material damage exists.

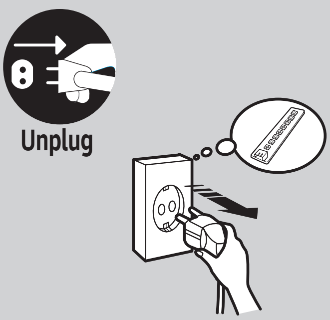
SYMBOLS

means “Prohibited”.
means “Do not disassemble”.
means “No contact”.
means ”Warning or Caution”.
means “Unplug the unit before preforming service”
means “Earth or Ground”.

Warning & Caution

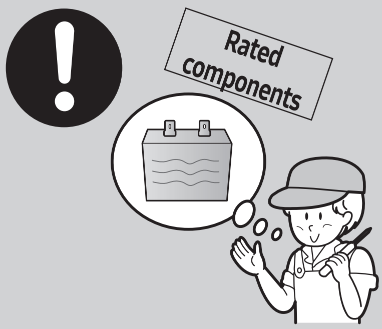
Plug out to exchange the interior lamp.

• It may cause electric shock.



Use the rated components on the replacement.

• Check the correct model, rated voltage, rated current, operating temperature and so on.



On repair, make sure that the wires such as harness are bundled tightly.

• Bundle tightly wires in order not to be detached by the external force and then not to be wetted.

On repair, remove completely dust or other things of housing parts, harness parts, and check parts.

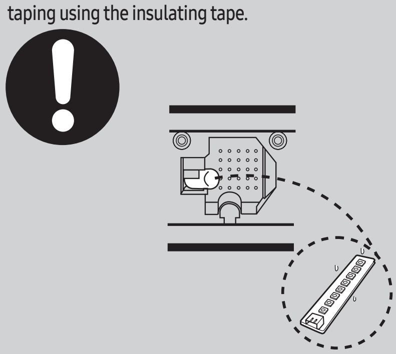
• Cleaning may prevent the possible fire by tracking or short.

After repair, check the assembled state of components.

• It must be in the same assembled state when compared with the state before disassembly.

Check if there is any trace indicating the permeation of water.

• If there is that kind of trace, change the related components or do the necessary treatment such as taping using the insulating tape.



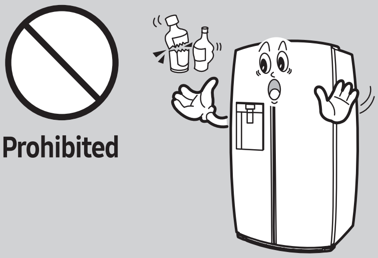
PRECAUTIONS(SAFETY WARNINGS)

Please let users know following warnings & cautions in detail.

⚠ Warning & Caution

Do not allow users to put bottles or kinds of glass in the freezer.

• Freezing of the contents may inflict a wound.



Do not allow users to store narrow and lengthy bottles or foods in a small multi-purpose room.

• It may hurt you when refrigerator door is opened and closed resulting in falling stuff down.

Do not allow users to store pharmaceutical products, scientific materials, etc., in the refrigerator.

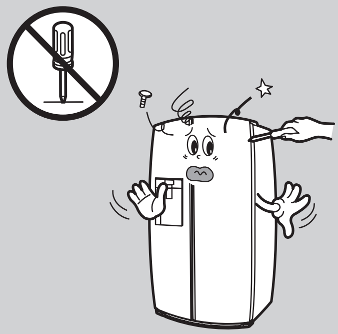
• The products which need precise temperature control should not be stored in the refrigerator.

Do not allow users to plug several appliances into the same power receptable.

• May cause abnormal generation of heat or fire.

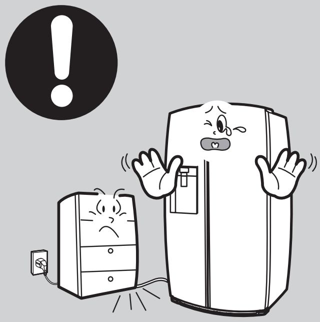
Do not allow users to disassemble, repair or alter.

• It may cause fire or abnormal operation which leads to injury.



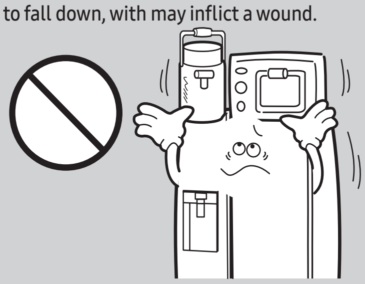
Do not allow users to bend the power cord with excessive force or do not have the power cord pressed by heavy article.

• May cause fire.



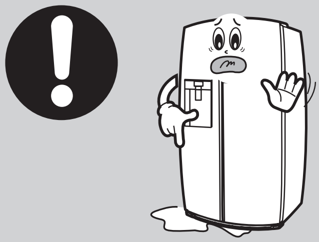
Do not allow users to store articles on the product.

• Opening or closing the door may cause things to fall down, with may inflict a wound.

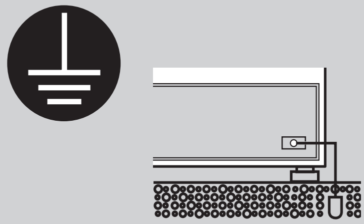


Do not allow users to install the refrigerator in the wet place or the place where water splashes.

• Deterioration of insulation of electric parts may cause electric shock or fire.

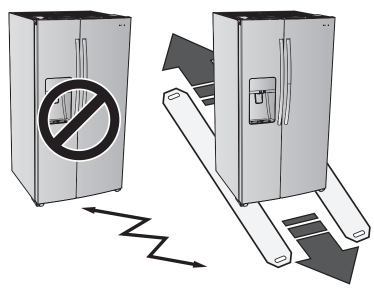


In order to reduce the risk of electric shock the appliance must be properly grounded.



⚠ CAUTION

When installing, servicing or cleaning behind the refrigerator, be sure to pull the unit straight out and push back in straight after finishing.

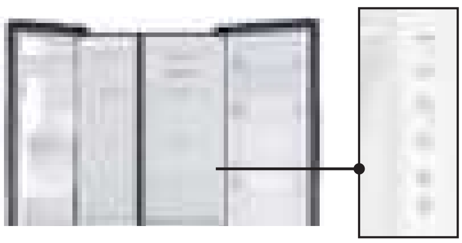


2. PRODUCT SPECIFICATIONS

2-1. Introduction of Main Function

Large Capacity with Space Max™

- The Samsung Side by side Refrigerator has a very spacious interior with a huge Capacity. Thinner wall with its special high-urethane insulation technology provides plenty of room in the drawers and shelves to store your weekly grocery shopping, including items of various shapes or sizes. And it's much easier to keep everything neatly and efficiently organized. So you can quickly find and take out the things you need whenever you need them.



Modern Design

- The Samsung Side by side Refrigerator's modern design seamlessly blends in with your kitchen interior. It has strikingly beautiful flat doors and recessed handles, so it won't protrude and get in the way; especially on Counter depth model, its built-in look is fully aligned with your existing appliances and cabinetry.
- The Ice & Water Dispenser is seamlessly integrated into the front of the door and only has the controls you need to enjoy a refreshing glass of chilled filtered water or fill a pitcher with cubed or crushed ice.
- All the other controls are discretely hidden inside, but easily accessible when you open the door. It is also compliant with the Americans with Disabilities Act (ADA).

PRODUCT SPECIFICATIONS

2-2. Model Specification

NOTE

This operation instruction covers various models. The characteristics of your appliance may differ slightly from those described in this manual.

- **Key features of your new refrigerator**
  Your Samsung Side-By-Side Refrigerator comes equipped with many space-saving features, stylishly integrated design and improved cooling performance.

- **All-around cooling**
  Provides even cooling throughout the refrigerator to maintain optimal temperatures to keep food fresh.

- **In-Door Ice Maker**
  The Icemaker is located in the freezer door, this ensures that all the shelf space can be fully utilized. Ice cubes are quickly produced and the clear ice bucket lets you easily see the amount of ice cubes produced.

- **Fingerprint Resistant Finish**
  Prevents surfaces from becoming covered in unsightly fingerprints and other marks and smudges.

- **LED Lighting**
  See everything in a new light with LED lighting.

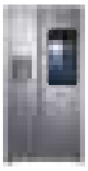
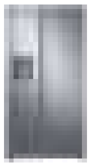
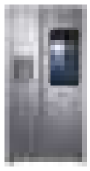
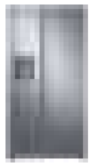
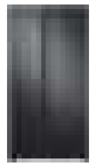
- **Energy Star ***
  Save energy and money and help protect the planet.

- **Embedded Wi-Fi ***
  With SmartThings App, you can easily check the temperature, receive alerts if the door is accidentally left open and control Rapid cooling(Power Cool, Power Freeze) and Sabbath mode.

* : Optional

PRODUCT SPECIFICATIONS

2-3. Basic Specification

Items		Specification				
Model		RS22T55**	RS22T52**	RS27T55**	RS27T52**	RS28T5B**
Image						
Capacity (AHAM 2008)	Net Total (cu.ft)	21.5	22	26.7	27.4	28.1
	Net for Freezer	7.4	7.4	9.5	9.5	10.2
	Net for Fridge	14.1	14.6	17.2	17.9	17.9
Capacity (ISO Storage)	Net Total (ℓ)	-	602	685	716	759
	Net for Freezer / 2-Star	-	193 (30)	206	217	254
	Net for Fridge	-	409	479	499	505
Dimensions (Width * Depth * Height)		912 * 726 * 1,780	←	912 * 851 * 1,780	←	←
Dimensions of Package (Width * Depth * Height)		958 * 764 * 1,915	←	958 * 889 * 1,915	←	←
Rated Voltage (V) / Rated Frequency (Hz)		220-240V/50Hz,60Hz , 110-127V/50Hz,60Hz				
Refrigerant		R-600a				
Refrigerant Charging Volume		92g	84g	92g	84g	84g
Product Weight (kg)		116	108	139	131	126
Weight of Packaged Product (kg)	Vinyl	120	112	143	135	130
	Box	125	117	149	141	136

PRODUCT SPECIFICATIONS

2-3-1. Electric Parts Specification

Items				Specification				
Model				RS22T5**1	RS22T5**1	RS27T5561	RS27T5**0	RS28T5*0
Freezing Component	Compressor		Type	NF54M7151ANTT3	NF54M5151ARTT3	NF54M7151ANTT3	NF54M5151ARTT3	NF54M5151ARTT3
			Drive mode	INVERTER				
			Charged oil	SUNUSO-2GSD(250cc)				
	Freezer		Freezer compartment	SPLIT FIN TYPE				
			Cold compartment	X				
	Condenser			Forced and spontaneous convection type				
	Desiccant			Molecular sieve XH-9				
	Capillary tube		Freezing compartment	F: ID0.85*L3300 ID0.75*L4300	F: ID0.85*L3300	F: ID0.85*L3300 ID0.75*L4300	F: ID0.85*L3300	F: ID0.85*L3300
			Cold compartment	None				
	Refrigerant			R-600a				
Components for defrosting	Freezer Component	Section		OFF		ON		
		THERMISTOR (F-SENSOR) 502AT	Strong (°C)	-24		-21		
			Intermediate (°C)	-21		-18		
			Weak (°C)	-17		-14		
	Cold compartment	Section		OFF		ON		
		THERMISTOR (R-SENSOR) 502AT	Strong (°C)	0		3		
			Intermediate (°C)	1.5		4.5		
			Weak (°C)	5		8		
	Defrosting Cycle	Initial defrosting cycle (simultaneous defrosting of F and R)		6 hours ± 10 minutes				
		Defrosting cycle of freezing component		12 to 77 hours (varying dependent upon operation conditions)				
		Defrosting cycle of cold component		None				
		Idle duration		-				
	Defrosting Sensor	F defrosting Sensor	Type	Thermistor (DTN-C502)				
			Specification	5.0 ㏀ at 25°C				
		R defrosting Sensor	Type	Thermistor (DTN-C502)				
			Specification	5.0 ㏀ at 25°C				
	Bimetal (F)		Type	Bimetal Thermostat				
			Operating temperature (°C)	On : 40°C, Off : 60°C				
	Bimetal (R)		Type	None				
			Operating temperature (°C)	None				

PRODUCT SPECIFICATIONS

2-3-2. Electric Component

Items			Specification				
Model			RS22T556*	RS22T520*	RS27T556*	RS27T5200	RS28T5B00
Electric Component	Defrosting heater in freezing compartment	Power supplying during freezing compartment defrosting	AC 230V, 300W or AC 120V, 300W				
	Defrosting heater in cold compartment	Power supplying during cold compartment defrosting	-				
	Home bar heater (cold compartment)	Ambient humidity sensor interlocking	-				
	Icemaker heater		-				
	Dispenser heater		AC 230V, 6W or AC 120V, 6W				-
	Water pipe heater		DC12V, 6W				-
	Water tank heater		-				
	Damper heater	Kept tumed on	DC12V, 3W				
	Bimetal for preventing overheating of defrosting heater in freezing compartment		BIMETAL THERMOSTAT - ON : 40℃, OFF : 60℃ THERMO FUSE - CUT OFF : 109~110℃				
	Bimetal for preventing overheating of defrosting heater in cold compartment		-				
	OLP	Type	-				
		Operating temperature (℃)	-				
		Shutdown temperature (℃)	-				
	Cooling fan motor in freezing compartment		DC12V, BLDC, F135, 1850rpm, Max 4.2W				
	Cooling fan motor in cold compartment		-				
	Compressor cooling fan motor		DC12V, BLDC, C160N, 1080rpm, Max 1.32W				
	Motor-driven damper (True-taste chamber)		DC12V, Step Motor, Heater 3W				
	Motor step valve (OPTIONAL)		DC 12V, Max 700mA	-	DC 12V, Max 700mA	-	-
	Freezing compartment lamp		LED Lamp, High intensity 3 Chips, DC12V / Max 2.39W, 1EA				
	Cold compartment lamp		LED Lamp, High intensity 5 Chips, DC12V / Max 4.22W, 1EA				
	Fridge Door Lamp		Reed Switch, Max DC 200V, 0.5A				
	Door reed switch		Reed Switch, Max DC 200V, 0.5A				
	Cam Cube Motor Switch		Reed Switch, Max DC 200V, 1A				-
	Guide Ice Route Motor Switch		Reed Switch, Max DC 200V, 0.5A				-
	Lever Switch		Micro Switch, AC125/250V, 1 or 16A				-
	LCD On/Off(Rocker) Switch		AC125/250V,12/6A	-	AC125/250V,12/6A	-	-
	Power cord		US : AC125V,10A / Other : AC250V,10 or 16A				
	Grounding screw		BSBN(BRASS SCREW)				

PRODUCT SPECIFICATIONS

2-4. Interior Views

2-4-1. Dispenser-featured models (RS22T5***, RS27T5***)

01 Freezer shelves
02 Multipurpose freezer bin
03 Freezer drawer
04 Refrigerator bin
05 Refrigerator shelves
06 Vegetable drawer
07 Multipurpose drawer
08 Ice maker *
09 Water filter

* Applicable models only

NOTE
The design of the egg container may differ with the model.

PRODUCT SPECIFICATIONS

2-4-2. Dispenser-featured models (RS28T5B**)

01 Freezer shelves
02 Multipurpose freezer bin
03 Freezer drawer
04 Refrigerator bin
05 Refrigerator shelves
06 Vegetable drawer
07 Multipurpose drawer
08 Ice maker *

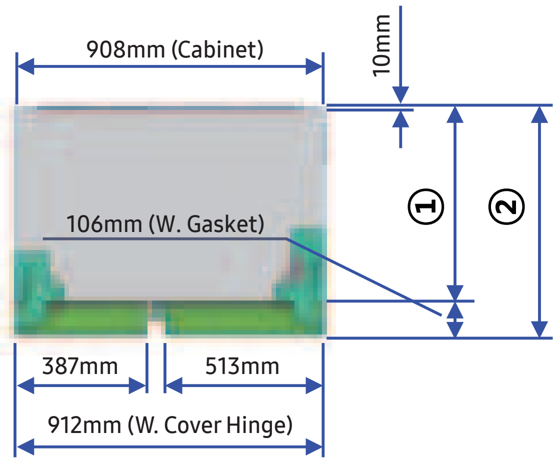
* Applicable models only

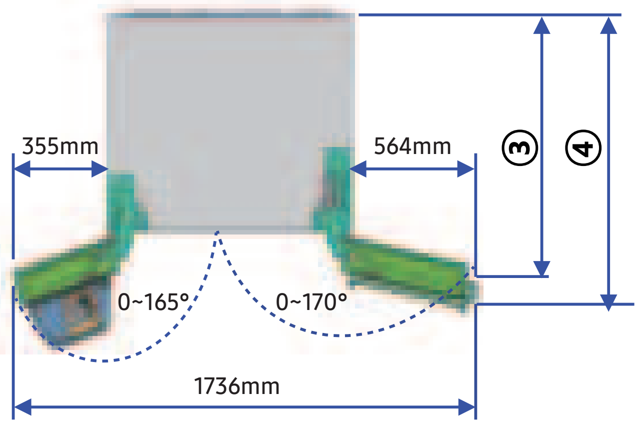
NOTE

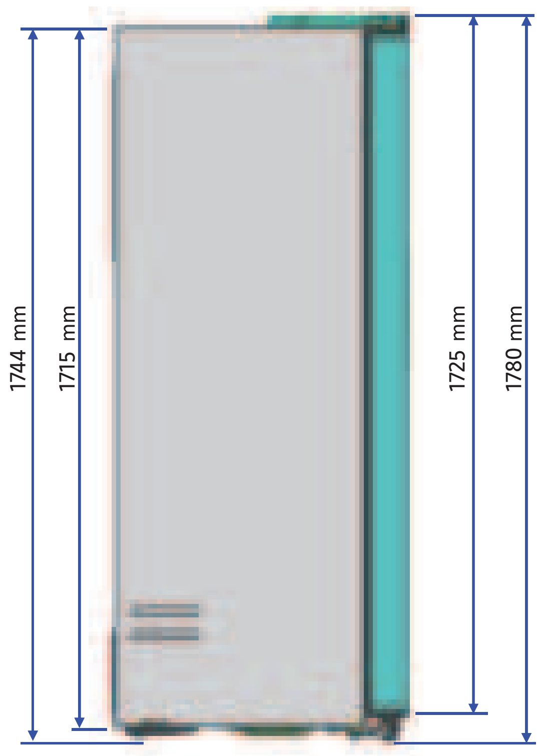
The design of the egg container may differ with the model.

PRODUCT SPECIFICATIONS

2-5. Dimensions (mm/inch)

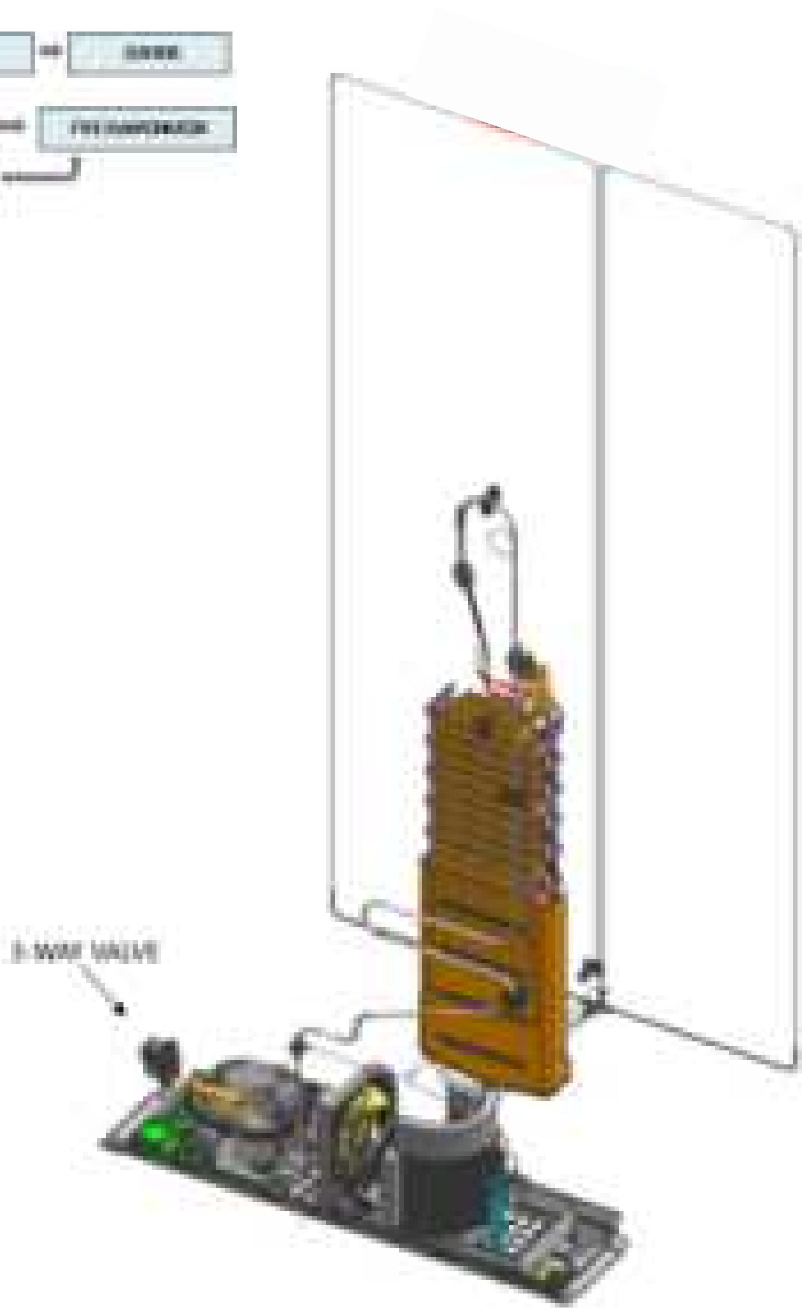




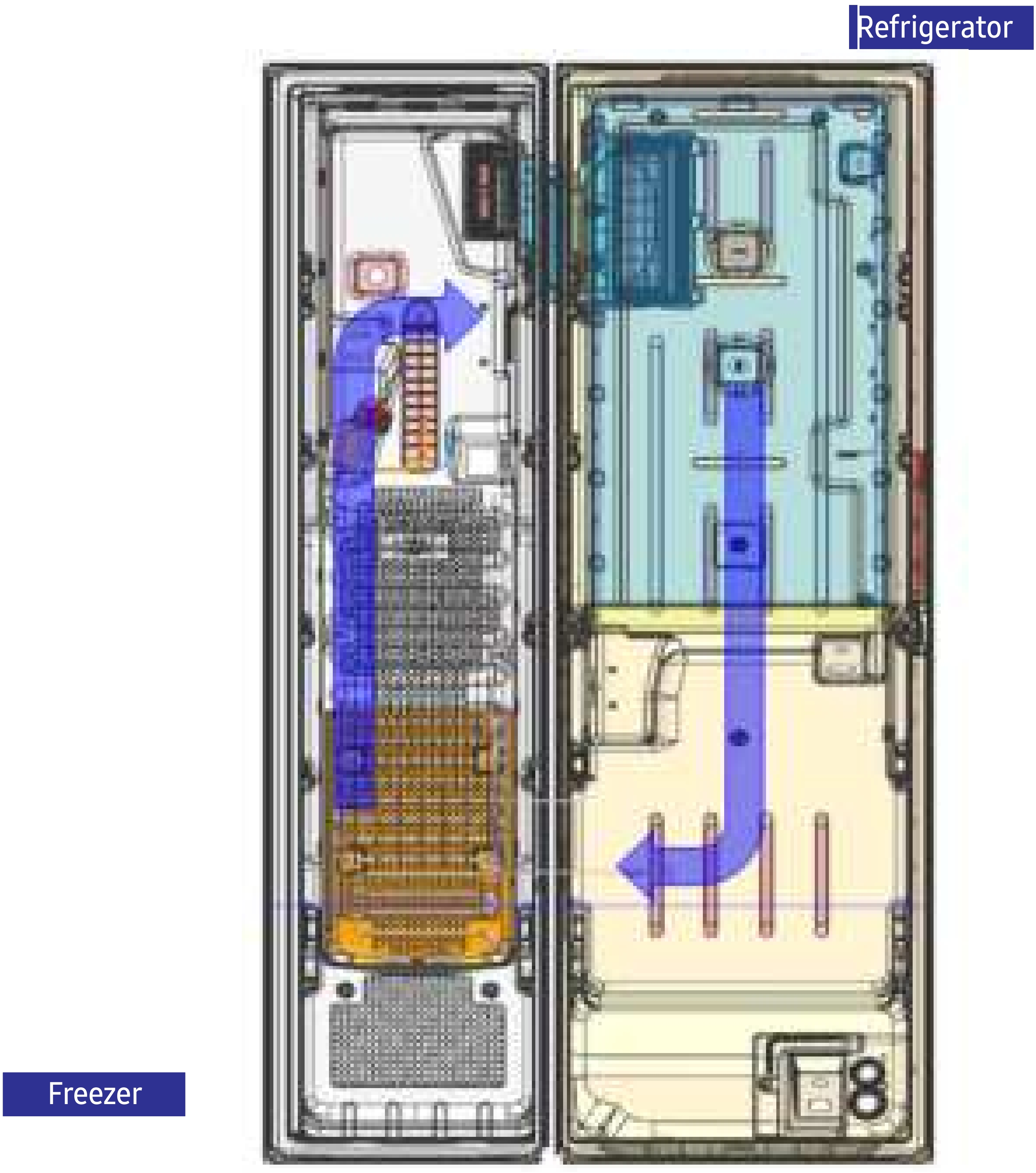


No.	RS22T55** RS22T52**	RS27T55** RS27T52** RS28T5B**
①	620mm	745mm
②	726mm	851mm
③	772mm	897mm
④	857mm	982mm

2-6) Refrigerant Route in Refrigeration cycle



2-7 Cooling Air Circulation



2-8. Fridge Manager

Fridge Manager

To access Fridge Manager, swipe the Home screen to the left, and then tap the **Fridge Manager** widget.

- The design of the **Fridge Manager** widget differs with the model.
- On the Fridge Manager widget, the current fridge and freezer temperatures are displayed.
- For detailed settings, tap widget to open the Fridge Manager.

NOTE

If you tap **Buy Filter** on the second page of the Fridge Manager widget, you will be directed to the website where you can purchase a water filter.

01 Temperature
Changes the set Temperature in the fridge or freezer.
You can also turn on and turn off the refrigerator's Power Cool function and the freezer's Power Freeze function.

02 Special Features
View a short explanation of some of the refrigerator's special features.

03 Fridge Settings
Change many of the fridge's settings including settings for ice making, the door alarm, and the temperature units (°F or °C).
See the next page for details.

04 External Conditions
Displays the external (ambient) temperature and humidity.

NOTE

The displayed conditions may differ from the actual temperature and humidity.

NOTE

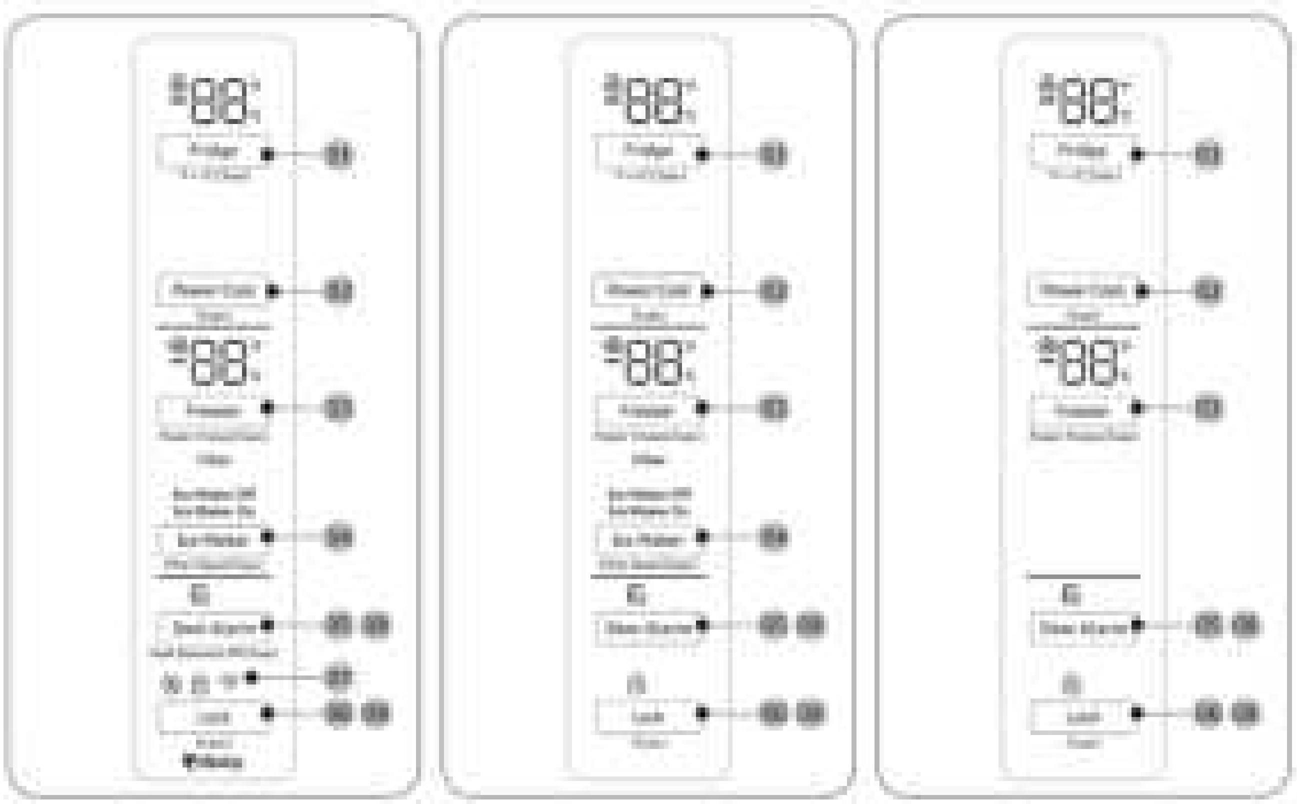
Also displayed is the **Ice Off** (ice making off) indicator at the top, center.
When the indicator displays **Ice Off**, the refrigerator's ice maker is turned off.

Ice Maker (applicable models only)	Turns the ice maker on and off. Tap and drag the button to turn on or off. Note that if the ice bucket is full of ice, the refrigerator does not start making ice when you tap and drag this button (turning the Ice Maker on), but displays the Ice Full indicator on the main screen. If you hold down the dispenser lever for 5 seconds, the Ice Maker Off status changes to Ice Maker On.
Dispenser Lock (applicable models only)	Turns the ice and water dispensers on and off. Tap to set Dispenser Lock on or off.
Door Alarm	The door alarm sounds if you leave the door open. Tap and drag the button to turn on or off.
Temp. Unit	Switch the temperature scale between Celsius and Fahrenheit. Touch ˚F or ˚C to change the scale.
Water Filter (applicable models only)	Provides a water fillet replacement tutorial and lets you reset the water filter replacement indicator. Tap to open. NOTE • After installing the water filter, tap **Water Filter**, and then tap **Reset**. Tapping **Reset** re-initlallzes the function that measures the time remaining until the water filter needs to be replaced again. • If you tap **Buy Filter**, you will be directed to the website where you can purchase a water filter.
Self Check	Self Check is a self diagnose function. Tap to open. Tap **Start** to run.
Demand Response (applicable models only)	Works with the Smart Grid energy saving manager. Tap to open. Tap and drag the button to turn on or off. See the Smart Grid section in this manual for more information.
Cooling Off	Cooling Off mode (also called shop mode), is designed for use by retailers when they are displaying refrigerators on the shop floor. In Cooling Off mode, the refrigerator's fan motor and lights work normally, But the compressors do not run, and the refrigerator and freezer do not get cold. If cooling off is turned on, all cooling controls will turn to OFF on the Fridge Manager • To activate Cooling Off, tap **Activate > Proceed** from **Cancel/Proceed**. • To deactivate Cooling Off, tap **Deactivate > Proceed** from **Cancel/Proceed**.

PRODUCT SPECIFICATIONS

2-9. Using the control panel

Type A Type B Type C



01 Fridge / °F ↔ °C (3 sec)

02 Power Cool (3 sec)

03 Freezer / Power Freeze (3sec)

04 Ice Maker / Filter Reset (3 sec) *

05 Door Alarm / Peak Demand Off (3 sec) *

06 Lock(3sec)

07 Network connection *

08 Sabbath Mode

* Applicable models only

PRODUCT SPECIFICATIONS

01 Fridge / °F ↔ °C (3 sec)

Fridge	You can use the Fridge button to set the fridge temperature. • Press Fridge repeatedly to select a desired temperature between 34°F (1°C) and 44°F (7°C). - The temperature indicator displays the currently set or selected temperature.
°F ↔ °C	You can also use the Fridge button to switch the temperature scale between Celsius and Fahrenheit. To switch the temperature scale, press and hold Fridge for 3 seconds to change the current temperature scale.

02 Power Cool (3 sec)

Power Cool	Power Cool speeds up the cooling process at maximum fan speed. This is useful to quickly cool food that spoils quickly, or after the door is left open for some time. The fridge keeps running at full speed for several hours and then returns to the previous temperature. Press and hold Power Cool for 3 seconds.

03 Freezer / Power Freeze (3 sec)

Freezer	You can use the Freezer button to set the freezer temperature. • Press Freezer repeatedly to select a desired temperature between 5°F (-15 °C) and -8°F (-23°C). - The temperature indicator displays the currently set or selected temperature.
Power Freeze	Power Freeze speeds up the freezing process at maximum fan speed.The freezer keeps running at full speed for 50 hours and then returns to the previous temperature. • To activate Power Freeze, press and hold Freezer for 3 seconds. The corresponding indicator ( ) lights up, and the refrigerator will speed up the freezing process for you. • To deactivate, press and hold Freezer for 3 seconds again. The freezer returns to the previous temperature setting. • To freeze large amounts of food, activate Power Freeze for at least 20 hours before putting food in the freezer. **NOTE** Using Power Freeze increases power consumption. Make sure you turn it off and return to the previous temperature if you do not intend to use it.

04 Ice Maker / Filter Reset (3 sec)

Ice Maker	Press and hold Ice Maker for more than 3 seconds to turn the ice maker on or off. The ice maker has 2 indicators (ON/OFF) to indicate the operating status. • When the ice maker operates, the ON indicator turns on. • When the ice maker is turned off, the OFF indicator turns on. In this case, ice making is disabled even if you press the Cubed Ice or Crushed Ice buttons on the dispenser panel. To enable ice making, you must turn the ice maker on. NOTE • If the ice maker is turned off, it will turn on if you press and hold the ice lever for more than 5 seconds. • After tuning off the ice maker by inside cabinet control panel, this function doesn't work for 1 minute.
Filter Reset	• After about 6 months (approximately 300 gallons) of using the original water filter, the Filter indicator turns red to remind you that the filter needs to be replaced. The Filter indicator blinks red for several seconds when you open the door, reminding you that the filter need to be replaced. If this happens, replace the filter, and press and hold Filter Reset for 3 seconds. The filter indicator will be reset, and the Filter indicator will be reset, and the filter indicator turn off. NOTE • Some regional areas have relatively large amounts of lime contained in the water. This may reduce the lifecycle of the filter. In these areas, you will have to replace the water filter more often than specified above. • If water is not dispensing properly, the water filter is most likely clogged. Replace the water filter.

05 Door Alarm / Peak Demand Off (3 sec)

Door Alarm	If the door is left open for more than 2 minutes, an alarm will sound and the alarm indicator will blink. You can toggle the alarm on and off by pressing Door alarm. The alarm is enabled by factory default. If the door is left open for 5 minutes, internal lights (in the fridge, freezer) will blink for 10 seconds and then turn on. This will be repeated every minute for 5minutes. This is to alert hearing-impaired users that a door is open and it is normal.
Peak Demand Off	The Peak Demand Off function activates/deactivates Smart Grid. Press Door Alarm for 3 seconds to set/clear the Peak Demand Off function.

06 Lock (3 sec)

Lock	To prevent accidental setting changes, press and hold Lock for 3 seconds. The main panel will be disabled and the Lock indicator ( ) will turn on. When Lock is on, the indicator blinks if any button on the main panel is pressed. If you press and hold the button again for more than 3 seconds, Lock will be deactivated. However, Lock will be reactivated if no button is pressed within 1 minute. NOTE Lock on the main panel does not affect the dispenser. The dispenser operates independently and so does the dispenser lever. To lock the dispenser and the dispenser lever, use the Lock function on the dispenser.

07 Network connection

Network connection	You can control and monitor your refrigerator on the SmartThings app. For more information, see the SmartThings section.

08 Sabbath Mode (LED Panel Model Only)

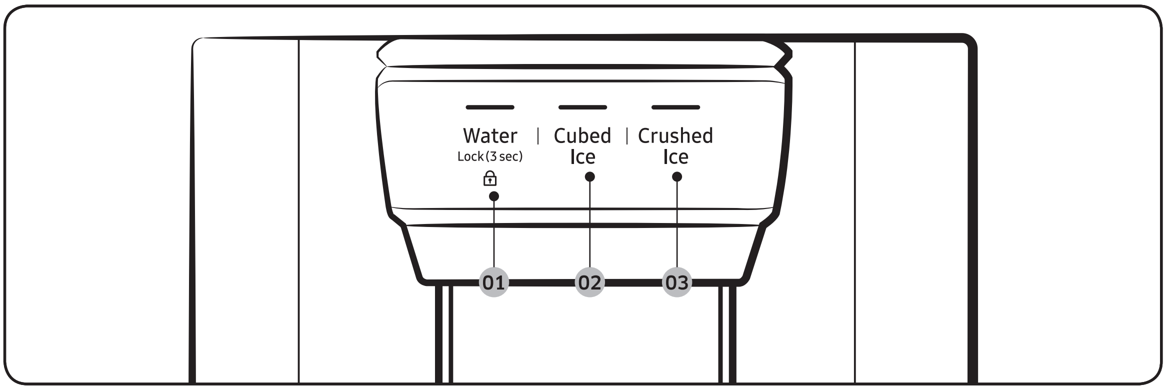
Sabbath Mode	The Sabbath mode stays active for 85 hours once it is activated. After that, it will be deactivated automatically. • To activate, press and hold Door Alarm and Lock simultaneously for 5 seconds to enter Sabbath mode. Then the refrigerator operates in Sabbath mode where the buttons, the display, and the room lamps are all under control. • To deactivate, press and hold the buttons again for 5 seconds to exit Sabbath mode. NOTE Even after the refrigerator powers off and restarts, the Sabbath mode remains active. To exit, you must deactivate it first.

09 Cooling Off (North America model)

Cooling Off	Cooling Off mode (also called Shop mode) is designed for use by retailers when they are displaying refrigerators on a retail floor. In Cooling Off mode, the refrigerator’s fan motor and lights operate normally, but the compressors do not run so that neither the refrigerator nor the freezer run the cooling process. To enter or exit the Cooling Off mode, press and hold Fridge and Power Cool simultaneously for 6 seconds. Then, when the display blinks, press Lock. When the Cooling Off mode is on, the refrigerator chime sounds, and the refrigerator shows “0” on the fridge temperature display and “FF” on the freezer temperature display.

PRODUCT SPECIFICATIONS

Dispenser panel (applicable models only)



01 Water / Lock **02** Cubed Ice **03** Crushed Ice

01 Water / Lock (3 sec)

Water	To dispense chilled water, press Water. The corresponding indicator turns on.
Lock (Dispenser panel / Dispenser lever)	To prevent use of the dispenser panel buttons and the dispenser lever, press and hold Water for more than 3 seconds. If you press and hold the button again for more than 3 seconds, the dispenser lock will be deactivated. When dispenser lock is on, the indicator blinks if any button on the dispenser panel is pressed or the dispenser lever is pressed. **NOTE** The Lock on the dispenser panel does not affect the controls on the main panel. To lock the main panel, use the Lock function on the main panel.

02 Cubed Ice

Cubed Ice	Press Cubed Ice to dispense cubed ice. The corresponding indicator turns on.

03 Crushed Ice

Crushed Ice	Press Crushed Ice to dispense crushed ice. The corresponding indicator turns on.

PRODUCT SPECIFICATIONS

2-10. Optional Material Speci ication

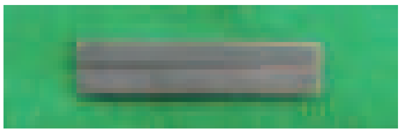
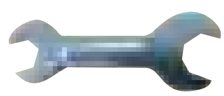
Photograph	Part Name	Part Code	Quantity	Remark
	FILTER-WATER	DA63-08839B	1	For Plumbing Model
	ASSY INSTALL-FILTER	DA97-01469W	1	For Plumbing Model (Except for North America)

3. DISASSEMBLY & REASSEMBLY

3-1. Precaution

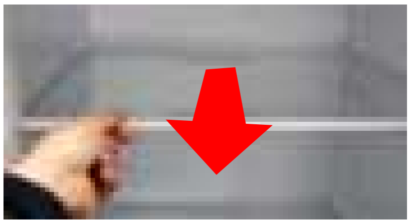
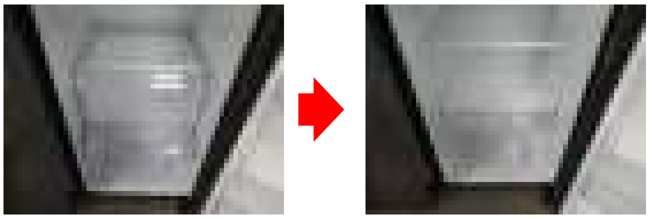
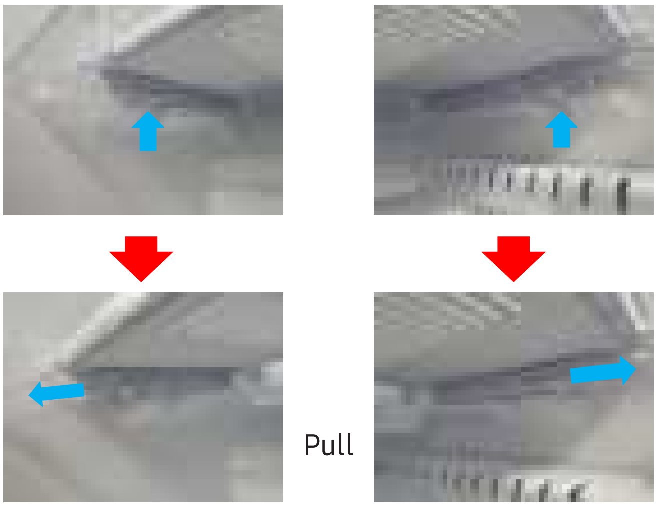
- Unplug the appliance before servicing or replacing electrical parts.
- Remove any foreign matter or dust from the power plug pins.
  - Otherwise there is a risk of fire.
- Do not use a cord that shows cracks or abrasion damage along its length or at either end.
- Do not plug several appliances into the same multiple power board. The refrigerator should always be plugged into its own individual electrical which has a voltage rating that matched the rating plate.
  - This provides the best performance and also prevents overloading house wiring circuits, which could cause a fire hazard from overheated wires.
- Do not install the refrigerator in a damp place or place where it may come in contact with water.
  - Deteriorated insulation of electrical parts may cause an electric shock or fire.
- In order to reduce the risk of electric shock the appliance must be properly grounded.
- Do not put bottles or glass containers in the freezer.
  - When the contents freeze, the glass may break and cause personal injury.
- Do not store volatile or flammable substances in the refrigerator.
  - The storage of benzene, thinner, alcohol, ether, LP gas and other such products may cause explosions.

- **NEED TOOL**

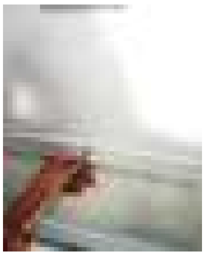
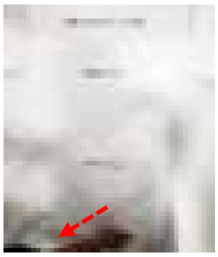
Item	How to use	Pictures
Phillips Head Driver	Use for assembling and disassembling of screw.	
Flat Head Driver	Use for assembling and disassembling of Beverage Station, Dispenser, Display, Cover Lamp etc...	
Magnet	Use for checking of the F/R Fan.	
Hexagon wrench (4mm diameter)	Use for adjust Door Hinge Ref Up.	
Box wrench (12mm)	Use for disassemble the compressor.	
Box wrench (10mm)	Use for disassemble the hinge lower.	
Spanner(19mm)	Use for adjust the height of the door.	

DISASSEMBLY & REASSEMBLY

3-2. Interior-Fridge

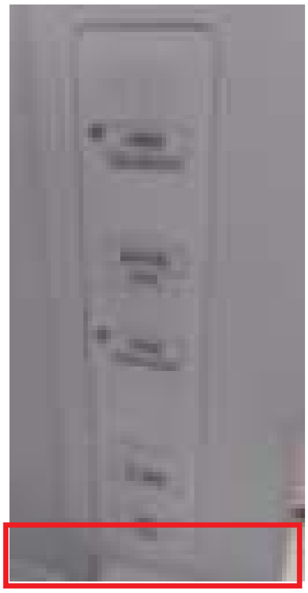
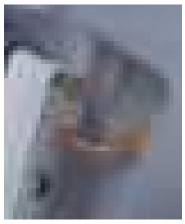
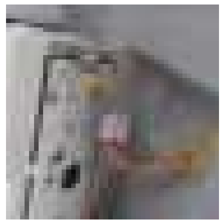
Part Name	Work sequence	Remarks
Shelf	Pull the Shelf out to the front.	
Drawer cover	Remove all Drawers before disassembling the cover.	
	After push up left, right hook, pull the filp cover out to the front.	 

3-3. R-Compartment Duct

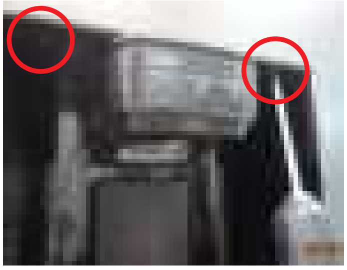
Part Name	Work sequence	Remarks
Multi Duct (Fridge)	1. Remove a screw first and pull the Multi Cover in the direction of the front.	 Cap Screw  Screw(1EA) 
	2. Remove a housing Connector.	

DISASSEMBLY & REASSEMBLY

3-4. Cover-Display

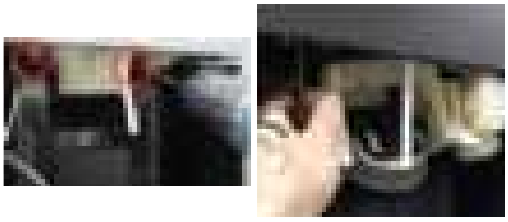
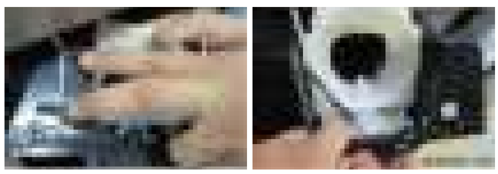
Part Name	Work sequence	Remarks
Cover-Display	Find the location of the serviceable part.	
	Separate using '-' driver. (If necessary, remove the inlay first and service it.)	
	Press and disconnect the wire connector connected to the cover display.	 

3-5. Dispenser Display & Assy Guide Ice Route

Part Name	Work sequence	Remarks
Assy Guide Ice Route	Remove 2 Screw bottom of the Display.	
	Disengage the the Display to lower direction by Pushing. ※ When decomposing, do not contact the lever.	  Protecting lever dispenser
	Disengage Housing of Display.	

DISASSEMBLY & REASSEMBLY

3-6. Dispenser Display & Assy Case Ice Route

Part Name	Work sequence	Remarks
Assy Case Ice Route	Pull out GUIDE ICE(white one) from Assy Case Ice Route. (Push L&R hooks like right picture.)	
	Remove 2 screws on Assy Case Ice Route and disconnect housing inside the Case Dispenser and remove Assy Case Ice Route.	
	■ **Assembly** Assembling housing of ASSY CASE ICE ROUTE Upper 2 Rib have to match with inside Hook and Screw assembling. Assembling GUIDE ICE to the ASSY CASE ICE ROUTE. CAUTION Upper Hook have to completely match with Hook for prevent the leak of Gasket Cover. Assy Case Ice Route 1 lib must be behind Case Dispenser lib. (Pic 2)	
	Display Housing assembling and Fixing the Harness with Hook .	
Assy Cover Dispenser	Input the water line to Display hose hole and assembling 2 screws.	

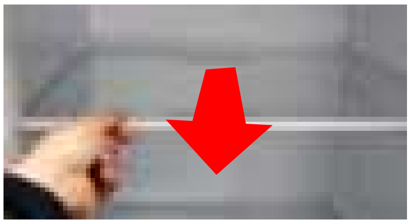
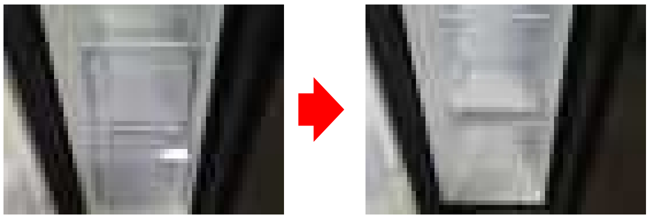
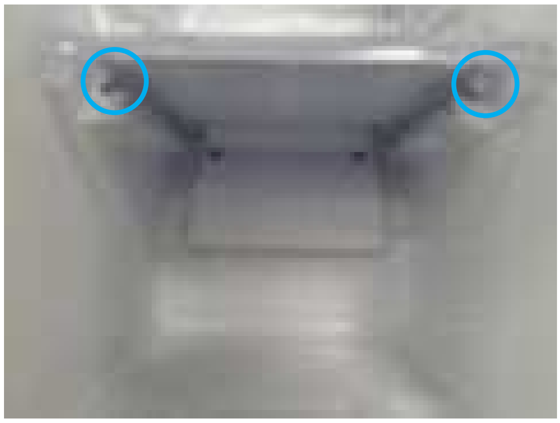
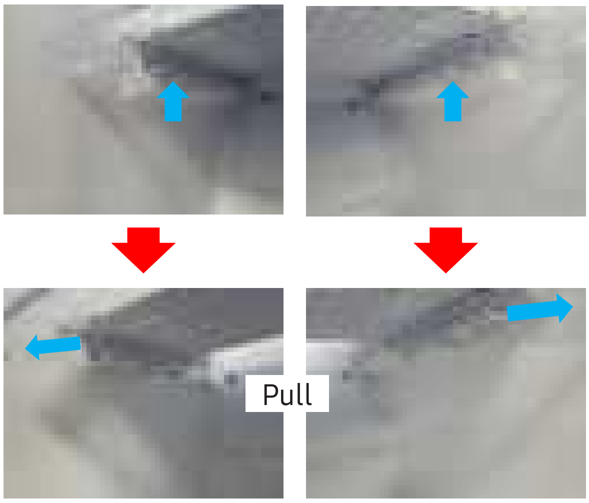
DISASSEMBLY & REASSEMBLY

3-7. Assy Lever Dispenser

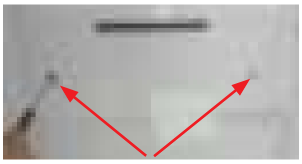
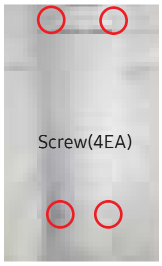
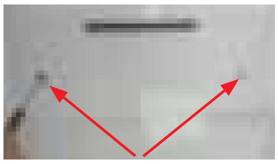
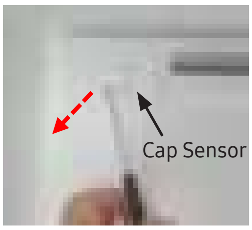
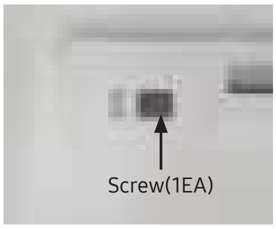
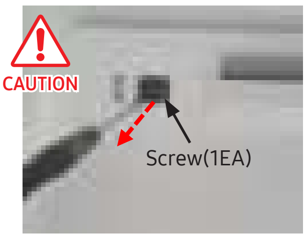
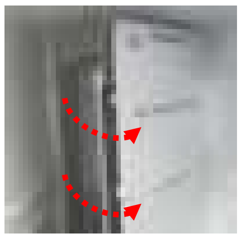
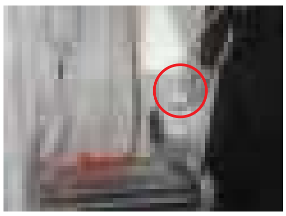
Part Name	Work sequence	Remarks
Assy Lever Dispenser	■ **Disassembly** 1. Put both fingers in the upper of lever. Pull and remover the lever slowly.	
	■ **Assembly** 2. Put the lever with adjusting to rectangle block slowly.	

DISASSEMBLY & REASSEMBLY

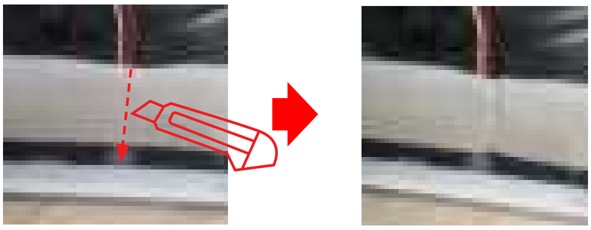
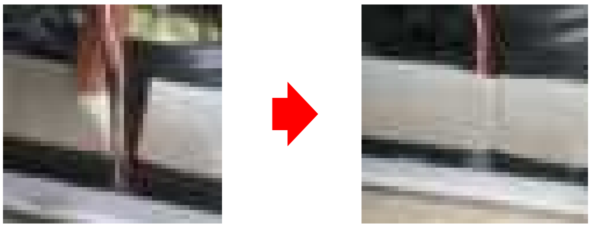
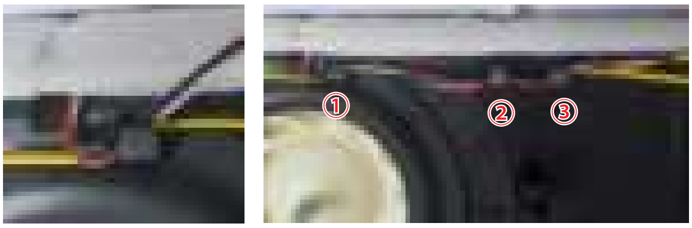
3-8. Interior-Freezer

Part Name	Work sequence	Remarks
Shelf	Pull the Shelf out to the front.	
Drawer cover	Remove all Drawers before disassembling the cover.	
	After push up left, right hook, pull the filp cover out to the front.	 

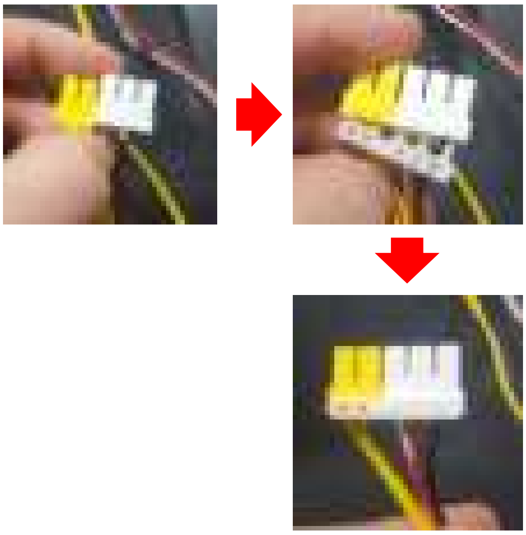
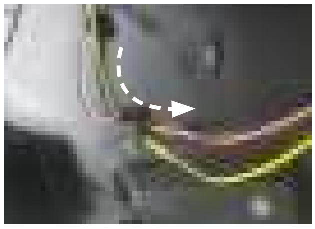
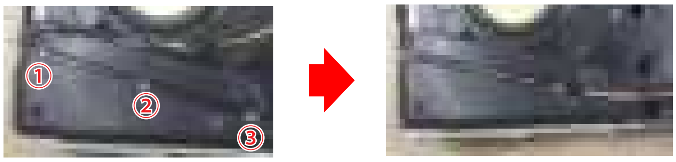
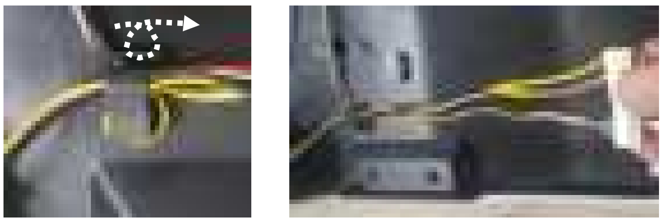
3-9. F-Compartment Duct

Part Name	Work sequence	Remarks
F-Compartment Duct & Evap Cover	1. Remove the screws (4ea) from the Evap Cover. (Before remove upper 2 screws, Remove 2 cap screws)	  Cap Screw(2EA) 
	2. Remove the Evap Cover from The cabinet. Pull the Cover in the direction of the front.	
Multi Duct (Freezer)	1. Remove two screws and Evap Cover First.	   Screw(2EA)
	2. Remove Cap Sensor using tools. CAUTION DO NOT force the Multi Cover off before next step. There is a screw on the back sideof this Cap sensor.	 
	3. Remove a screw first and pull the Multi Cover in the direction of the front.	 
	4. Disconnect two Housing Connectors.	

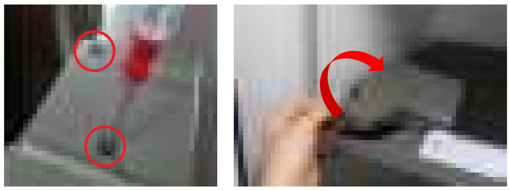
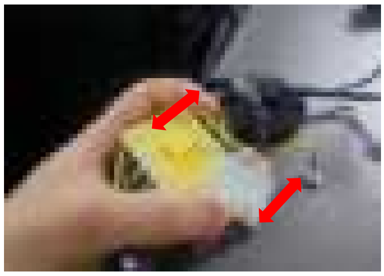
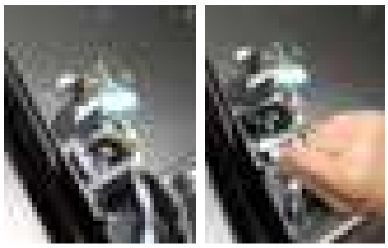
3-9-1. F-Compartment Duct (Wire Fixing Instruction in detail)

Part Name	Work sequence	Remarks
Multi Duct (Freezer)	[Wire disassembly] Cut the seal along the side of wire line.	
	[Wire re-assembly] Fix wires first and then cover the seal.	
	[Sensor wire fixing] Insert the wire into each 4 hooks.	
	[Fan wire fixing] Insert the wire into each 3 hooks.	

3-9-2. F-Compartment Duct (Wire Fixing Instruction in detail)

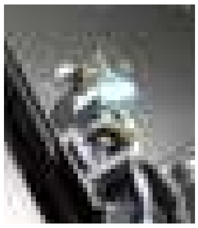
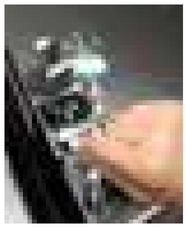
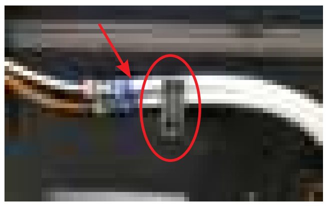
Part Name	Work sequence	Remarks
Multi Duct (Freezer)	[Sensor&Fan Wire fixing] 1. Sensor wire connector and Fan wire connector have to be assembled together with a Fixer.	
	[Sensor&Fan Wire fixing] 2. Pass the Sensor wire and Fan wire into a finish hook.	
	[Damper motor wire fixing] 1. Insert the wire into each 3 hooks.	
	[Damper motor wire fixing] 2. Wind the wire twice into a finish hook All remained wires have to be same length	

3-10. Assy Door Disassembling (FreezerDoor)

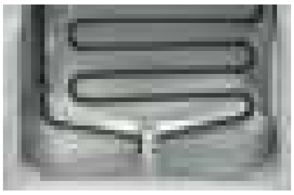
Part Name	Work sequence	Remarks
Door	1. Open the Door and remove 2 fitting of the Hose at the bottom. Remove the fitting by pulling the hose while hold the left cap of the fitting in the direction of the arrow.	
	2. Disconnect the power cord, and remove 2 cover screws using a Phillips screwdriver. Then open the Freezer door and pull the hooks on the sides to loosen the cover. Lift up the cover towards you to detach it.	
	3. Disconnect the two top housings.	
	4. Remove the three hinge-link holding the hinge.	

Part Name	Work sequence	Remarks
Door	5. Disassemble the top hinge from the front of the door.	
	Remove the door from the lower hinge by lifting up the door straight. CAUTION Be careful not to pinch the water tubing and the wire harness on the door Lift up the door straight in upper direction until the water hose is eliminated from lower hinge. Do not pull the door forward to remove it. It cause injury of water hose and tubing.	

3-11. Assy Door Disassembling (Fridge Door)

Part Name	Work sequence	Remarks
Door	1. Disconnect the power cord, and remove 2 cover screws using a Phillips screwdriver. Then pull the the hooks on the sides to loosen the cover. Lift up the cover towards you to detach it.	 
	2. Disconnect the top housing.	
	3. Remove the three hinge-link holding the hinge.	 
	4. Disassemble the top hinge from the front of the door.	
	5. Disassemble the Fridge door.	
Reassembly doors	• The Water Line must be fully inserted to the center of transparent coupler (Type A) or the line (Type B) to prevent water leakage from the dispenser. • Check that it holds the line firmly. • The cover hose must be fully covered water hose. (Make the cover hose fit to end of the fitting) • Bottom hook must fix the cover hose to prevent loosen the hose line.	 

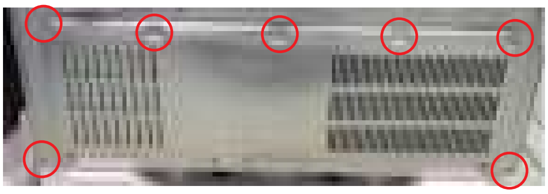
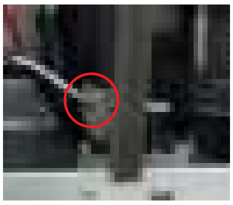
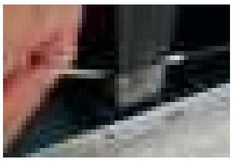
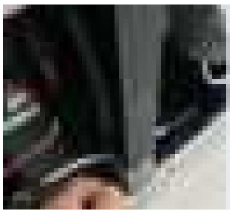
3-12. Evaporator

Part Name	Description	Figure	
		Freezer	Fridge
Evaporator	Disengage the housing connector. (right)		-
	Remove the evaporator by pulling the lower part of the evaporator while lifting it up.		-
	Disengage the housing connector. (right)		-
	Remove the evaporator by pulling the lower part of the evaporator while lifting it up.		-

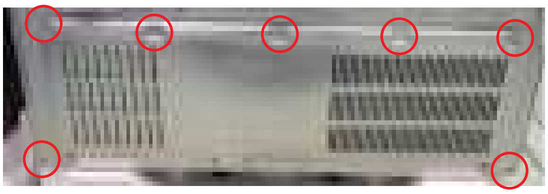
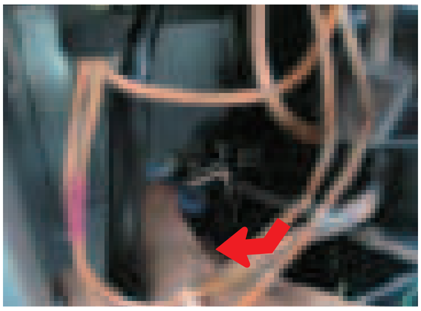
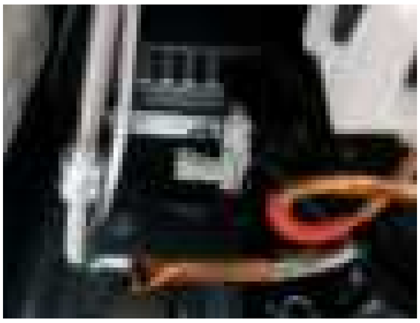
3-13. Main PCB and Inverter PCB Disassembling (whole)

Work sequence	Remarks
1. Pull the refrigerator forward to make space for servicing.	
2. Disassemble the four screws. - New issued model use 5 screw.	
3. Disassemble the fourteen connectors of the housing(Number of the housing connectors may vary dependent upon models and functions). While pressing the upper hook on the location shown on the figure, pull the main main PBA and remove it.	

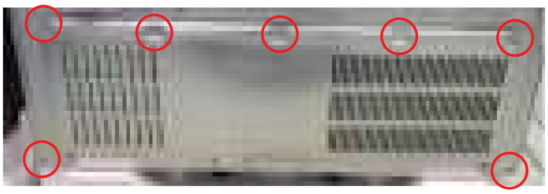
3-14. Comp Cooling Fan

Work sequence	Remarks
1. Unscrew 7 screws of COVER COMP.	
2. Remove screw from the Assy Support Circuit Motor.	
3. Separate Assy Support Circuit Motor connect.	
4. Cutting seal.	
5. Pipe moved down side.	
6. Assy Support Circuit Motor turn out and assy in reverse.	

3-15. Relay Cover disassembling

Work sequence	Remarks
1. Remove 7 screws from the Comp Cover.	
2. Remove the cover relay with a flat screwdriver. CAUTION Pipe bending or exercise of excessive force may pinching on the hands when removing the cover.	
3. User a flat driver to disassemble the relay protector from the compressor.	 

3-16. Step Valve Disassembling (whole)

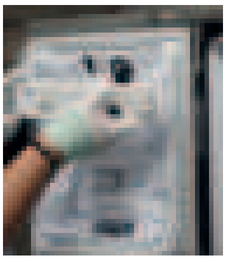
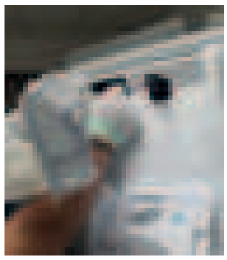
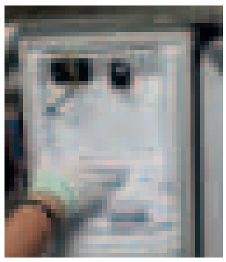
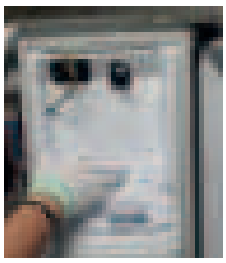
Work sequence	Remarks
1. Remove 7 screws from the Comp Cover.	
2. Remove 1 screw.	
3. Remove 1 screw.	
4. Remove the refrigerant, and disassemble the step valve from the connection pipe. CAUTION Exerting excessive force when forming the pipe causes bending of the pipe.	

CAUTION

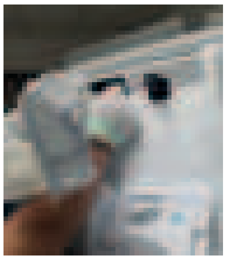
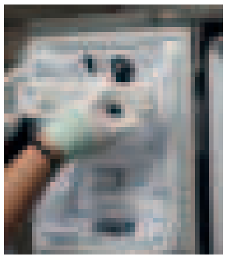
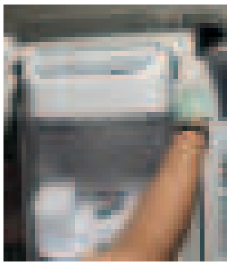
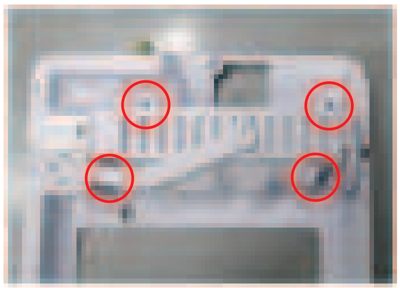
Cautions for R-600a : Environment-friendly gas (R-600a) is used as refrigerant at this refrigerator, which is combustible gas though in volume kept small. If gas is leaked by serious damage to the refrigerator during transportation, installation and operation may cause fire or burn when spark makes contact with leaked gas.

DISASSEMBLY & REASSEMBLY

3-17. Ice Maker Compartment (Indoor Auger Motor)

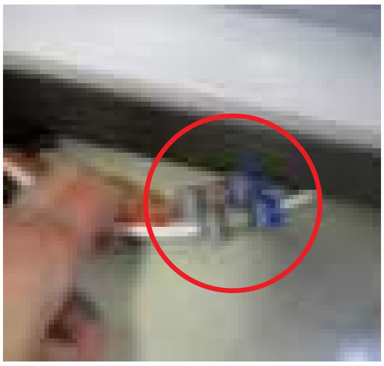
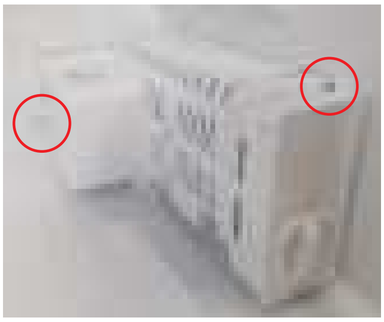
Part Name	Description	Figure
Assy Tray Ice * Indoor only	Disassemble the Assy Tray Ice by holding the handle and pulling up The bottom part slightly.	
Assy Case Auger Motor (Disassemble) * Indoor only	Lift up the Ice Maker Cover and gently pull out.	
	Unscrew the 5 screws	
	Remove the Assy Ice Maker-Mech (Left Picture) Disconnect the housing connectors by gently pulling.	 
	Disconnect the housing connectors by gently pulling. Remove the Assy Case Auger Motor (Right Picture)	 
Assy Case Auger Motor (Assemble) * Indoor only	Place the Assy Case Auger Motor in Panel Door. Connect the housing connector.	 

DISASSEMBLY & REASSEMBLY

Part Name	Description	Figure
Assy Case Auger Motor (Assemble) * Indoor only	Connect the housing connector. Place the Assy Ice Maker-Mech in the Panel Door (Right Picture)	 
	Screw the 5 screws.	
	Place the Ice Maker Cover and gently push.	
Assy Ice Maker * Indoor only	Lift up the Ice Maker Cover and gently pull out.	
	Unscrew the 4 screws.	
	Disassemble the Ice Maker Assy.	

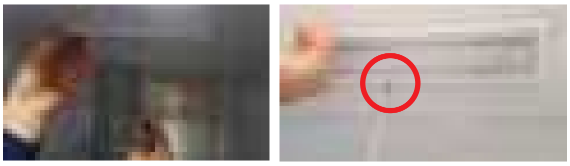
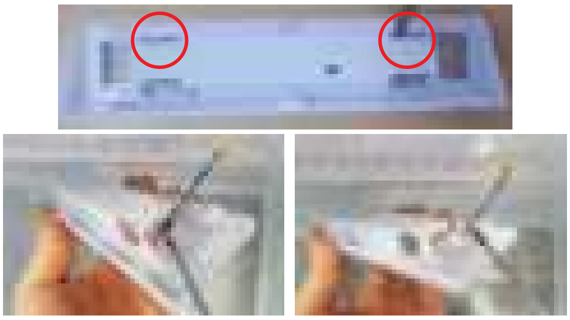
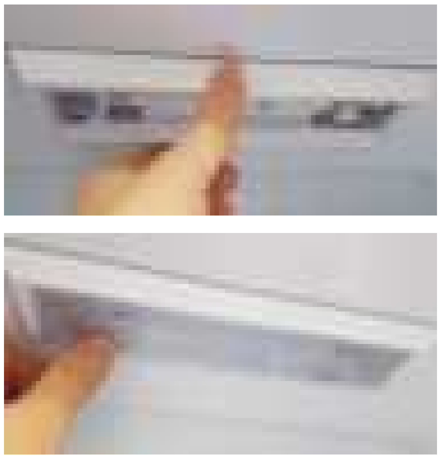
DISASSEMBLY & REASSEMBLY

3-18. Water tank

Part Name	Description	Figure
Water tank ※indoor plumbing model	1. Separate the hose and fitting from the bottom of the fridge door.	
	2. Disassembly 2 screws.	
	3. Disconnect the housing and Pull out the Hose. CAUTION To prevent the folding hose for assembling : Input the OJC-Seal into the hole over ½ or fully.	
	4. Check the Water Dripping. (Have to remove Air bubbles in the Water tank)	

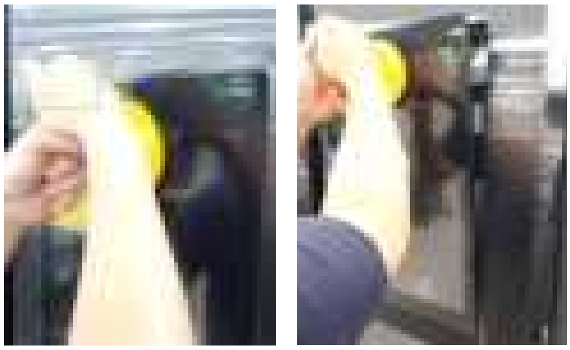
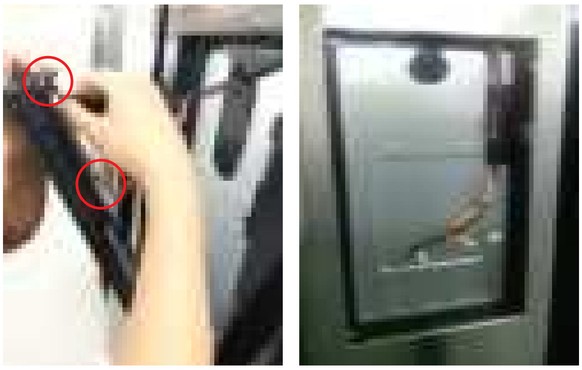
DISASSEMBLY & REASSEMBLY

3-19. LED Lamp

Part Name	Description	Figure
LED Lamp	1. Disassemble the case lamp using a flat head driver. Disassemble from the back of the case lamp.	
	2. Remove the housing by pressing the housing connection using a flat head driver. - IF you remove COVER LAMP before, you can do it easily.	
	3. Assemble the new lamp in the case to be replaced and connect it to the top housing.	
	4. Assemble the case at the top and assemble the new cover.	

DISASSEMBLY & REASSEMBLY

3-20. LCD

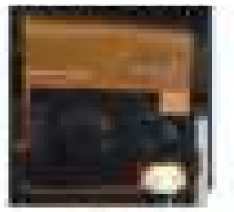
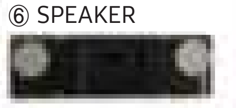
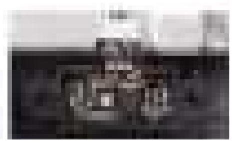
Part Name	Description	Figure
Disassembling LCD	1. Remove a cap screw and a screw.	
	2. Attach a suction cup to the top of the display and pull out the display. Make sure you hold the display at the bottom.	
	3. Disconnect carefully the wiring connectors. CAUTION Never lay the panel face down on a flat surface. Place the display on a blanket or mat	

DISASSEMBLY & REASSEMBLY

3-21. Assy Panel-Camera

Part Name	Description	Figure
Disassembling CAMERA	1. Remove the top door bin of right fridge door.	
	2. Remove a cap screw and a screw.	 
	3. Disconnect carefully the wiring connectors.	 

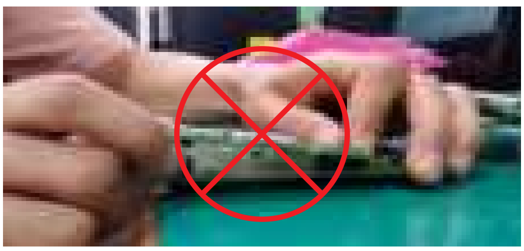
3-22. How to remove the subcomponents from the Assy Case Display

How To Do	Descriptive Picture
1. Remove 6 screws, and release the latches to remove the Case Housing from the Assy Case Display. ※ Componets ① LCD PBA ② Touch FPCB Cable ③ LCD FLAT Cable ④ Wi-Fi/BT Module ⑤ MIC PBA Cable ⑥ SPEAKER ⑦ Backlight cable ⑧ Proximity sensor	
※ Dismantling components ① LCD PBA Remove the 4 housings first and then push the latch to remove the bracket. ② Touch FPCB Cable Remove a housing. ③ LCD FLAT Cable Remove a housing. ④ Wi-Fi/BT Module Separate from the LCD PBA. ⑤ MIC PBA Cable Remove a housing. ⑥ SPEAKER Remove a housing and Separate from LCD Assy. ⑦ Backlight cable Remove a housing. ⑧ Proximity sensor Remove a housing and Separate from LCD Assy.	① LCD PBA  ② Touch FPCB Cable  ③ LCD FLAT Cable  ④ Wi-Fi/BT Module  ⑤ MIC PBA Cable  ⑥ SPEAKER  ⑦ Backlight cable  ⑧ Proximity sensor 

※ Note when separating Wi-Fi Module

⚠ **CAUTION**

- When separating modules, give a uniform force to both sides to separate.
- Concerns about connector breakage when separating module only by one side.



DISASSEMBLY & REASSEMBLY

3-23. Removing the USB

Part Name	Description	Figure
USB	1. Push the USB Cover upwards, which is on the top left of the right fridge door.	 
	2. Pull the USB to the right and remove it. * Assembly is the reverse of disassembly.	

4. CHECK THE INSTALLATION STATUS

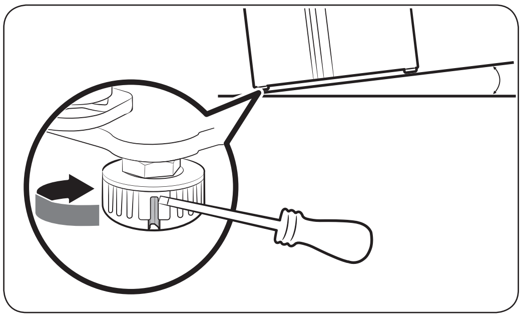
4-1. Adjust the leveling feet and door height

4-1-1. Adjust the leveling feet

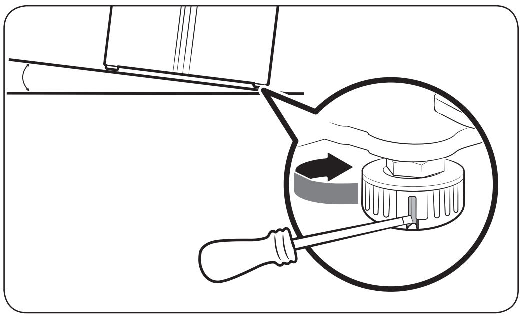
⚠ CAUTION

- The refrigerator must be levelled on a flat, solid floor. Failing to do so can cause damage to the refrigerator or physical injury.
- Levelling must be performed with an empty refrigerator. Make sure no food items remain inside the refrigerator.
- For safety reasons, adjust the front side a little higher than the rear side.

The refrigerator can be levelled using the front legs that have a special screw (leveller) for levelling purposes. Use a flat-head screwdriver for levelling.



■ **To adjust the height of the freezer side:**
Insert a flat-head screwdriver into the leveller of the freezer-side front leg. Turn the leveller clockwise to raise, or turn it counter clockwise to lower.

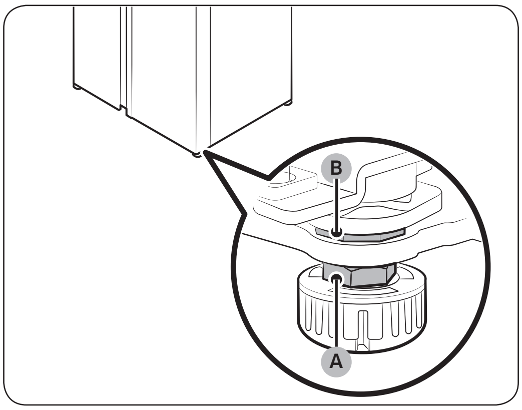


■ **To adjust the height of the fridge side:**
Insert a flat-head screwdriver into the leveller of the fridge-side front leg. Turn the leveller clockwise to raise, or turn it counter clockwise to lower.

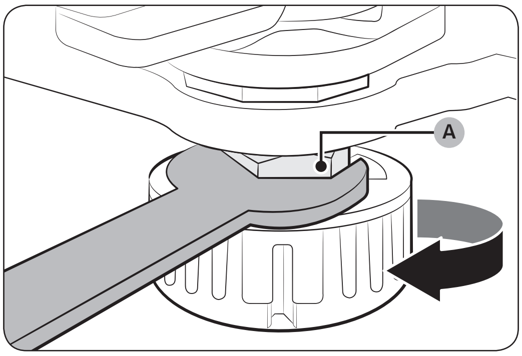
4-1-2. Adjust the door height as well as the door gap

To adjust the height of a door

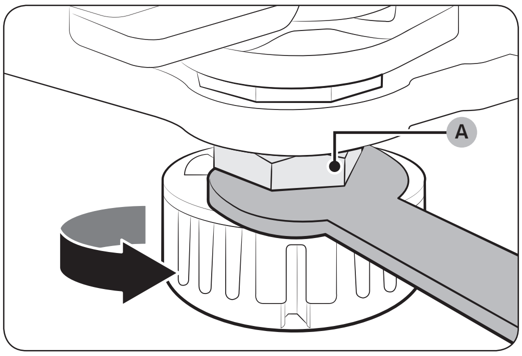
The height of fridge door can be adjusted using the clamp nut and the height nut on the front bottom of each door.



1. Open the door to adjust, and locate the two nuts (A) and (B) on the front bottom of the door.



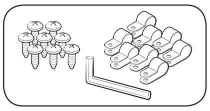
2. With a 19 mm spanner, turn the clamp nut (A) clockwise to loosen. Then, open the door, and do the following inside the door.
   - To raise the door, turn the height nut (B) counter clockwise.
   - To lower the door, turn the height nut (B) clockwise.



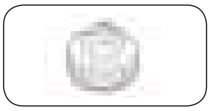
3. When complete, tighten the clamp nut (A) by turning it counter clockwise.

4-1-3. Connect the water dispenser line (applicable models only)

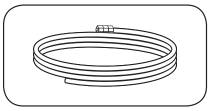
Installation (external dispenser line)



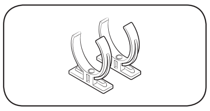
Water line fixer and screws*



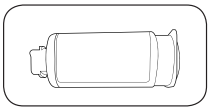
Coupler*



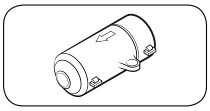
Water line*



Water filter lock-clip*



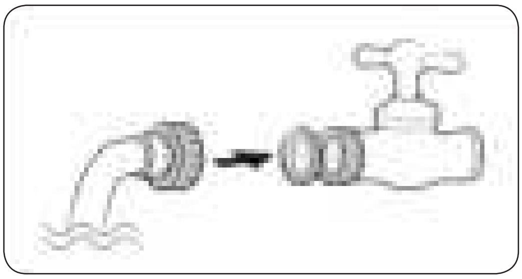
Water filter



Check valve*

* : Except for North America

Parts and tools required



1. Assemble the water pipe line using couplers.
2. Close the main water supply, and then close the water tap.
3. Connect the coupler to the cold water tap.

⚠ CAUTION

- Make sure you connect the coupler to the cold water tap. Connection to the hot water tap may cause the water filter to be clogged and fail to operate normally.
- The warranty for your refrigerator does not cover the water line installation. The water line installation will be performed at your own costs unless the installation fee is included in the retailer’s price.
- Samsung takes no responsibility for the water line installation. If water leaks occur, contact the installer of the water line.
- The water line must be repaired by a qualified professional. If you encounter a water leak, contact a local Samsung service centre or the installer of the water line.

4-1-4. Connect to a water source (applicable models only)

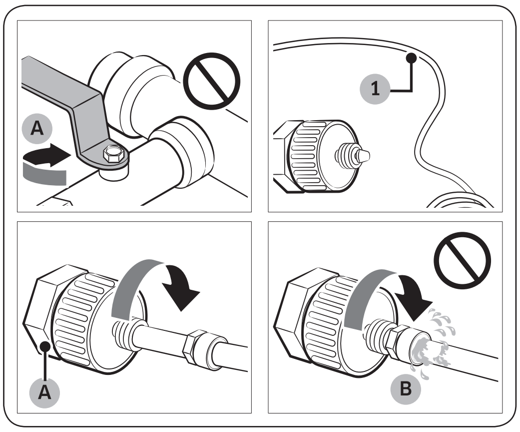
CAUTION

- The water line must be connected by a qualified technician.
- The warranty for your refrigerator does not cover the water line installation. The water line installation will be performed at your own costs unless the installation fee is included in the retailer's price.
- Samsung takes no responsibility for the water line installation. If water leaks occur, contact the installer of the water line.

To connect the cold water pipe to the water filtering hose

CAUTION

- Make sure the water filtering hose is connected to a cold, potable water source pipe.
Connecting to the hot water pipe may cause the water filter to malfunction.



A. Close Main Water pipe
B. No gap

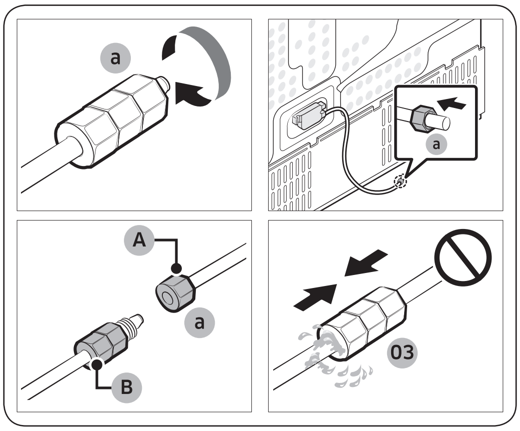
1. Shut off the water supply by closing the main water valve.
2. Locate the cold, potable water pipe (1).
3. Follow the Water Line Installation to connect to the water pipe.

NOTE

New hose-sets supplied with the appliance are to be used and old hose-sets should not be reused.

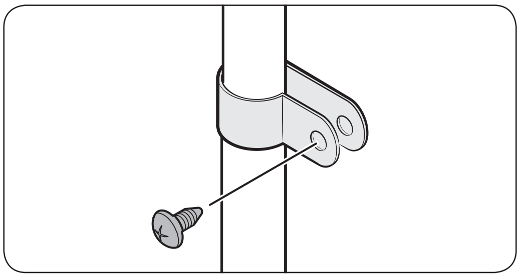
CHECK THE INSTALLATION STATUS

To connect the water filtering hose to the water line



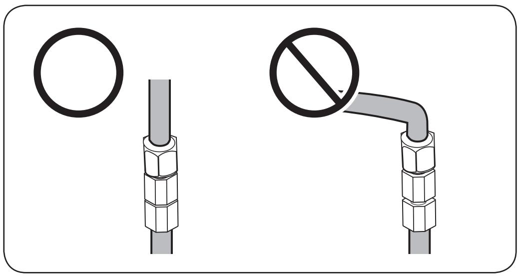
A. Water Line from unit
B. Water Line from kit

1. Loosen and remove the compression nut (a) from the water line of the refrigerator, and insert it to the water filtering hose.
2. Tighten up the compression nut (a) to connect the water filtering hose and the water line.
3. Open the main water valve and check for any leaks.
4. If there are no leaks, dispense about 10 litres of water or dispense the water for 6 - 7 minutes before actually using the refrigerator to remove impurities inside the water filtering system.
5. Secure the water line to a sink or a wall using a clip.



CAUTION

- Do not tighten the water line excessively. Make sure the water line is not bent, pinched, or squashed.
- Do not mount the water line on any part of the refrigerator. This may damage the refrigerator.



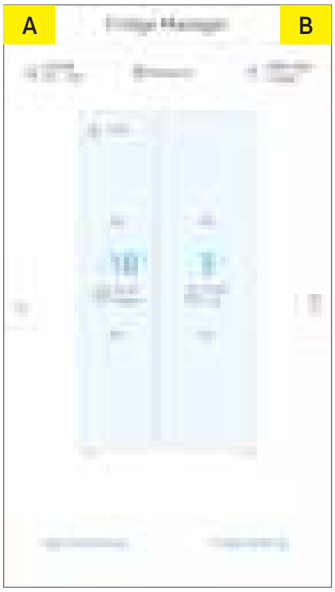
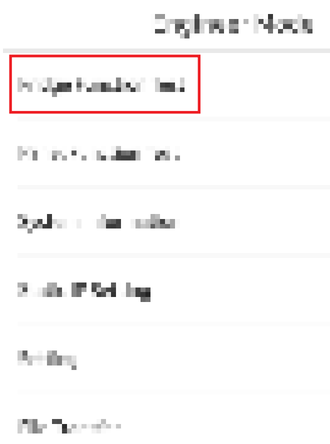
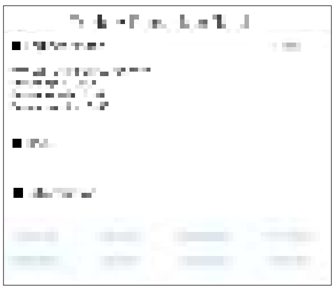
NOTE

If you have to relocate the refrigerator after connecting the water line, make sure the joined section of the water line must is straight.

CHECK THE INSTALLATION STATUS

4-2. Function for failure diagnosis

※ **How to launch the Fridge Function Test Menu.**

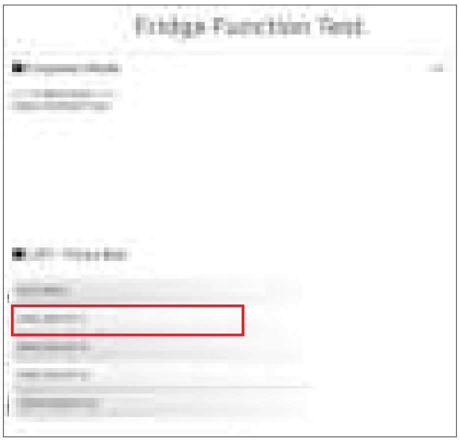
Sequence	Note
① Within 3 seconds, touch the “A” and “B” part as “A-B-A-B-A-B” (“Engineer mode” massage will be displayed as toast popup.)	
② Choose “Fridge Function Test”.	
③ Fridge Function test launched.	

4-2-1. Test mode (manual operation / manual defrost function)

1) Manual operation function (Force Run)

Sequence	Note
① Choose “Force Run”.	
② To run compressor manually, press the force run button on the list. All test function will be canceled by click “TEST CANCEL” button on the top of the force test list.	

CHECK THE INSTALLATION STATUS



FORCE RUN 1 (FF 1)

If Force Run 1(FF 1) selected, compressor will run at once without 7 minutes delay in any mode. If the refrigerator is on the defrost cycle at the moment, defrost will be finished and Force Run will begin.
(Be careful if Force Run get started at the moment of compressor off, over load could be occurred)

When Force Run 1(FF 1) operation runs, Compressor & F-fan(Freezer) operate continuously for 24 hours, F-Valve will maintain Open status, and F- Cooling speed valve will maintain Close status. And Fridge compartment will be controlled by the setting temperature.
And setting temperature will be selected automatically as below: freezer compartment -8°F(-23°C), fresh food compartment 34°F(1°C). During Force Run, Power Freeze & Power Cool function will not be work. If a function is selected, the power function icon of the selected function will be off automatically after 10 seconds

FORCE RUN 2 (FF 2)

If Force Run 2(FF 2) selected, compressor will run at once without 7 minutes delay in any mode. If the refrigerator is on the defrost cycle at the moment, defrost will be finished and Force Run will begin. (Be careful if Force Run get started at the moment of compressor off, over load could be occurred)

When Force Run 2(FF 2) operation runs, Compressor & F-fan(Freezer) operate continuously for 24 hours, F-Valve will maintain Close status, and F- Cooling speed valve will maintain Open status. And Fridge compartment will be controlled by the setting temperature.
And setting temperature will be selected automatically as below: freezer compartment -8°F(-23°C), fresh food compartment 34°F(1°C). During Force Run, Power Freeze & Power Cool function will not be work. If a function is selected, the power function icon of the selected function will be off automatically after 10 seconds.

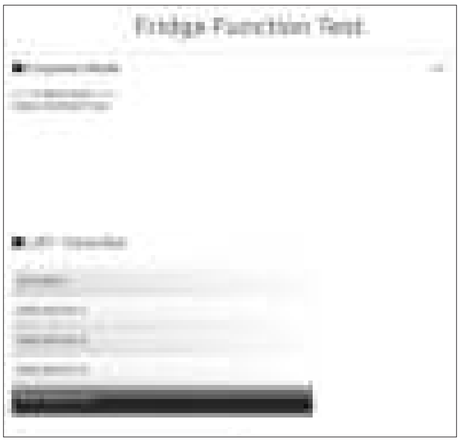
FORCE RUN 3 (FF A)

If Force Run 3(FF A) selected, compressor will run at once without 7 minutes delay in any mode. If the refrigerator is on the defrost cycle at the moment, defrost will be finished and Force Run will begin.
(Be careful if Force Run get started at the moment of compressor off, over load could be occurred)

When Force Run 3(FF A) operation runs, Compressor & F-fan(Freezer) operate continuously for 24 hours, both F-Valve and F-Cooling speed valve will maintain Open status. And Fridge compartment will be controlled by the setting temperature.
And setting temperature will be selected automatically as below: freezer compartment -8°F(-23°C), fresh food compartment 34°F(1°C). During Force Run, Power Freeze & Power Cool function will not be work. If a function is selected, the power function icon of the selected function will be off automatically after 10 seconds.

CHECK THE INSTALLATION STATUS

2) Manual defrost (Force Defrost)



FORCE DEFROST (Fd)

Forced Freezer and Fridge compartment defrosting begins.

3) Test cancel mode

All test function will be canceled by click “TEST CANCEL” button on the top of the force test list. Or, all test functions will be canceled by turning main power ON and OFF.

4-2-2. Self-diagnostic function

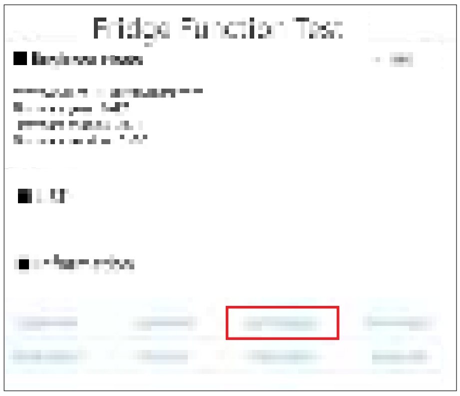
1) Self-diagnostic function in the Initial power ON.

1-1) Micom operates self-diagnostic function to check the temperature sensor condition within 1 second when the refrigerator turned On initially.

1-2) If bad sensor is detected by the self-diagnostic function, the error message will pop-up with error number. At this moment, there is no beep sound.(Refer to self-diagnostic CHECK LIST)

1-3) Self-diagnostic button is recognized only when the error is displayed by the bad sensor.Display does not operate normally but temperature control will be controlled by the emergency operation.

1-4) When the error is detected by self-diagnosis, the error can be canceled automatically if all troubled sensors are corrected or press the error skip key. (Press the 'OK' Button for 3 sec on pop-up message.)

Sequence	Note
① Choose "Self Diagnosis".	
② Current errors will be shown on the box. When the error on the List box is selected, the detailed failure mode is displayed. The Self Diagnosis function stays on for 60 seconds.	

CHECK THE INSTALLATION STATUS

4-2-3. Display function of Load condition

Sequence	Note
① Choose “Load Status”.	
② Load Status mode displays current output signals from MICOM. It is not the actual operating load status but output signals from MICOM. That is to say, even though the display shows any refrigerator mode is on, the actual status can of different with the display due to the errors of load and relay on PCB etc. The Load Status function status function stays on for 30 seconds.	

Load Status mode displays current output signals from MICOM. It is not the actual operating load status but output signals from MICOM. That is to say, even though the display shows any refrigerator mode is on, the actual status can be different with the display due to the errors of load and relay on PCB etc. The Load Status function stays on for 30 seconds.

CHECK THE INSTALLATION STATUS

Load mode checklist

NO	Part	Display	Operation contents
1	R Damper Heater	R DAMPER HEATER ON	When R Damper Heater operates
2	Overload condition	AMBIENT TEMP HIGH	When ambient temperature is more than 34°C
3	Low Temperature condition	AMBIENT TEMP LOW	When ambient temperature is less than 21°C
4	Normal condition	AMBIENT TEMP is not displayed.	When ambient temperature is between 22°C and 33°C.
5	Exhibition mode	SHOW ROOM MODE	Display mode
6	Ice Pipe Heater (DISPENSER model)	ICE PIPE HEATER ON	When Ice Pipe Heater operates
7	ICE MAKER FULL (DISPENSER model)	FULL ICE	When the Ice Maker's Bucket is full
8	COMP	COMPRESSOR	When COMP operates
9	F-FAN HIGHEST	FREEZER-FAN HIGHEST	When F-FAN HIGHEST operates
10	F-FAN HIGH	FREEZER-FAN HIGH	When F-FAN HIGH operates
11	F-FAN-LOW	FREEZER-FAN LOW	When F-FAN LOW operates
12	F-DEF HEATER	FREEZER-DEFROST-HEATER ON	When F-DEF HEATER operates
13	C-FAN HIGHEST	C-FAN HIGHEST	When C-FAN HIGHEST operates
14	C-FAN HIGH	C-FAN HIGH	When C-FAN HIGH operates
15	C-FAN LOW	C-FAN LOW	When C-FAN LOW operates
16	F Valve	FREEZER VALVE OPEN	When freezer valve open
17	F Cooling Speed Valve	FREEZER COOLING SPEED VALVE OPEN	When freezer cooling speed valve open
18	DISPENTER HEATER (DISPENSER model)	DISPENSER HEATER ON	When DISPENSER HEATER operates
19	R DAMPER	R DAMPER OPEN	When R DAMPER open

4-2-4. DEMO MODE 1 : Exhibition mode setting function (North America Model)

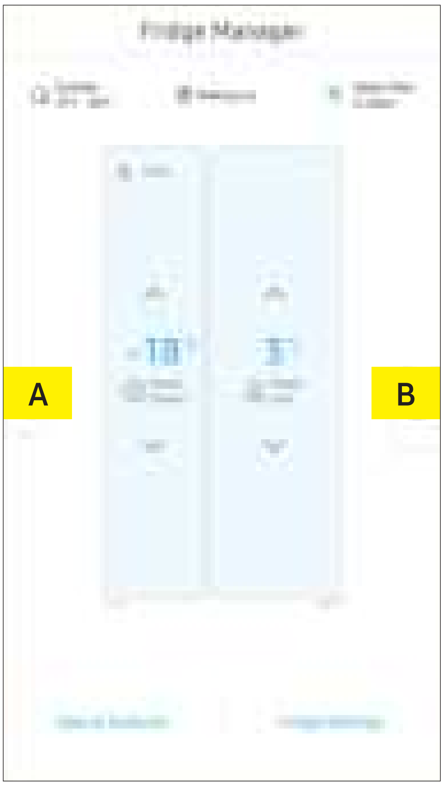
1) Cooling Off mode (also called Shop mode), is designed for use by retailers when they are displaying refrigerators on the shop floor. In Cooling Off mode, the refrigerator's fan motor and lights work normally, but the compressors do not run, and the refrigerator and freezer do not get cold. If Cooling Off is turned on, all cooling controls will turn to OFF on the Fridge Manager.
   - To activate Cooling Off, tap Activate > Proceed from Cancel/Proceed.
   - To deactivate Cooling Off, tap Deactivate > Proceed from Cancel/Proceed.

2) If Cooling Off mode is selected, display "Off" on the temperature setting display of the fridge manager and it indicates the refrigerator has entered the Cooling Off mode.

3) During Cooling Off mode, Fridge or Freezer compartments sensors are higher than 149°F (65°C) Cooling Off mode will be canceled automatically and freezing operation will be returned. (There is no buzzer sound when the Cooling Off mode is canceled by the temperature)

4) Operation contents of Cooling Off mode
   - Display, Fan motor and etc operate normally, not to operate compressor only.
   - Defrost is not operated. (including dispenser heater, ice pipe heater)
   - Display function of the initial real temperature is finished.
   - Under the condition of Cooling Off mode, Cooling Off mode will be operated when Power On after Power OFF.

4-2-5. DEMO MODE 2 : Exhibition Mode setting function (All regions except for North America)



1) Exhibition Mode setting function

1-1) To activate the Exhibition Mode, Move Side Navi Key(back button, Help Button) UP or DOWN. and then, touch the "A" and "B" part as "A-B-A-B-A-B" within 3 seconds.

1-2) If Exhibition mode is selected, display "Off" on the temperature setting display of the fridge manager and it indicates the refrigerator has entered the Exhibition mode.

1-3) During Exhibition mode, if fridge or freezer compartments sensors are higher than 149°F (65°C) Exhibition mode will be canceled automatically and freezing operation will be returned. (There is no buzzer sound when the Exhibition mode is canceled by the temperature)

1-4) Operation contents of Exhibition mode
   - Display, Fan motor and etc operate normally, not to operate compressor only.
   - Defrost is not operated. (including dispenser heater, ice pipe heater)
   - Display function of the initial real temperature is finished.
   - Under the condition of Exhibition mode, Exhibition mode will be canceled when Power On after Power OFF.

4-2-6. Option setting function

Sequence	Note
① Choose "Fridge Option". **NOTE** Basically, all the data in option has cleared from the factory. Therefore, almost all setting value are "0".But, some setting values could be changed for the purpose of improving performmance.NOTE You need to check the product manual and/or specification.	
② Change the option 00,01,02,03,04,12,19 and click the "Save Option" button on the end of the option list to save the options. **NOTE** Option setting function exists in the other items.We will skip the explanation of the other functions by the option because it is associated with refrigerator control function and is not needed at SERVICE.NOTE (Please do not set the other options except above SERVICE Manual.)	

4-2-7. Option TABLE

1) Temperature changing table of freezer compartment

Set item	Freezer Temp Shift
Reference	Fridge Room 7-SEG
Value	0

Setting value FZ compartment Code	Temp. compensation
0	0.0°C
1	-0.5°C
2	-1.0°C
3	-1.5°C
4	-2.0°C
5	-2.5°C
6	-3.0°C
7	-3.5°C
8	+0.5°C
9	+1.0°C
10	+1.5°C
11	+2.0°C
12	+2.5°C
13	+3.0°C
14	+3.5°C
15	+4.0°C

2) Temperature changing table of Fridge compartment

Set item	Fridge Temp Shift
Reference	Fridge Room 7-SEG
Value	1

Setting value FZ compartment Code	Temp. compensation
0	0.0°C
1	-0.5°C
2	-1.0°C
3	-1.5°C
4	-2.0°C
5	-2.5°C
6	-3.0°C
7	-3.5°C
8	+0.5°C
9	+1.0°C
10	+1.5°C
11	+2.0°C
12	+2.5°C
13	+3.0°C
14	+3.5°C
15	+4.0°C

3) Ice Tray water supply of freezer compartment Ice Maker (In DISPENSER MODEL with flow sensor Only features)

Set item	FZ-Room ICE TRAY
Reference	Fridge Room 7-SEG
Value	2

Setting value FZ compartment Code	Water Settings
0	92cc
1	105cc

4) Eject waiting time changing table of freezer compartment Ice Maker (In DISPENSER MODEL Only features)

Set item	FZ-Room Ice Maker Eject waiting time Shift
Reference	Fridge Room 7-SEG
Value	3

Setting value FZ compartment Code	Setting time
0	85min
1	84min
2	83min
3	82min
4	81min
5	80min
6	79min
7	78min
8	77min
9	76min
10	75min
11	74min
12	73min
13	72min
14	86min
15	87min

5) Eject temperature changing table of freezer compartment Ice Maker (In DISPENSER MODEL Only features)

Set item	FZ-Room Ice Maker Eject temperature Shift
Reference	Fridge Room 7-SEG
Value	4

Setting value FZ compartment Code	Temp. compensation
0	-17.0°C
1	-16.0°C
2	-15.0°C
3	-14.0°C
4	-13.0°C
5	-12.0°C
6	-18.0°C
7	-19.0°C

6) Minimum Comp RPM shifting.

- This option is rising minimum Comp RPM.
  As this option is applied, Comp operation.

Set item	Minimum Comp RPM setting
Reference	Fridge Room 7-SEG
Value	12

Setting value FZ compartment Code	Comp RPM
0	No RPM Change
1	Minimum 2450RPM
2	Minimum 2450RPM
3	Minimum 2450RPM

7) Operation rate changing table of dispenser heater. (In DISPENSER MODEL Only features)

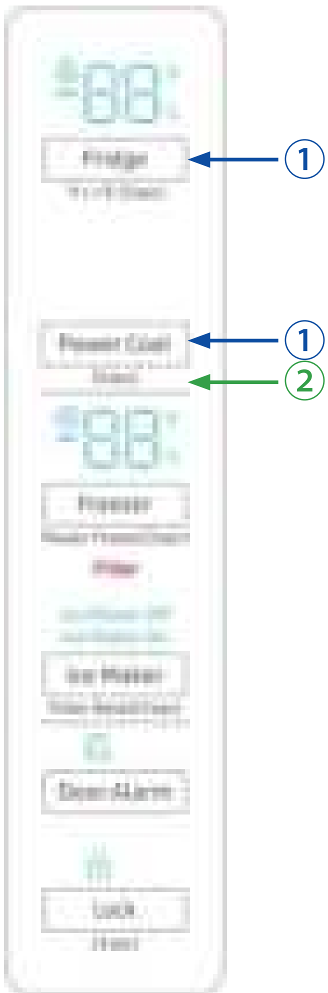
Set item	Minimum Comp RPM setting
Reference	Fridge Room 7-SEG
Value	19

Setting value FZ compartment Code	Temp Display Option
0	0.00%
1	+ 20% operation (up to 100%)

4-2-8. Test mode (manual operation / manual defrost function)

- Press the Fridge and Power Cool buttons at the same time for at least 6 seconds. The Control Display will blink with an interval of 0.5 seconds. In that case, you may take your fingers off both buttons and press the Power Cool button to enter the Test Mode.
- If any key on the front of panel is pressed within 15 seconds after the test mode, it will be operated as below sequence : Manual operation(FF) -> manual defrost of freezer compartment(Fd) -> cancel(Display all off)
- If any key on the front of panel is not pressed within 15 seconds after the test mode, the test mode will be canceled and it will be returned to previous mode.

1) Manual operation function



Fridge Key + Power Cool Key are pressed simultaneously for 6 seconds. And the Control Display will blink with an interval of 0.5 seconds. In that case, you may take your fingers off both buttons and press the Power Cool button to enter the Test Mode.

CHECK THE INSTALLATION STATUS

1-1) If any key is pressed once in test mode, blinks "FF" on the display and it indicates the refrigerator has entered the manual operation. At this moment, buzzer beeps as an alarm.

1-2) If manual operation is selected, compressor will run at once without 7 minutes delay in any mode. If the refrigerator is on the defrost cycle at the moment, defrost will be finished and manual operation will begin.(Be careful if manual operation get started at the moment of compressor off, over load could be occurred)

1-3) If manual operation works, compressor & f-fan operate continuously for 24 hours and fridge compartment will be controlled by the setting temperature.

1-4) When the manual operation runs, setting temperature will be selected automatically as below: freezer compartment -23°C, fridge compartment 1°C.

1-5) During manual operation, Power Freezer & Power Cool function will not be work.If a function is selected, the power function icon of the selected function will be off automatically after 10 seconds.

1-6) Manual operation can be canceled by removing power from the unit, then resupplying power.

1-7) Alarm(0.25 sec ON/ 0.75 sec OFF) will beep continuously until manual operation is completed and there is no function to make the sound stop.

CHECK THE INSTALLATION STATUS

2) Forced Defrost

2-1) When you press any key one more time at the manual operation status, Fd lights up on the Display Panel. At this time, the Forced Operation stops immediately and F-Defrost will be performed at the same time.

2-2) At this time, it will send out “Beep” sound for 2 seconds and then it will perform Forced F Defrost while sending out “0.5 sec On and 0.5 sec Off” sound.

3) Test cancel mode

3-1) During the defrosting of freezer compartment if the display panel change to the test mode and test button is pressed one more time, defrosting of freezer compartment will be canceled and the unit will return to the normal operation.
Or, all test functions will be canceled by turning main power ON and OFF.

CHECK THE INSTALLATION STATUS

4-2-9. Self-diagnostic function

1) Self-diagnostic function in the Initial power ON

① If Fridge Key + Power Cool Key are pressed simultaneously for 10 seconds, the error mode by self-diagnosis will be canceled.

1-1) Micom operates self-diagnostic function to check the temperature sensor condition within 1 second when the refrigerator turned On initially.

1-2) If bad sensor is detected by the self-diagnostic function, the applicable display LED will blink for 0.5 sec. At this moment, there is no beep sound.(Refer to self-diagnostic CHECK LIST)

1-3) Self-diagnostic button is recognized only when the error is displayed by the bad sensor.Display does not operate normally but temperature control will be controlled by the emergency operation.

1-4) When the error is detected by self-diagnosis, the error can be canceled automatically if all troubled sensors are corrected or Self-diagnostic function key (Fridge Key + Power Cool Key) are pressed simultaneously for 10 seconds.
(Return to normal display mode)

CHECK THE INSTALLATION STATUS

2) Self-diagnostic function during normal operation

② **If Fridge Key + Power Cool Key are pressed simultaneously For 10 seconds, the self-diagnosis function will be selected.**

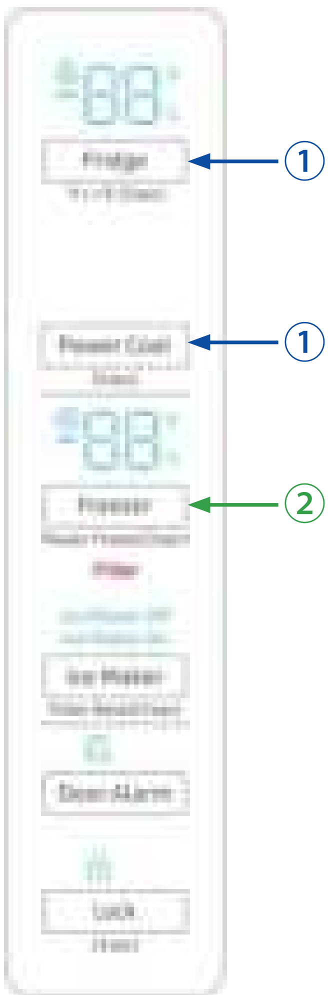
2-1) If Fridge Key + Power Cool Key are pressed simultaneously for 6 seconds during normal operation, the temperature setting display will operate for 4 seconds (ON/OFF 0.5sec each).If Fridge Key + Power Cool Key are pressed simultaneously for 10 seconds (including above 4seconds), self-diagnostic function will be selected.

2-2) At this moment, self-diagnostic function will be returned with buzzer sound.If there is an error, display of error will be operated for 30 seconds and then return to normal condition whether problem is corrected or not. (Refer to self-diagnosis CHECK LIST)

2-3) Input by button is not accepted during self-diagnostic function.

CHECK THE INSTALLATION STATUS

4-2-10. Display function of Load condition



① **If Fridge Key and Power Cool Key are pressed simultaneously for 6 seconds, ALL ON/OFF will blink with 0.5interval for 4 seconds.**
② **If take the finger off from above keys and press Freezer Key, load condition mode will be started.**

1) If Fridge Key and Vacation Key are pressed simultaneously for 6 seconds during normal operation, the temperature setting display of Fridge and freezer compartments will blink ALL ON/OFF with 0.5 for 4 seconds.

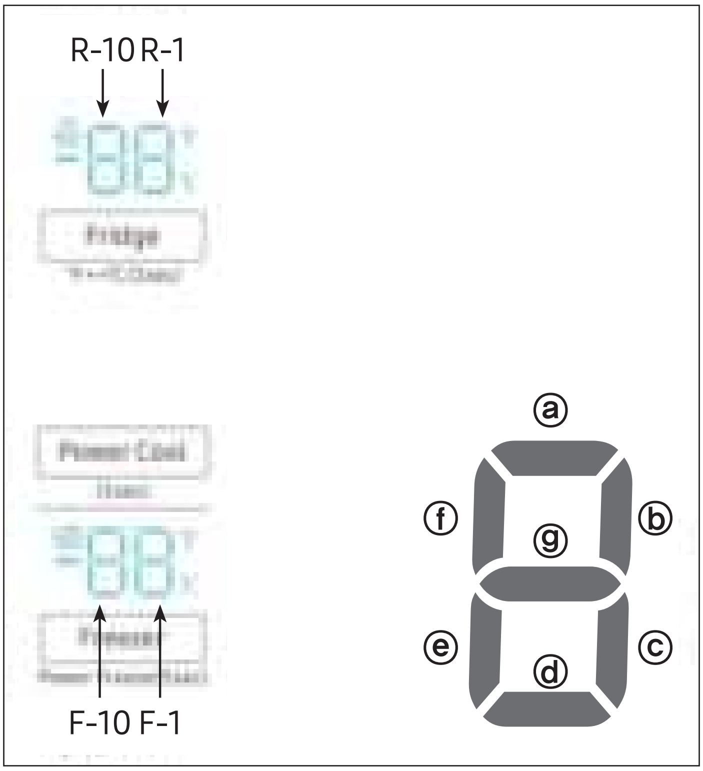
2) At this moment, If Freezer Key after Fridge Key and Vacation Key is pressed, load condition display mode will be returned with alarm. At LED all on state, only load condition display will blink ON/OFF with 0.5 seconds interval.

3) Load condition display mode shows the load that micom signal is outputting.
However, It means that micom signal is outputting, it does not mean whether load is operating or not. That is to say that though load operation is displayed, load could not be operated by actual load error or PCB relay error etc. (This function would be applied at A/S.)

4) Load condition display function will maintain for 30 seconds and then normal condition will be returned automatically.

5) Load condition display is as below. Only the load control LED will blink with 0.5interval in "Display LED".

CHECK THE INSTALLATION STATUS



1) If Fridge Key and Power Cool Key are pressed simultaneously for 6 seconds during normal operation, the temperature setting display of Fridge and freezer compartments will blink ALL ON/OFF with 0.5 for 4 seconds.

2) At this moment, If Freezer Key after Fridge Key and Power Cool Key is pressed, load condition display mode will be returned with alarm. At LED all on state, only load condition display will blink ON/OFF with 0.5 seconds interval.

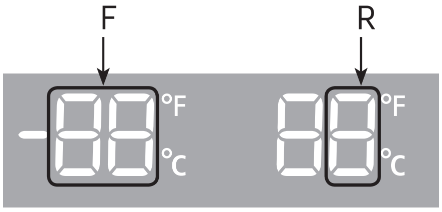
3) Load condition display mode shows the load that micom signal is outputting. However, It means that micom signal is outputting, it does not mean whether load is operating or not. That is to say that though load operation is displayed, load could not be operated by actual load error or PCB relay error etc. (This function would be applied at A/S.)

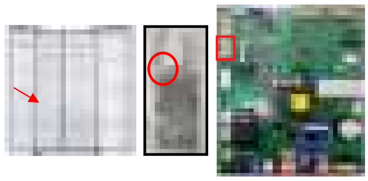
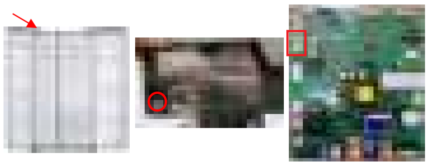
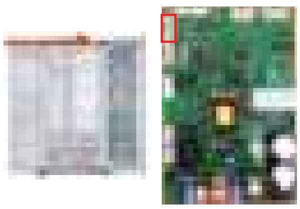
4) Load condition display function will maintain for 30 seconds and then normal condition will be returned automatically.

5) Load condition display is as below. Only the load control LED will blink with 0.5interval in "Display LED"

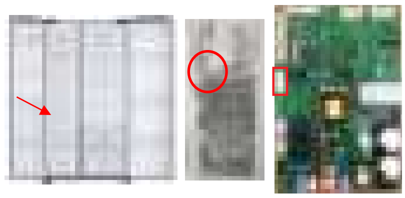
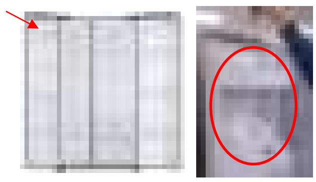
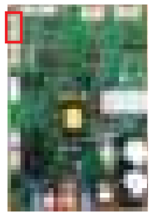
CHECK THE INSTALLATION STATUS

■ Self-diagnosis checklist

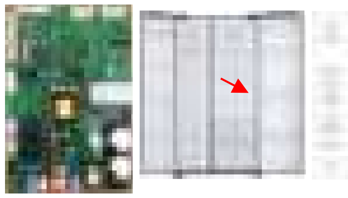
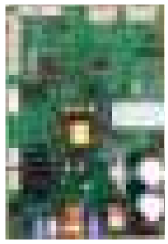
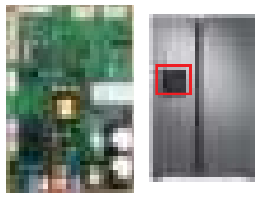
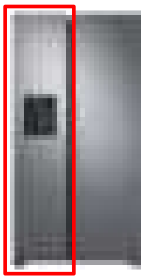
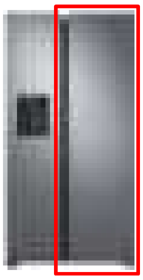


LED		Item	Trouble contents	Diagnostic method	Image
F	R				
1	C	F-Sensor Error	Display error :separation of sensor housing part, contact error, disconnection, short circuit. Display error of detecting temperature of sensor :more than 65°C or less than-50°C	When measuring the voltage between the Main PCB CN20 1PIN - 3PIN, it should read between 4.5V~1.0V.	
2		R-Sensor Error		When measuring the voltage between the Main PCB CN20 2PIN - 4PIN, it should read between 4.5V~1.0V.	
4		F-DEF-Sensor Error		When measuring the voltage between the Main PCB CN20 5PIN - 7PIN, it should read between 4.5V~1.0V.	
6		Ambient-Sensor Error		The voltage of MAIN PCB CN40 18PIN-20PIN shall be between 4.5V~1.0V.	
8		Ice Maker Sensor Error	Display error :separation of sensor housing part, contact error, disconnection, short circuit. Display error of detecting temperature of sensor:more than 65°C or less than- 50°C	The Voltage of MAIN PCB CN90 11PIN - 13PIN Shall be between 4.5V ~ 1.0V.	
13		Humidity-Sensor Error	Separation of sensor housing part, contact error, disconnection, short circuit.	When measuring the voltage between the Main PCB CN40 14PIN - 20PIN, it should read between 4.5V~1.0V.	

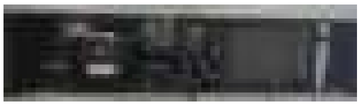
CHECK THE INSTALLATION STATUS

LED		Item	Trouble contents	Diagnostic method	Image
F	R				
21	C	F-FAN Error	Display error during operation of applicable fan motor :Feed back signal line contact error, motor wire separation, motor error.	The voltage of MAIN PCB CN20 15PIN - 17PIN shall be between 7V~12V.	
23		C-FAN Error	Display error during operation of applicable fan motor :Feed back signal line contact error, motor wire separation, motor error.	The voltage of MAIN PCB CN20 22PIN - 24PIN shall be between 7V~12V.	
24		F-DEF Error	Separation of freezer compartment defrost heater housing part, contact error, disconnection, short circuit or temperature fuse error. Display error :the defrosting does not finish though freezer compartment defrost is heating continuously for more than 100 minutes.	After separating Main PCB CN70& CN85 wire from PCB, Resistance value between CN70 5PIN - CN85 3PIN Shall be 63(230) ohm ± 7%(Resistance value is varied by input power) 0 ohm : heater short, ∞ ohm : wire/bimetal open (Must power off)	
26		Ice Maker Function Error	When the Freezer Ice Maker error occurs more than 3 times, the error will be displayed.	After replacing the Ice Maker, check if it operates normal.	
27		Damper Heater Error	Display error when open error is detected by damper heater :separation of Damper Heater housing part, contact error, disconnection, short circuit.	The voltage of MAIN PCB CN40 25PIN - 27PIN shall be 7V~12V.	
33		Ice Pipe Heater Error	Display error when open error is detected by Ice Pipe heater :separation of Ice Pipe Heater housing part, contact error, disconnection, short circuit.	The voltage of MAIN PCB CN90 1PIN - 5PIN shall be 7V~12V.	

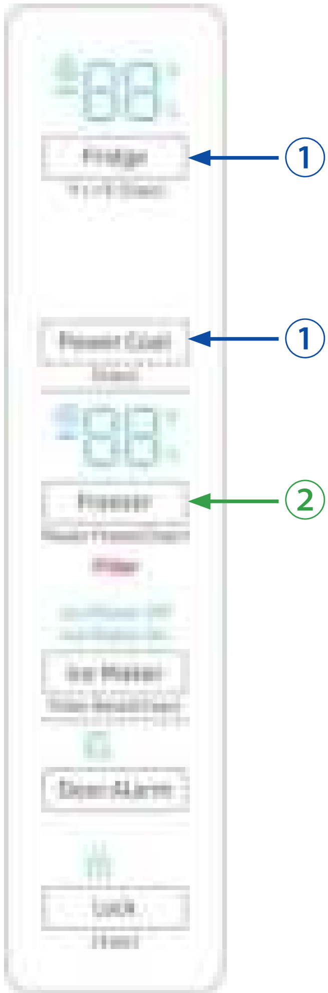
CHECK THE INSTALLATION STATUS

LED		Item	Trouble contents	Diagnostic method	Image
F	R				
41		MAIN ↔ Panel Communication Error	Display 41Er in the panel with alarm : MICOM ↔ MAIN PANEL communication error.	Actually, If there is not a problem, it is desirable to replace Main and Panel PCB after a cable problem confirming.	
44		Main - Inverter Communication Error	Display 44Er in the panel :Inverter MICOM ↔ Main MICOM communication error.	Actually, If there is not a problem, it is desirable to replace Main PCB after a cable problem confirming.	
46		I/O Expander Communication Error	Display 46Er in the panel :I/O Expander ↔ Main MICOM communication error.	It is desirable toreplace Main PCB.	
47		Main ↔ Dispenser Panel Communication Error	Display 47Er in the panel with alarm : MAIN MICOM ↔ DISPENSER PANEL communication error.	Actually, If there is not a problem, it is desirable to replace Main and Dispenser Panel PCB after a cable problem confirming.	
52	C	Main ↔ WIFI Communication Error	Display 52Er in the panel with alarm : MAIN MICOM ↔ WIFI communication error.	Actually, If there is not a problem, it is desirable to replace Main and WIFI after a cable problem confirming.	
71		The F Compartment Abnormal High temperature indicator blinks	When the freezer temperature is abnormally high or the freezer door is open for a certain period of time and the freezer temperature increases, the freezer display blinks.	The temperature has been abnormally increased. Check if the door has been open for a long time or if hot food has been stored in the compartment. If the reason for the error is removed, the error code disappears after a predetermined period of time.	
72		The R Compartment Abnormal High temperature indcator blinks	When the fridge temperature is abnormally high or the fridge door is open for a certain period of time and the fridge temperature increases, the fridge display blinks.	Check if door has been open for a long time or if hot food has been stored in the compartment. If the reason for the error is removed, the error code disappears after a predetermined period of time.	

CHECK THE INSTALLATION STATUS

LED F	LED R	Item	Trouble contents	Diagnostic method	Image
81	C	Comp starting Failure Error	When the Compressorfails starting.	Check if there is a short between compressor terminals. Check IPM Voltage [Under13.5V] Check if there is a short between IPM Pins [#1~33] Check the Compressor and the Cycle.	 
82		IPM Fault Error	When there is a IPM Fault error.		
83		Comp Abnormal current Detection Error	When there is abnormal current detected at the Compressor.	Check the compressor wire connections. Check the soldering status of the inverter PCB. (Check if any parts have short-circuited). Check the Comp and Cycle.	
84		Motor Locked Over RPM error	When there is a Compressor restriction error.	Check if the compressor and the Cycle is normal. Check the input voltage. Check the soldering of the inverter PCB. (Check if any parts have short-circuited.)	
86		Comp over voltage Error	The error code is displayed when the AC Input Voltage is too high.	Check the input voltage. (This error occurs when theinput voltage is AC 310V or higher.)	
96		Comp IPM ShutDown Error	When there is a comp IPM shutdown error.	Check the IPM temperature(over12) status of the inverter PCB. Check the heat-sink assembly status of the IPM.	

4-2-11. Display function of Load condition



1) If Fridge Key + Power Cool Key are pressed simultaneously for 6 seconds during normal operation, the temperature setting display of fridge and freezer compartments will blink ALL ON/OFF with 0.5 for 4 seconds.

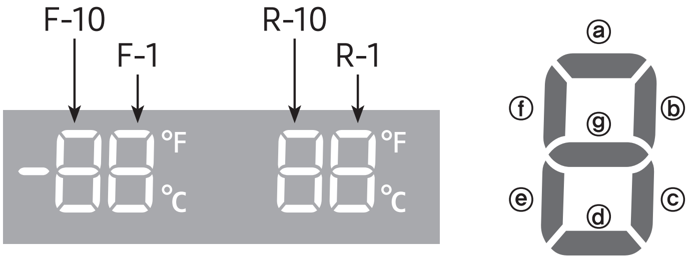
2) If take the finger off from above keys and press Freezer key load status will display. At LED all on state only load condition display will blink ON/ OFF with 0.5 seconds interval.

3) Load condition display mode shows the load that micom signal is outputting.However, It means that micom signal is outputting, it does not mean whether load is operating or not. That is to say that though load operation is displayed, load could not be operated by actual load error or PCB relay error etc. (This function would be applied at A/S.)

4) Load condition display function will maintain for 30 seconds and then normal condition will be returned automatically.

5) Load condition display is as below. Only the load control LED will blink with 0.5 interval in "Display LED".

CHECK THE INSTALLATION STATUS



※ Load mode Check list

NO	Part	Display (LED)	Operation contents
1	R Damper Heater	R-1-ⓓ	When R Damper Heater operates, applicable LED blinks.
2	Overload condition	R-1-ⓔ	When ambient temperature is more than 34°C,LED blinks.
3	Low Temperature condition	R-1-ⓕ	When ambient temperature is less than 21°C,LED blinks.
4	Normal condition	R-1-ⓔ, ⓕ (Not blinks ALL LED)	When ambient temperature is between 22°C and 33°C.
5	Exhibition mode	R-1-ⓖ	LED Blinks at the display mode.
6	Ice Pipe Heater (DISPENSER model)	R-10-ⓐ	When Ice Pipe Heater operates, applicable LED blinks.
7	Ice Maker Full (DISPENSER model)	R-10-ⓓ	When the Ice Maker's Bucket is full, applicable LED blinks.
8	COMP	F-1-ⓐ	When COMP operates, applicable LED blinks.
9	F-Fan Highest	F-1-ⓑ, ⓒ	When F-FAN HIGHEST operates, applicable LED blinks.
10	F-Fan High	F-1-ⓑ	When F-FAN HIGH operates, applicable LED blinks.
11	F-Fan Low	F-1-ⓒ	When F-FAN LOW operates, applicable LED blinks.
12	F-DEF Heater	F-1-ⓓ	When F-DEF HEATER operates, applicable LED blinks.
13	C-Fan Highest	F-1-ⓔ, ⓕ	When C-FAN HIGHEST operates, applicable LED blinks.
14	C-Fan High	F-1-ⓔ	When C-FAN HIGH operates, applicable LED blinks.
15	C-Fan Low	F-1-ⓕ	When C-FAN LOW operates, applicable LED blinks.
16	Dispenter Heater (DISPENSER model)	F-10-ⓔ	When DISPENSER HEATER operates, applicable LED blinks.
17	R Damper	F-10-ⓖ	When R DAMPER opens, applicable LED blinks.
18	WIFI Status (WIFI model)	WIFI Icon	When AP and Internet Disconnect : LED OFF. When AP connect and Internet Disconnect : LED Blinks. When AP and Internte connect : LED ON.

4-2-12. Exhibition mode setting function (North America Model)

1) If Fridge Key + Power Cool Key are pressed simultaneously for 6 seconds, ALL ON/OFF will blink with 0.5 interval for 4 seconds. If take the finger off from above keys and press Lock key, Cooling Off mode will be started with buzzer sound.

2) If the cooling off mode key is pressed once more time during the cooling off operation, Cooling Off mode will be canceled.

3) If Cooling Off mode is selected, blinks "O-FF" on the temperature setting display of the panel and it indicates the refrigerator has entered the Cooling Off mode.

4) During Cooling Off mode, Fridge or Freezer compartments sensors are higher than 149°F (65°C) Cooling Off mode will be canceled automatically and freezing operation will be returned. (There is no buzzer sound when the Cooling Off mode is canceled by the temperature)

5) Operation contents of Exhibition mode
   - Display, Fan motor and etc operate normally, not to operate compressor and Door Open Alarm.
   - Defrost is not operated. (including dispenser heater, ice pipe heater)
   - Display function of the initial real temperature is finished.
   - Under the condition of Cooling Off mode, Cooling Off mode will be operated when Power On after Power OFF.

4-2-12. Exhibition mode setting function (All Regions Except for North America)

1) If Fridge Key + Power Cool Key are pressed simultaneously for 6 seconds, ALL ON/OFF will blink with 0.5 interval for 4 seconds. If take the finger off from above keys and press Lock key, Exhibition mode will be started with buzzer sound.

2) If above Exhibition entering Key are pressed one more time, Exhibition mode will be canceled.

3) If Exhibition mode is selected or canceled by key, displays "on" or "oF" for 5 seconds on the temperature setting display of the panel and it indicates the refrigerator has entered or exited the Exhibition mode. After 5 seconds, the display panel will work normally.

4) During Exhibition mode, if fridge and freezer compartments sensors are higher than 149°F(65°C) Exhibition mode will be canceled automatically and freezing operation will be returned. (There is no buzzer sound when the Exhibition mode is canceled by the Temperature.)

5) Operation contents of Exhibition mode
- Display, Fan motor and etc operate normally, not to operate compressor and Door Open Alarm.
- Defrost is not operated. (including dispenser heater, ice pipe heater)
- Display function of the initial real temperature is finished.
- Under the condition of Exhibition mode, Exhibition mode will be canceled when Power On after Power OFF.

CHECK THE INSTALLATION STATUS

4-2-13. MAC Address Display Mode

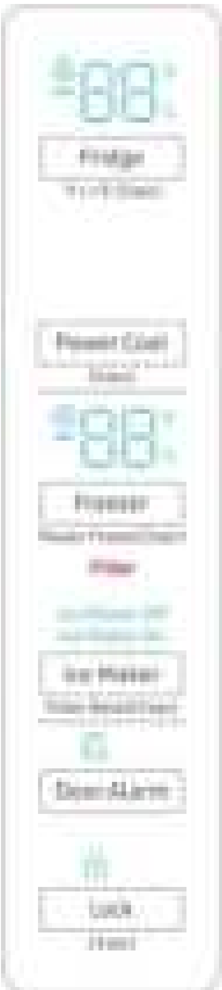
1) This function is to display the MAC address of the Refrigerator Wi-Fi module, and consecutively displays the MAC address on the Freezer & Fridge Temperature Display Part for one minute.
   - If you press the Freezer button and the Fridge button simultaneously for six seconds or longer, the Display will blink.Here, if you press the Freezer button, a buzzer sound will be made and the product will enter the MAC Address Display Mode.

2) If the MAC address is "11-22-33-44-55-66“, "-- / --" → "11 / 22" → "33 / 44" → "55 / 66" → "-- / --" will be repeatedly displayed on the Freezer & Fridge Temperature Display Part for one minute. When one minute has passed, the MAC Address Display Mode will be canceled with a buzzer sound, and the product will return to the normal Display Mode.

※ If the MAC address is not valid or the Wi-Fi module is not connected, "-- / --" will be displayed on the Freezer & Fridge Display Part for one minute.

4-2-14. Option setting function

If Fridge Key + Power Cool Key are pressed simultaneously for 6 seconds during normal operation, ALL ON/OFF will blink with 0.5 interval for 4 seconds. If take the finger off from above keys and press Fridge, Fridge and Freezer compartments temperature display will be changed to operation setting mode.



■ Key control in option mode

Power Cool Key	Reference Value down key
Fridge Key	Reference Value Up key
Lock Key	Code Up / Down Key (Rotation)

- If the display changes to option setting mode, all displays will be off except freezer and fridge compartments temperature display as below. (Fridge and freezer compartments case will be explained only because all options are operated with the same method according to the option table.)

4-2-14. Option setting function

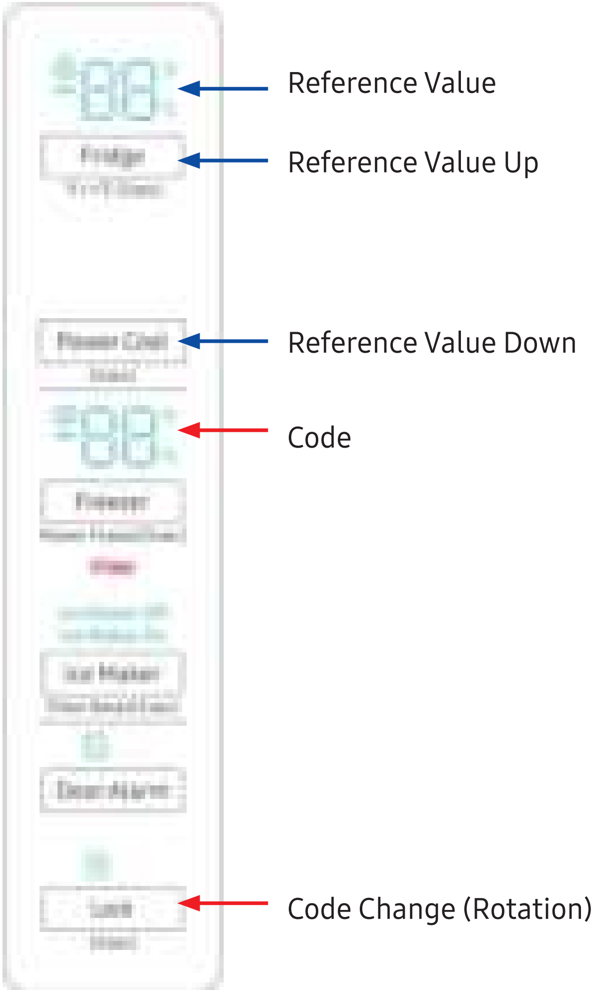
- Press the Fridge and Power Cool buttons at the same time for at least 6 seconds.The Control Display will blink with an interval of 0.5 seconds. In that case, you may take your fingers off both buttons and press the Fridge button to enter the Option Setting Mode.

KEY operation method for changing to option mode

① Press the Fridge and Power Cool buttons simultaneously for 6 seconds. All On/Off will blink with 0.5sec interval 4seconds.
② If take the finger off from above keys and press Fridge Key, Option setting mode will be started.

CHECK THE INSTALLATION STATUS

KEY control method after converting to option mode

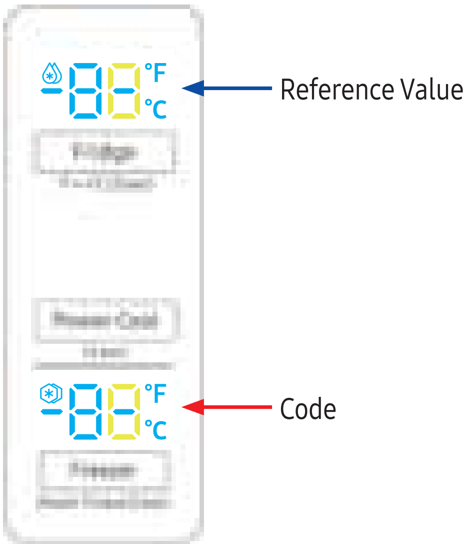


■ Key control in option mode

Lock Key	Code Change Key(Rotation)
Power Cool Key	Reference Value down key
Fridge Key	Reference Value Up key

CHECK THE INSTALLATION STATUS

- If the display changes to option setting mode, all displays will be off except freezer and fridge compartments temperature display as below.(Fridge and freezer compartments case will be explained only because all options are operated with the same method according to the option table.)



1) For example, if you want to change freezer compartment standard temperature to -4°F(-2°C) by operating option, do as below. This function is for changing the standard temperature. In -2°F(-19°C) of current temperature of freezer compartment, if you make the temperature lower to -4°F (-2°C) by the option, the standard temperature would be controlled -6°F(-21°C)
Therefore, if you change the setting of temperature option to -2°F(-19°C) on the panel, the appliance will be operated with -6°F(-21°C). It means that standard temperature is controlled -4°F(-2°C) less than setting temperature in the display.

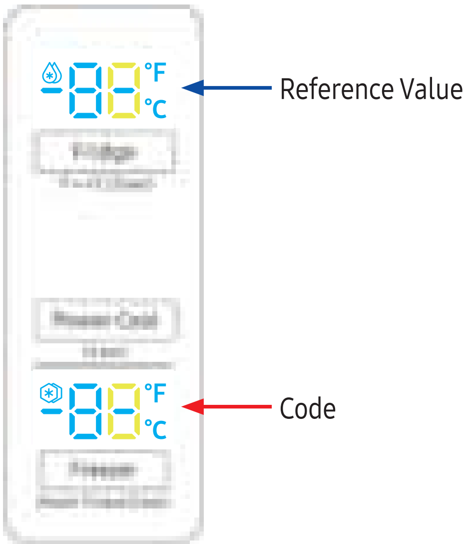
Note

Basically, all the data in option has cleared from the factory.Therefore, almost all setting value are "0".But, some setting values could be changed for the purpose of improving performmance. NOTE You need to check the product manual and/or specification.

2) After changing to the option mode, fresh food compartment "0" , freezer compartment "0" will be displayed.
(Basically fresh food compartment "0", freezer "0" would be set at shipping process, but setting value could be changed for the purpose of improving product at mass producing process.)
- If fresh food compartment "0" shows only, temperature reference value of freezer compartment will be set and current freezer compartment temperature code will be displayed on the freezer temperature display.

3) If freezer compartment "4" is set as below freezer compartment code after fresh food compartment "0 is set, standard temperature of freezer compartment will be lower than -4°F(-2.0°C). (Refer to the picture "changing the freezer compartment temperature")

CHECK THE INSTALLATION STATUS



: If you wait for 20 seconds after completing the setting, MICOM will save the setting value to the EEPROM and normal display will be returned and the option setting mode will be canceled.

4) By the same method as above, it is possible to control the fridge compartment temperature, water supply, ice-maker harvest temperature/time, defrost return time etc.

5) Option function is set in the EEPROM at shipping process in the factory.
You would better not to change the option of your own.
Completing the setting is that option function return to normal display after 20 seconds.
Do not turn off the appliance before returning to the normal display mode.

Note

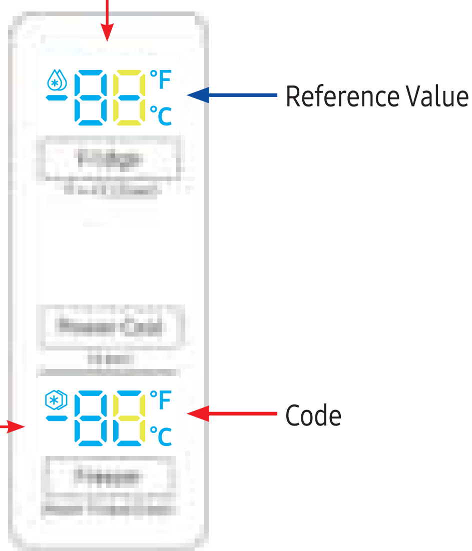
Option setting function exists in the other items.We will skip the explanation of the other functions by the option because it is associated with refrigerator control function and is not needed at SERVICE. NOTE (Please do not set the other options except above SERVICE Manual.)

4-2-15. Option TABLE

1) Temperature changing table of freezer compartment

Set item	Freezer Temp Shift
Reference	Fridge Room 7-SEG
Value	1

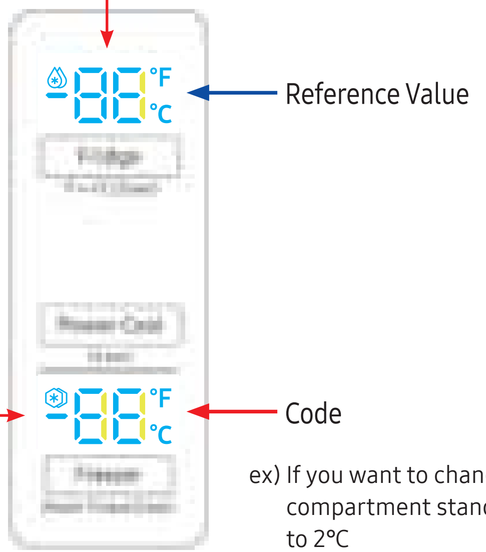
Setting value FZ compartment Code	Temp. compensation
0	0.0℃
1	-0.5℃
2	-1.0℃
3	-1.5℃
4	-2.0℃
5	-2.5℃
6	-3.0℃
7	-3.5℃
8	+0.5℃
9	+1.0℃
10	+1.5℃
11	+2.0℃
12	+2.5℃
13	+3.0℃
14	+3.5℃
15	+4.0℃



2) Temperature changing table of Fridge compartment

Set item	fridge Temp Shift
Reference	Fridge Room 7-SEG
Value	1

Setting value FZ compartment Code	Temp. compensation
0	0.0℃
1	-0.5℃
2	-1.0℃
3	-1.5℃
4	-2.0℃
5	-2.5℃
6	-3.0℃
7	-3.5℃
8	+0.5℃
9	+1.0℃
10	+1.5℃
11	+2.0℃
12	+2.5℃
13	+3.0℃
14	+3.5℃
15	+4.0℃



ex) If you want to change the Fridge compartment standard temperature to 2℃

3) Ice Tray water supply of freezer compartment Ice Maker (In DISPENSER MODEL with flow sensor Only features)

Set item	FZ-Room ICE TRAY
Reference	Fridge Room 7-SEG
Value	2

Setting value FZ compartment Code	Water Settings
0	92cc
1	105cc

4) Eject waiting time changing table of freezer compartment Ice Maker (In DISPENSER MODEL Only features)

Set item	FZ-Room Ice Maker Eject waiting time Shift
Reference	Fridge Room 7-SEG
Value	3

Setting value FZ compartment Code	Setting time
0	85min
1	84min
2	83min
3	82min
4	81min
5	80min
6	79min
7	78min
8	77min
9	76min
10	75min
11	74min
12	73min
13	72min
14	86min
15	87min

5) Eject temperature changing table of freezer compartment Ice Maker (In DISPENSER MODEL Only features)

Set item	FZ-Room Ice Maker Eject temperature Shift
Reference	Fridge Room 7-SEG
Value	4

Setting value FZ compartment Code	Temp. compensation
0	-17.0°C
1	-16.0°C
2	-15.0°C
3	-14.0°C
4	-13.0°C
5	-12.0°C
6	-18.0°C
7	-19.0°C

6) Minimum Comp RPM shifting.

- This option is rising minimum Comp RPM.
  As this option is applied, Comp operation.

Set item	Minimum Comp RPM setting
Reference	Fridge Room 7-SEG
Value	12

Setting value FZ compartment Code	Comp RPM
0	No RPM Change
1	Minimum 2450RPM
2	Minimum 2450RPM
3	Minimum 2450RPM

7) Temp Display Option.

- This option is for temp display all on mode.
  If user wants temp display always on, this option is solution.

Set item	Temp Display Option
Reference	Fridge Room 7-SEG
Value	16

Setting value FZ compartment Code	Temp Display Option
0	Normal Display (Temp display temporally on at use)
1	Always Temp Display On
2	Always Temp Display On

8) Operation rate changing table of dispenser heater.
(In DISPENSER MODEL Only features)

Set item	Minimum Comp RPM setting
Reference	Fridge Room 7-SEG
Value	19

Setting value FZ compartment Code	Temp Display Option
0	0.00%
1	+ 20% operation (up to 100%)

4-3. Diagnostic method according to the trouble symptom (Flow Chart)

DATA1. F/R/F-DEF Temperature table

T(°F)	T(°C)	Rmin (kΩ)	Rcent (kΩ)	Rmax (kΩ)	T(°F)	T(°C)	Rmin (kΩ)	Rcent (kΩ)	Rmax (kΩ)	T(°F)	T(°C)	Rmin (kΩ)	Rcent (kΩ)	Rmax (kΩ)
-40	-40	84.44	87.81	91.27	17.6	-8	18.28	18.72	19.16	75.2	24	5.046	5.197	5.351
-38.2	-39	80.13	83.28	86.52	19.4	-7	17.51	17.92	18.33	77	25	4.863	5.011	5.162
-36.4	-38	76.06	79.01	82.04	21.2	-6	16.77	17.16	17.55	78.8	26	4.689	4.833	4.98
-34.6	-37	72.22	74.98	77.82	23	-5	16.07	16.43	16.8	80.6	27	4.521	4.662	4.805
-32.8	-36	68.6	71.18	73.84	24.8	-4	15.4	15.74	16.08	82.4	28	4.36	4.498	4.638
-31	-35	65.18	67.6	70.08	26.6	-3	14.76	15.08	15.41	84.2	29	4.206	4.34	4.477
-29.2	-34	61.95	64.21	66.54	28.4	-2	14.16	14.46	14.76	86	30	4.058	4.189	4.322
-27.4	-33	58.89	61.02	63.2	30.2	-1	13.58	13.86	14.14	87.8	31	3.916	4.044	4.174
-25.6	-32	56.01	58	60.04	32	0	13.02	13.29	13.56	89.6	32	3.78	3.904	4.032
-23.8	-31	53.28	55.15	57.06	33.8	1	12.49	12.75	13.01	91.4	33	3.649	3.771	3.895
-22	-30	50.7	52.45	54.24	35.6	2	11.97	12.23	12.48	93.2	34	3.523	3.642	3.763
-20.2	-29	48.26	49.9	51.58	37.4	3	11.49	11.73	11.98	95	35	3.403	3.518	3.637
-18.4	-28	45.95	47.49	49.06	39.2	4	11.02	11.26	11.51	96.8	36	3.287	3.4	3.515
-16.6	-27	43.77	45.21	46.68	41	5	10.57	10.81	11.05	98.6	37	3.175	3.286	3.399
-14.8	-26	41.7	43.05	44.43	42.8	6	10.15	10.38	10.62	100.4	38	3.068	3.176	3.286
-13	-25	39.74	41.01	42.3	44.6	7	9.745	9.973	10.2	102.2	39	2.965	3.07	3.178
-11.2	-24	37.88	39.07	40.28	46.4	8	9.359	9.581	9.805	104	40	2.866	2.969	3.074
-9.4	-23	36.12	37.24	38.38	48.2	9	8.989	9.207	9.425	105.8	41	2.771	2.871	2.974
-7.6	-22	34.45	35.5	36.57	50	10	8.637	8.849	9.063	107.6	42	2.68	2.778	2.878
-5.8	-21	32.87	33.85	34.86	51.8	11	8.3	8.507	8.716	109.4	43	2.592	2.687	2.785
-4	-20	31.37	32.29	33.23	53.6	12	7.977	8.18	8.384	111.2	44	2.507	2.6	2.696
-2.2	-19	29.94	30.81	31.69	55.4	13	7.67	7.867	8.067	113	45	2.426	2.517	2.61
-0.4	-18	28.59	29.41	30.24	57.2	14	7.375	7.568	7.763	114.8	46	2.347	2.436	2.527
1.4	-17	27.31	28.08	28.85	59	15	7.094	7.282	7.473	116.6	47	2.272	2.359	2.448
3.2	-16	26.09	26.81	27.54	60.8	16	6.824	7.008	7.194	118.4	48	2.199	2.284	2.371
5	-15	24.93	25.61	26.29	62.6	17	6.567	6.746	6.928	120.2	49	2.129	2.212	2.297
6.8	-14	23.83	24.47	25.11	64.4	18	6.32	6.495	6.673	122	50	2.062	2.143	2.226
8.6	-13	22.79	23.39	23.99	66.2	19	6.084	6.255	6.429	123.8	51	1.997	2.076	2.157
10.4	-12	21.8	22.36	22.92	68	20	5.858	6.025	6.195	125.6	52	1.934	2.011	2.091
12.2	-11	20.85	21.38	21.91	69.8	21	5.642	5.805	5.97	127.4	53	1.874	1.949	2.027
14	-10	19.95	20.45	20.95	71.6	22	5.435	5.594	5.755	129.2	54	1.816	1.889	1.965
15.8	-9	19.1	19.56	20.03	73.4	23	5.236	5.391	5.549	131	55	1.76	1.832	1.906

DATA2. Ext-Temperature/Ext-Humidity Sensor table

- Temperature

T(°C)	Vout	T(°C)	Vout	T(°C)	Vout	T(°C)	Vout
-40	0.62	-12	1.316	16	2.66	44	3.956
-39	0.644	-11	1.364	17	2.708	45	3.98
-38	0.668	-10	1.412	18	2.756	46	4.004
-37	0.692	-9	1.46	19	2.804	47	4.028
-36	0.716	-8	1.508	20	2.852	48	4.052
-35	0.74	-7	1.556	21	2.9	49	4.076
-34	0.764	-6	1.604	22	2.948	50	4.1
-33	0.788	-5	1.652	23	2.996	51	4.124
-32	0.812	-4	1.7	24	3.044	52	4.148
-31	0.836	-3	1.748	25	3.092	53	4.172
-30	0.86	-2	1.796	26	3.14	54	4.196
-29	0.884	-1	1.844	27	3.188	55	4.22
-28	0.908	0	1.892	28	3.236	56	4.244
-27	0.932	1	1.94	29	3.284	57	4.268
-26	0.956	2	1.988	30	3.332	58	4.292
-25	0.98	3	2.036	31	3.38	59	4.316
-24	1.004	4	2.084	32	3.428	60	4.34
-23	1.028	5	2.132	33	3.476	61	4.364
-22	1.052	6	2.18	34	3.524	62	4.388
-21	1.076	7	2.228	35	3.572	63	4.412
-20	1.1	8	2.276	36	3.62	64	4.436
-19	1.124	9	2.324	37	3.668	65	4.46
-18	1.148	10	2.372	38	3.716	66	4.484
-17	1.172	11	2.42	39	3.764	67	4.508
-16	1.196	12	2.468	40	3.812		
-15	1.22	13	2.516	41	3.86		
-14	1.244	14	2.564	42	3.908		
-13	1.268	15	2.612	43	3.932		

- Humidity

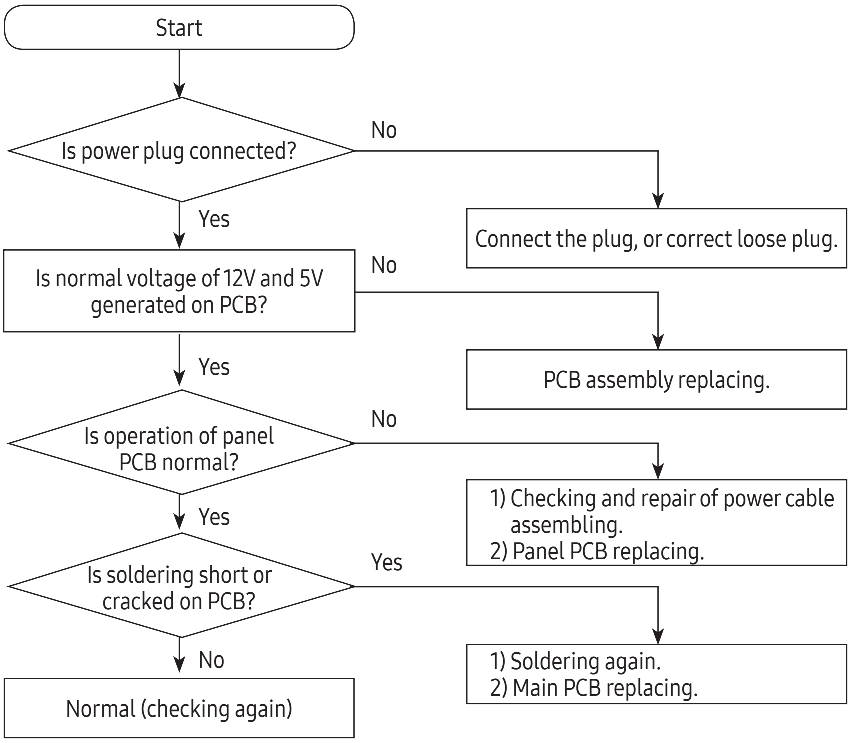
Relative Humidity	Vout	Relative Humidity	Vout	Relative Humidity	Vout	Relative Humidity	Vout
0	0.985	28	1.741	56	2.497	84	3.253
1	1.012	29	1.768	57	2.524	85	3.28
2	1.039	30	1.795	58	2.551	86	3.307
3	1.066	31	1.822	59	2.578	87	3.334
4	1.093	32	1.849	60	2.605	88	3.361
5	1.12	33	1.876	61	2.632	89	3.388
6	1.147	34	1.903	62	2.659	90	3.415
7	1.174	35	1.93	63	2.686	91	3.442
8	1.201	36	1.957	64	2.713	92	3.469
9	1.228	37	1.984	65	2.74	93	3.496
10	1.255	38	2.011	66	2.767	94	3.523
11	1.282	39	2.038	67	2.794	95	3.55
12	1.309	40	2.065	68	2.821	96	3.577
13	1.336	41	2.092	69	2.848	97	3.604
14	1.363	42	2.119	70	2.875	98	3.631
15	1.39	43	2.146	71	2.902	99	3.658
16	1.417	44	2.173	72	2.929	100	3.685
17	1.444	45	2.2	73	2.956		
18	1.471	46	2.227	74	2.983		
19	1.498	47	2.254	75	3.01		
20	1.525	48	2.281	76	3.037		
21	1.552	49	2.308	77	3.064		
22	1.579	50	2.335	78	3.091		
23	1.606	51	2.362	79	3.118		
24	1.633	52	2.389	80	3.145		
25	1.66	53	2.416	81	3.172		
26	1.687	54	2.443	82	3.199		
27	1.714	55	2.47	83	3.226		

5. SELF DIAGNOSIS & TROUBLE SHOOTING

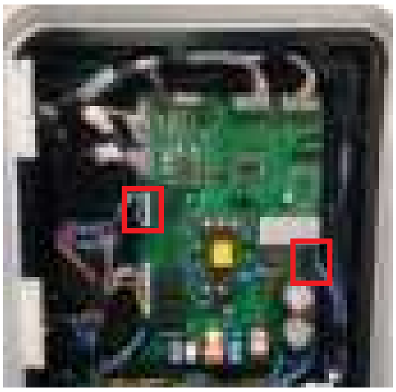
5-1. Power Not Supplied

CAUTION

AC input power and high voltage of DC 310V or higher are generated at PCB. Be cautious during repair and measurement.



5-2. Unable to Defrost



Start

F-DEF Sensor is normal (with Self Diagnosis) — NO → Replace the sensor.

YES

F-DEF Heater is normal — NO → Check Thermistor, Heater and their connectors

YES

Defrost Sensor Temp is lower than -5°C — NO → Forced Operation for a certain time.

YES

F-DEF-HEATER : Read resistance between CN85 #3 and CN70 #5

Defrost Sensor Voltage is lower than 3.1V

Press the FREEZE AND ALARM buttons at the same time for 6 sec and then Touch the Freeze button.

Forced F Defrost

F-DEF-SENSOR : Read resistance between CN20 #5 and #7.

Power is supplied to Defrost Heater. — NO → MAIN PBA connectorsare normal.

YES

It goes back to Cooling Operation after heating — NO → Check all the sensors

YES

Normal

Note

When F-DEF Sensor is higher than +10°C and +17°C, it will stop heating and go back to Cooling Operation.

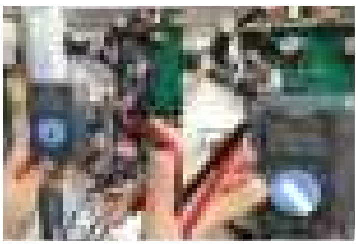
MAIN PBA connectorsare normal. — NO → Repairing connection terminal

YES

Check Thermistor, Heater and their contacts.

Repair or Replace defective relays, or PBA Ass'y

- **Checking method of resistance for F-DEF-Sensor**
  - Check the Resistance of CN20 #5 and #7 on the PBA
  - Compare the temperature table after measurement



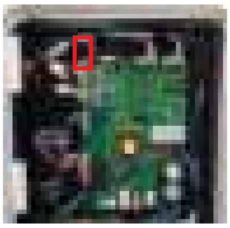
- **Sensor Check Point**
  1. EXT-ENSOR
  2. F-SENSOR
  3. F-DEF-SENSOR
  4. R-SENSOR
  5. R-DEF-SENSOR
  6. CV-SENSOR
  7. CV-DEF-SENSOR
  8. HUM1-SENSOR
  9. HUM2-SENSOR
  10. ICE-ROOM-SENSOR
  11. PANTRY-SENSOR
  12. ICE-MAKER-SENSOR
  13. IR-SENSOR

5-3. Self-Diagnosis Error (Defective Sensor)

When there is sensor error, display panel show that. When it occurs during the initial power on time, display panel keeps blinking relevant 7-SEG and the refrigerator go into the emergency operation mode.
Under the emergency operation mode, the refrigerator can not do its normal operation. So check out with the Self Diagnosis in this manual.

1) When the Ambient Sensor is defective

- **Checking method of voltage for Ext-Sensor**
  - Compare the temperature table after measurement. (Same method is applied to other sensors.)



Ambient-Sensor : Read the voltage between CN40 #18 and #20

Compare the ext-temp. with sensed temp. using Temp-Resistance Table.

Start

MAIN PBA connector (CN40) is inserted properly.

NO → Repair the defective contact of the connector or insert the connector properly.

YES → Ambient Temp Sensor is ok.

NO → Replace Thermistor.

YES → PBA & Thermistor are ok. Check the contact of theconnector again.

2) When the Fridge Temp Sensor is defective (also applied to other sensors)

Start

Fridge Sensor is good.

NO → Replace Thermistor.

YES → MAIN PBA connector (CN20) is inserted properly.

NO → Repair the defective contact of the connector or insert the connector properly.

YES → The order of Connector(CN20) is the same as the one in the circuit diagram

NO → Check the circuit diagram and correct any wrong inserted pins.

YES → The voltage between CN20 #9 and #11 is within the normal range.

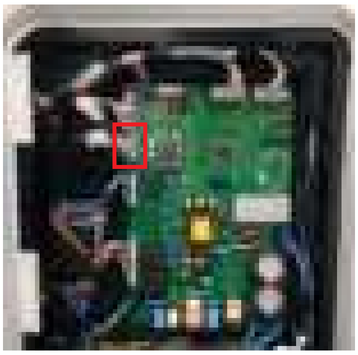
Refer to Circuit Diagram and F-Def-Temp Sensor Troubleshooting in this manual.

NO → Check MAIN PBA Cold Solder or Short

YES → PBA & Thermistor are good. Check the contact of the connector again.

5-4. When Alarm Sound continues (Buzzer Sound)

1) When "DingDong" sound continues



F-Door sensing : CN20 #1-#3
R-Door sensing : CN20 #10-#12

Nomal State
-Door Open : 5 Vdc
-Door Close : 0 Vdc

Separate the Door S/W and check its resistance change according to s/w on(0Ω) and off(∞Ω)

Start

- Door is closed completely?
  - Door is ajar → Romove causes after comprehending the condition of interference by door gasket or food etc.
  - Closed well ↓
- Is water penetrated into the door switch?
  - YES → Replace Door switch.
  - NO ↓
- Door sensing voltage changes as the door gets open or closed.
  - NO → Repair wire connection and door switch
  - YES ↓
- The connector(CN20) is Inserted properly.
  - NO → Repair the defective contact of the connector or insert the connector properly.
  - YES ↓
- Door S/W is good.
  - NO → Replace REED S/W.
  - YES → MAIN PBA & DOOR S/W are good

2) When "Beeping" sound continues

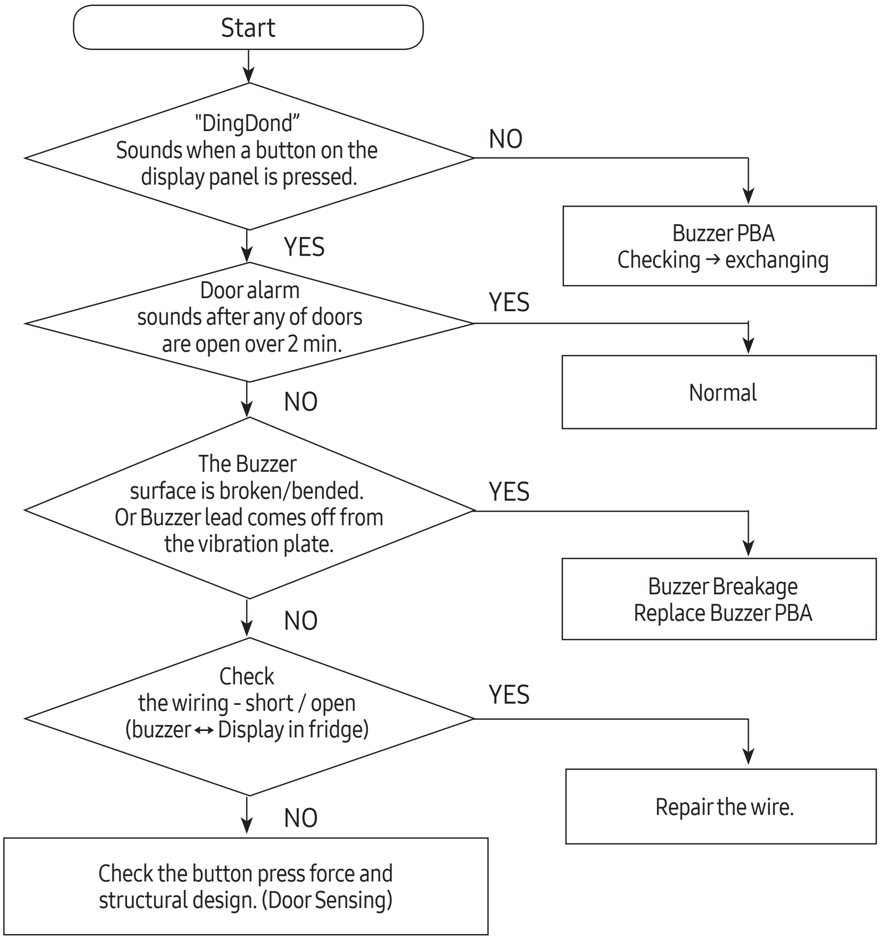
Start

- Forced Operation or Forced Defrost is selected.
  - YES → Cancel Forced Operation or Forced Defrost, or Re-plug the unit.
  - NO ↓
- After plugging in again, it sounds continuously.
  - NO → Panel PBA is normal.
  - YES → Replace Panel PBA

3) If a buzzer does not sound

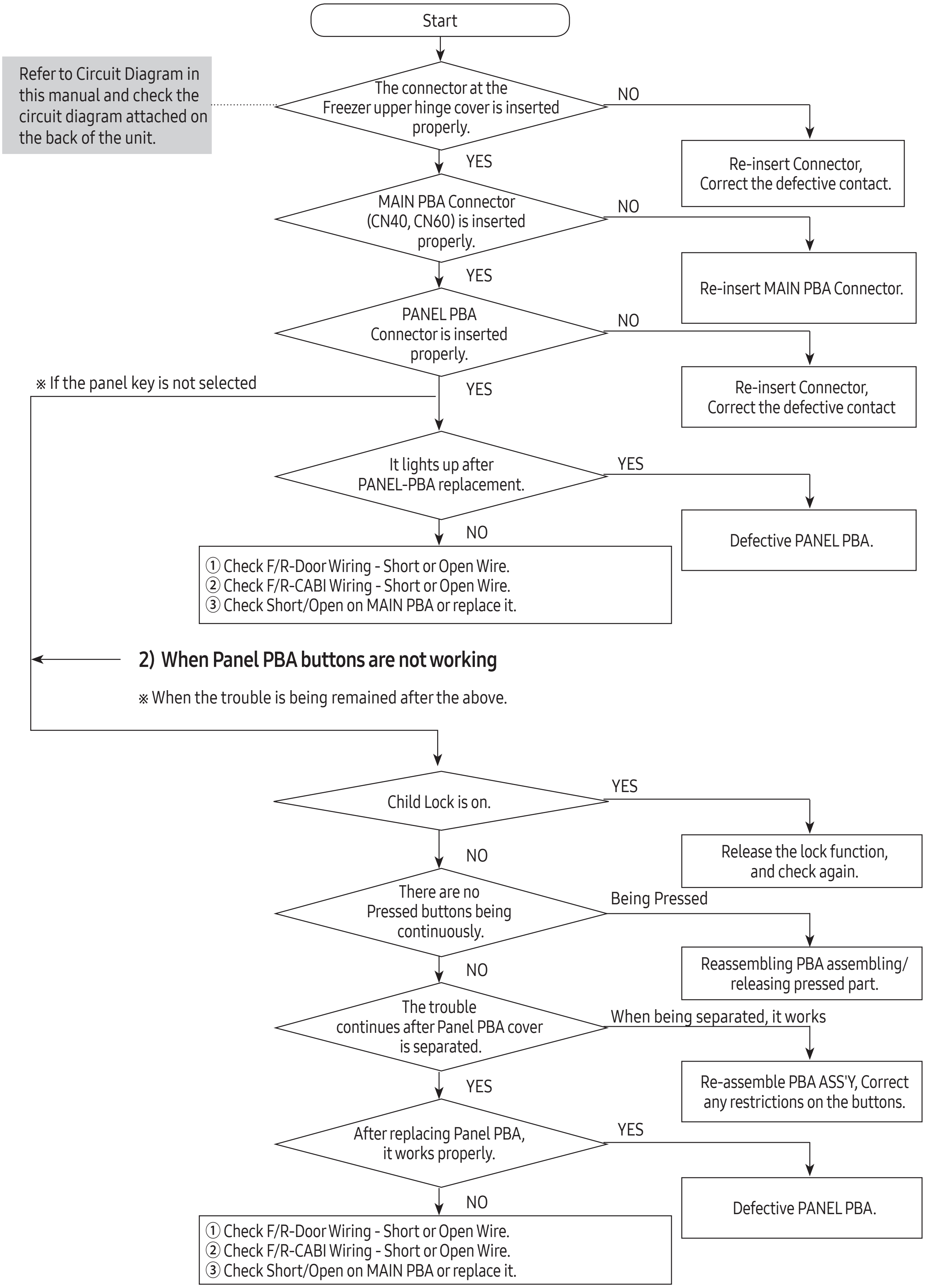
This model has a buzzer PBA (Except F-HUB Model). If the buzzer does not sounds when button is pressed, Forced Operation, door open and disconnection with Panel PBA, check out defective soldering on PBA or buzzer PBA damage. (It is recommended replacing Buzzer PBA if components are damaged)

※ It could be unable to hear in case there is heavy noise or built-in environment.



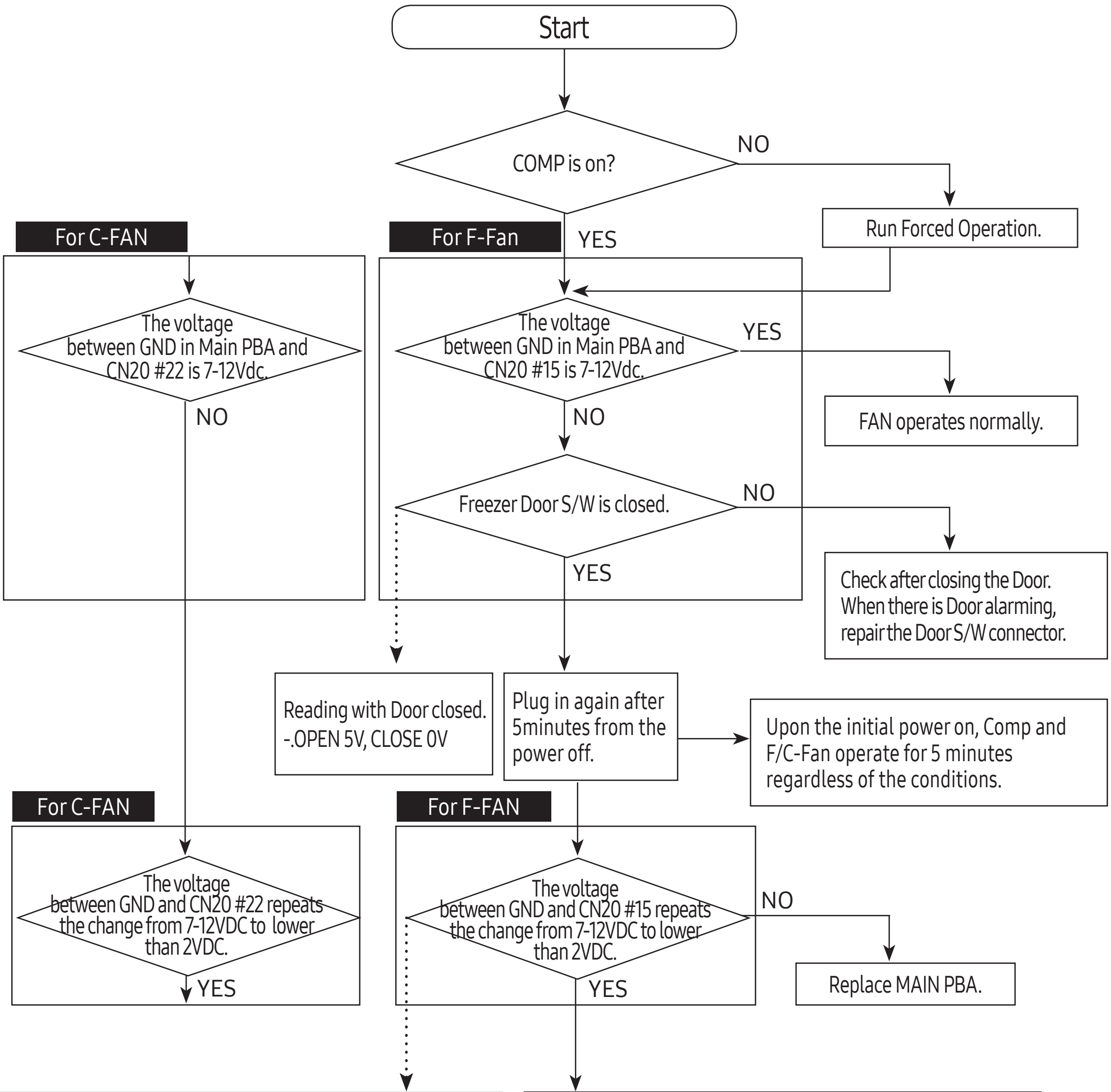
5-5. When PANEL PBA operates abnormally

1) When PANEL PBA does not light up or partially does



5-6. When Fan does not operate

- This model has BLDC FAN motor. BLDC motor is driven by DC7~12V
- F-Fan motor usually runs together with the Compressor. Once the door is open and closed at high ambient temperature, F-Fan motor put off its operation for one minute. Therefore, you are advised not to take it as an error.
- When fridge door is open, the F-Fan motor stops.



Note

Pulse signal(FG) generated from the Fan motor rotation is transmitted through CN20 #13(F), #20(C)
Micom senses these signal.
If Micom senses no signal, Micom make Fan Motor stop for 10 sec and try to operate it again. Micom keeps doing the same procedure 4 times until Micom receive the signal.
If it still fails, MICOM stop trying for 10 minutes.
This procedure prevents the motor from overload when foreign substances such as ice built up around the motor restrict motor rotation.

◆ Possible Causes
1. Defective FAN-MOTOR
2. Contact problem at the terminal
3. FG signal error(Refer to Fan-Motor circuit in this manual)

☞ Checking method of C-Fan motor voltage
- Read the voltage between MAIN PBA GND and CN74 #2
- If it has 7~12 voltage, It's normal.
(Same method is applied to other Fan Motor)

5-7. When the internal lamp of the freezer/fridge does not light up

Model with light bulb applied

Caution

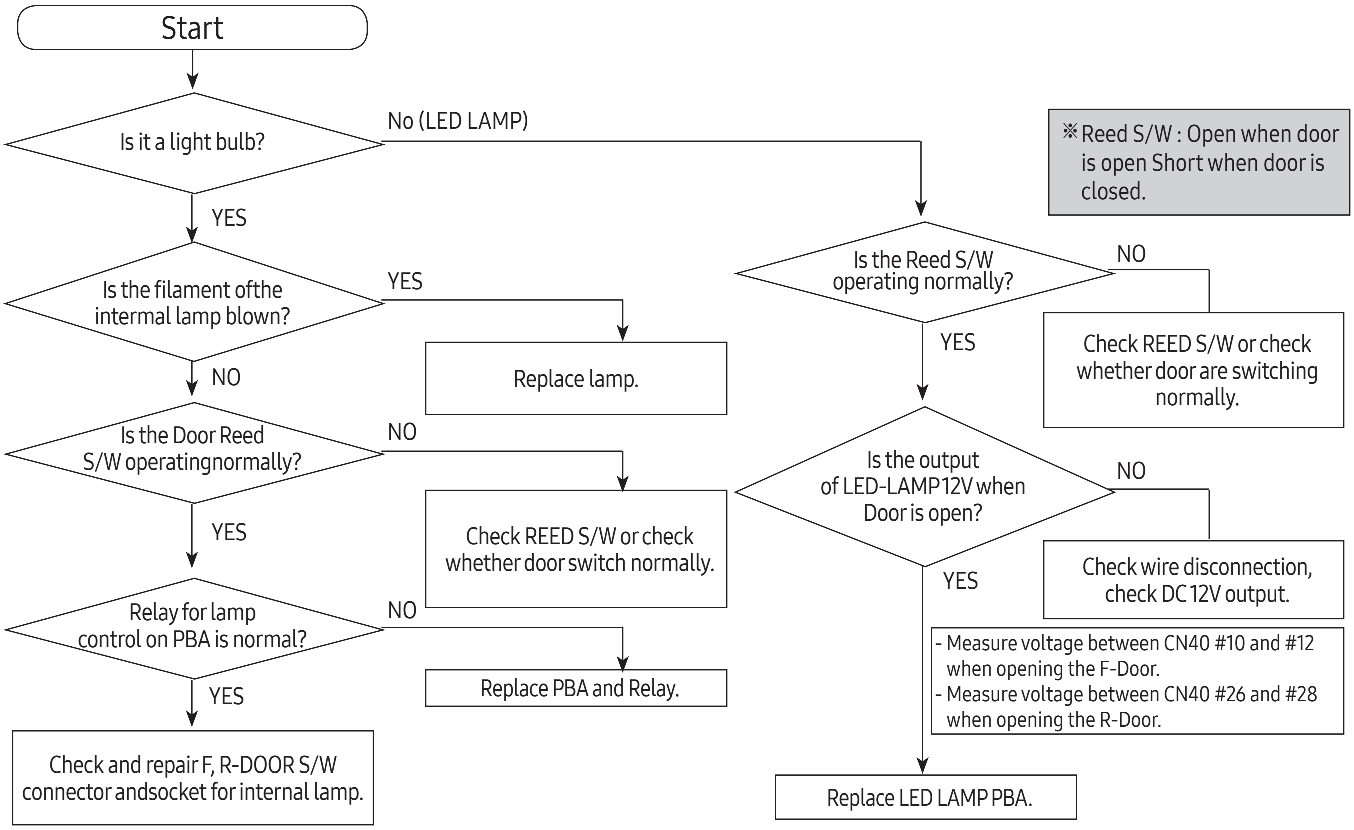
1. Turn off the power for repair because there is risk of electric shock when replacing the internal lamp.
2. Please keep in mind you could get burnt by the excessive heating of an incandescent light bulb.

Reference

Internal lamp RELAY switches ON/OFF as the DOOR is OPEN/CLOSE.
When the internal lamp is not on, check the operating noise of the RELAY.
RELAY operation noise is generated every time as the DOOR is OPEN/CLOSE.

Reference

1. When the DOOR is open, the DOOR S/W becomes OPEN. Micom senses 5V from Door S/W and recognizes that the Door is open. If 5V has sensed for more than 2 minutes, buzzer will sounds “DingDong” every 1 minute.Therefore, if the DOOR S/W is defective, it can generate the “Ding Dong” sound every 1 minute.
2. ON/OFF control of the internal lamp is linked with the DOOR S/W.



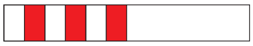
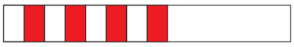
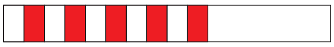
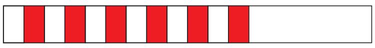
<Light bulb control method>

	Fridge internal lamp	Freezer internal lamp
Model with light bulb applied	On : Fridge door is open Off : Fridge door is closed	On : Freezer door is open Off : Freezer door is closed

5-8. LED blinking frequency depending on protecting functions

If Failure Condition is detected during compressor is operating, immediately stop Compressor operating and stand by 5 minutes. During this 5 minutes, RPM command signal is not available. It means, even if available RPM command signal is applied to the compressor, it does not work and keep standing by.

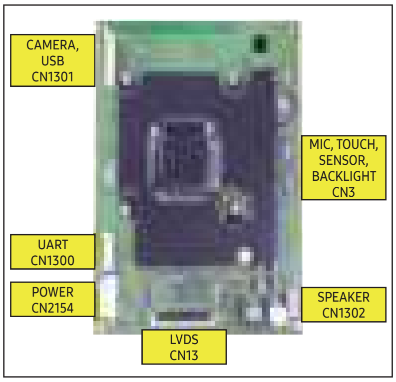
Blinking time is 1 second and dwell time is 2 seconds.

LED Blinking Frequency	Protecting Functions	Remarks
	Normal Operation	N/A
	Starting Failure	1. Check the COMP terminals short (U, V, W) 2. Check IPM Pins short of inverter PBA 3. Check IPM operating Voltage (Under DC 13.5V) 4. Other cases, check the COMP, cycle, etc
	IPM Fault	
	Abnormal Current Detection	1. Check COMP wire connections (U, V, W) 2. Check PCB Bottom side soldering state. 3. Other cases, check the COMP, cycle, etc
	Motor Locked / Over RPM	1. Check PCB Bottom side soldering state. 2. Check Input Voltage oscillation. 3. Other cases, check the COMP, cycle, etc
	Under Voltage	1. Check Input voltage under AC 53V (Input power AC 110~127V) or AC 106V (Input power AC 220~240V) 2. Check PCB Bottom side soldering state.
	Over Voltage	1. Check Input voltage under AC 155V(Input power AC 110~127V) or AC 310V(Input power AC 220~240V) 2. Check PCB Bottom side soldering state.

LED blinking frequency depending on protecting functions.
If the same blinking, After 5 minutes, Follow the Remarks

SELF DIAGNOSIS & TROUBLE SHOOTING

5-9. When the LCD does not work properly



Start

SMPS connector is inserted properly. (CN1, CN3)
- NO → Insert the connector properly or Repair any defective contact.
- YES ↓

The connector at the Fridge upper hinge cover is inserted properly.
- NO → Insert the connector properly or Repair any defective contact.
- YES ↓

LCD PBA connector is inserted properly. (CN13, CN2154, CN3)
- NO → Insert the connector properly or Repair any defective contact.
- YES ↓

Is normal voltage of 12V generated on SMPS output terminal?

Read the voltage between SMPS: CN1 #1,3 - #5,7.

- NO → Replace the SMPS.
- YES ↓

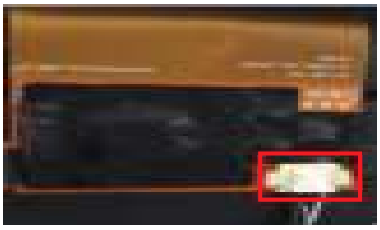
Is normal voltage generated from LCD PBA?

Read the voltage a between LCD PBA : CN2154 #1,2 - #4,5 (12V)
Read the voltage a between CN13 #1,2,3 - #7 (5V)
Read the voltage a between CN3 #17,18 - #14,15,19,20 (40~51V)

- NO → Replace the LCD PBA.
- YES ↓

Replace the LCD Assy.

5-10. When it does not respond to touch



Start

LCD PBA Touch connector is inserted properly.(CN3)

NO → Insert the connector properly or Repair any defective contact.

YES

TSP FPCB connector is inserted properly. (J3)

NO → Insert the connector properly or Repair any defective contact.

YES

Read the voltage between LCD PBA: CN3 #14 - #10

Is normal voltage of 3.3V generated from LCD PBA?

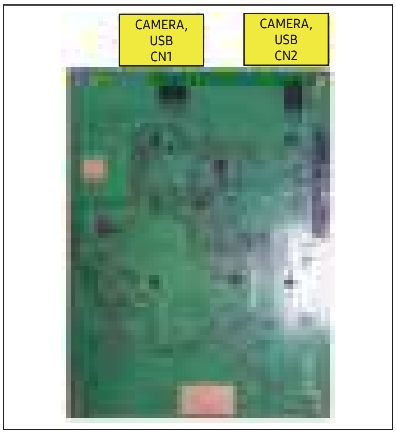
NO → Replace the LCD PBA.

YES

Replace the LCD Assy.

SELF DIAGNOSIS & TROUBLE SHOOTING

5-11. When the Wi-Fi does not work properly



Start

LCD PBA connector is inserted properly.(CN1,CN2)

NO → Insert the connector properly or Repair any defective contact.

YES ↓

Wi-Fi Module connector is inserted properly.

NO → Insert the connector properly or Repair any defective contact.

YES ↓

Read the voltage between LCD PBA: CN2 #1 - #3

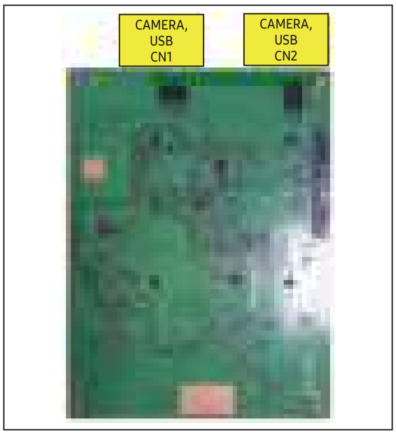
Is normal voltage of 5V generated from LCD PBA?

NO → Replace the LCD PBA.

YES ↓

Replace the Wi-Fi / BT Module.

5-12. When the BT does not work properly



Start

LCD PBA connector is inserted properly.(CN1,CN2)

NO → Insert the connector properly or Repair any defective contact.

YES ↓

BT Module connector is inserted properly.

NO → Insert the connector properly or Repair any defective contact.

YES ↓

Read the voltage between LCD PBA: CN2 #1 - #3

Is normal voltage of 5V generated from LCD PBA?

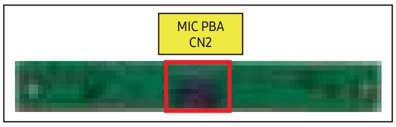
NO → Replace the LCD PBA.

YES ↓

Replace the Wi-Fi / BT Module.

SELF DIAGNOSIS & TROUBLE SHOOTING

5-13. When the microphone does not work properly



Start

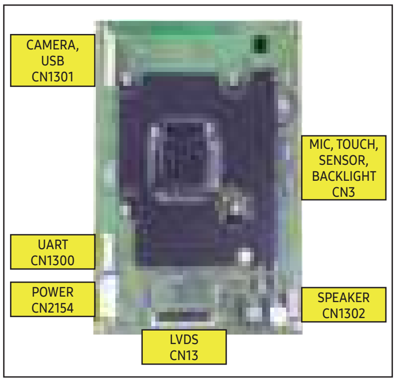
LCD PBA connector is inserted properly.(CN3)
- NO → Insert the connector properly or Repair any defective contact.
- YES ↓

MIC PBA connector is inserted properly.
- NO → Insert the connector properly or Repair any defective contact.
- YES ↓

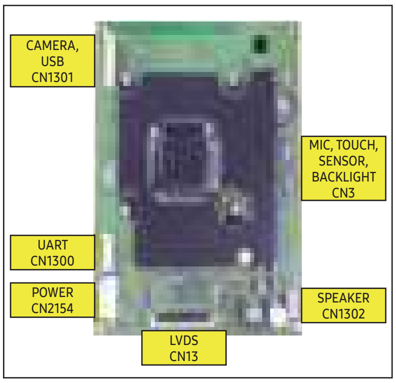
Read the voltage between LCD PBA: CN3 #6 - #8

Is normal voltage of 1.8V generated from LCD PBA?
- NO → Replace the LCD PBA.
- YES ↓

Replace the MIC PBA.



5-14. When the Glaze Camera does not work properly



Start

LCD PBA connector is inserted properly. (CN1301)
- NO: Insert the connector properly or Repair any defective contact.
- YES: ↓

The connector at the Fridge upper hinge cover is inserted properly (F/R-Door).
- NO: Insert the connector properly or Repair any defective contact.
- YES: ↓

French Assy connector is inserted properly.
- NO: Insert the connector properly or Repair any defective contact.
- YES: ↓

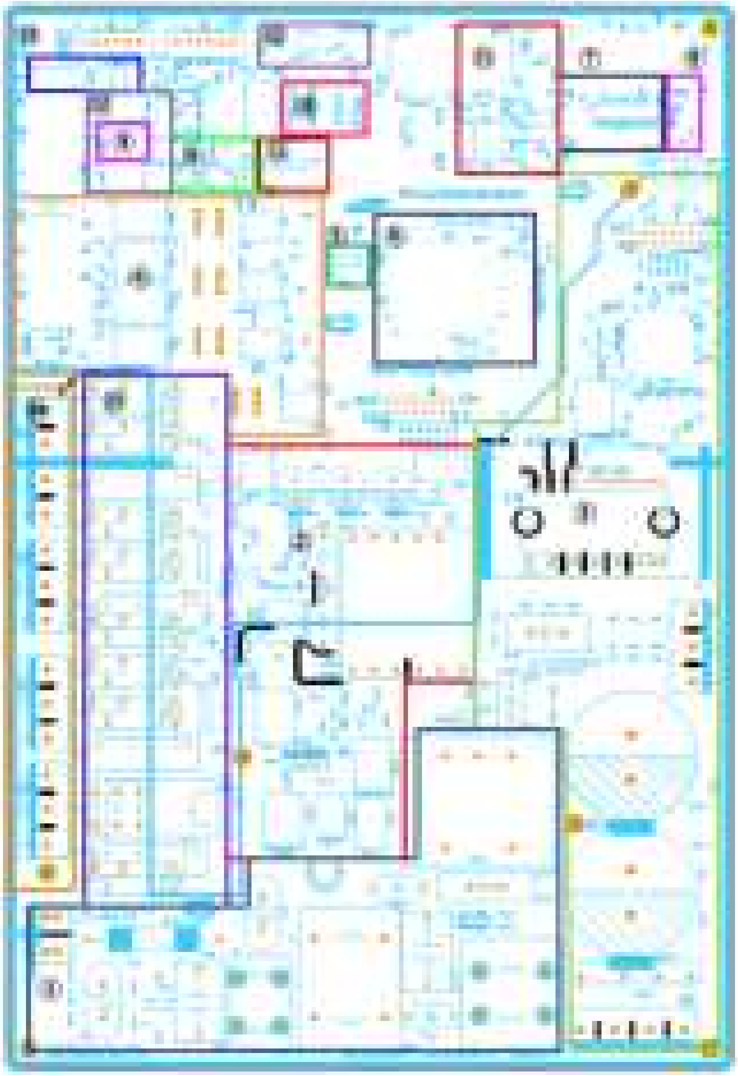
Is normal voltage of 5V generated from LCD PBA?

Read the voltage between LCD PBA: CN1301 #6,7 - #10,11

- NO: Replace the LCD PBA.
- YES: Replace the French Assy.

6. PCB DIAGRAM

6-1. PBA Layout with part position



1. EMI FILTER part
2. DC 15V, 12V, 5V, GND supplied from SMPS PCB
3. Inverter circuit part
   -.COMP Driving / Feedback Circuit
   -.BOOTSTRAP Charger : It is an independent power circuit for the driving of the IMP High-Phase IGBT.
   -. Current Pickup Circuit : It pickups the currents taken by the Shunt resistance and does the PWM DUTY control.
4. FAN MOTOR control part : To supply the power from 7V ~ 12V according to the motor types. (F,R,C,ICE)
5. EEPROM : Save and record every kinds of data.
6. Micom : control the regrigerator Ceramic resonator : generate the basic frequency of Micom operation.
   Reset IC : make Micom reset if input voltage of Micom is detected less than the specified voltage
7. Operate ICE-MAKER, supply power to MOTOR, and sense the variation of switch.
8. Main Micom ↔ Panel Micom serial communication circuit - Dispenser option input part (Water & Cover Ice route switch)
9. Auto Fill control part.
10. Control Room damper & Damper heater
11. Water Tank Heater Controls (also controls other options)
12. LED LAMP Control Circuit (F,R room Lamp)
13. Relay parts that controls AC load and receives Micom operating signal through Sink IC.
14. Connector with AC load
15. Diode option setting area

PCB DIAGRAM

6-2. Connector Layout with part position (Main Board)

CN40
1. STEP VALVE A
2. MAIN DISPLAY TX
3. STEP VALVE B
4. MAIN DISPLAY RX
5. 12V
6. 12V
7. STEP VALVE A/
8. GND
9. STEP VALVE B/
10. F-LED
12. GND
13. MAIN DISPLAY VCC
14. HUMIDITY-SENSOR
16. GND
17. R-ROOM DAMPER B
18. AMBIENT-SENSOR
19. R-ROOM DAMPER B/
20. GND
21. R-ROOM DAMPER A/
23. R-ROOM DAMPER A
25. R-ROOM DAMPER HEATER
26. R-LED
27. 12V
28. GND

※ N/A : 11, 15, 22, 24

CN05
1. RX
2. GND
3. TX
4. RESET

5, 12V

6. 5V

CN60
1. 5V
2. 12V

3~4. GND

8. ICE&WATER LEVER
10. CUBE MOTOR S/W
12. ICE ROUTE S/W
14. 5V
16. GND
18. 12V
20. DISPENSER DISPLAY RX
22. DISPENSER DISPLAY TX
23. 12V
24. GND

※ N/A : 5, 6, 7, 9, 11, 13, 15, 17, 19, 21, 25, 26

CN20
1. F-DOOR S/W
2. R-DOOR S/W

3~4. GND

5. F-DEF-SENSOR
6. 5V

7~8. GND

9. F-SENSOR
10. R-SENSOR

11~12. GND

13. F-FAN F/G
15. F-FAN

17~18. GND

20. C-FAN F/G
22. C-FAN
23. 12V
24. GND

※ N/A : 14, 16, 19, 21

CN90
1. 12V
2. GND
5. WATER PIPE HEATER
7. ICE MAKER MOTOR(CW)
9. ICE MAKER MOTOR(CCW)
11. ICE MAKER SENSOR
13. GND
15. ICE MAKER TEST S/W
17. 5V
18. 12V
19. HORIZONTAL HALL IC
21. GND
22. 5V

24~26. GND

28. 5V

※ N/A : 3, 4, 6, 8, 10, 12, 14, 16, 20, 23, 27

CN70
3. DISPENSER HEATER
5. F-DEF HEATER
7. COM(L)

※ N/A : 1

CN71
1. ICE ROUTE MOTOR
3. WATER SOLENOIDE VALVE
5. ICE SOLENOIDE VALVE

CN71-1
1. CUBE MOTOR
3. AUGER MOTOR
5. COM(N)

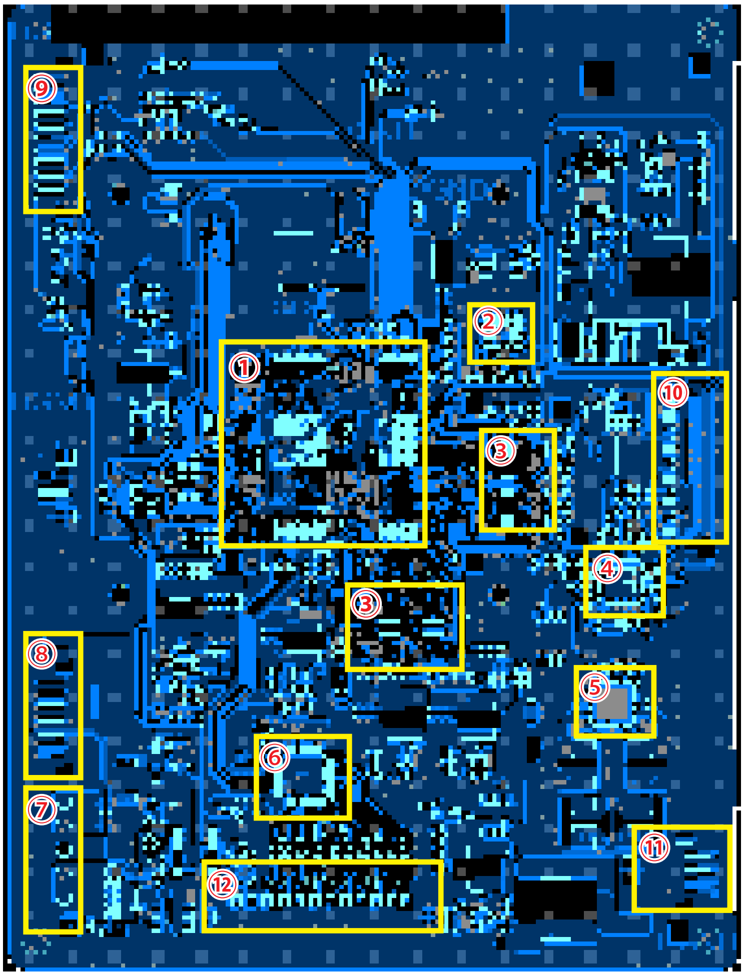
CN81
1. AC POWER(N)
3. AC POWER(L)

CN102
1. COMP (W)
3. COMP (V)
5. COMP (U)

CN85
1. AC POWER(L)
3. AC POWER(N)
5.
7. SET GND

PCB DIAGRAM

6-3. LCD PBA Layout with part positions



1. CPU (KANT-MS) – It controls performance of all loads
2. eMMC – Memory for operating CPU
3. LPDDR – Memory for operating CPU
4. CODEC – Circuit for sound input
5. AMP – Circuit for sound output
6. LVDS Circuit – LCD signal (Vx1 to LVDS)
7. DC 12V Power – Supply 12V to LCD PBA
8. Connector – UART
9. Connector – USB, Camera
10. Connetor – Touch, Backlight, Sensor, Mic
11. Connector – Speaker
12. Connector –LVDS Signal for LCD

PCB DIAGRAM

6-4. Connector Layout with part positions

CAMERA, USB
CN1301
1. 5V
2. USB_DN
3. USB_DP
4. GND
5. GND
6. 5V
7. 5V
8. CAM_DN
9. CAM_DP
10. GND
11. GND

MIC, TOUCH, BACKLIGHT, SENSOR
CN3
1. SENSOR 5V
2. GND
3. SENSOR
4. GND
5. MIC_R
6. MIC 1.8V
7. MIC_L
8. GND
9.
10. GND
11. TOUCH_INT
12. TOUCH SDA
13. TOUCH SCL
14. TOUCH 3.3V
15.~20. BACK LIGHT

SET UART
CN1300
1. SET TX
2. SET RX
3. GND

POWER
CN2154
1, 2 : 12V
4, 5 : GND

SPEAKER
CN13
1~3 : 5V
7, 14, 17, 24 : GND

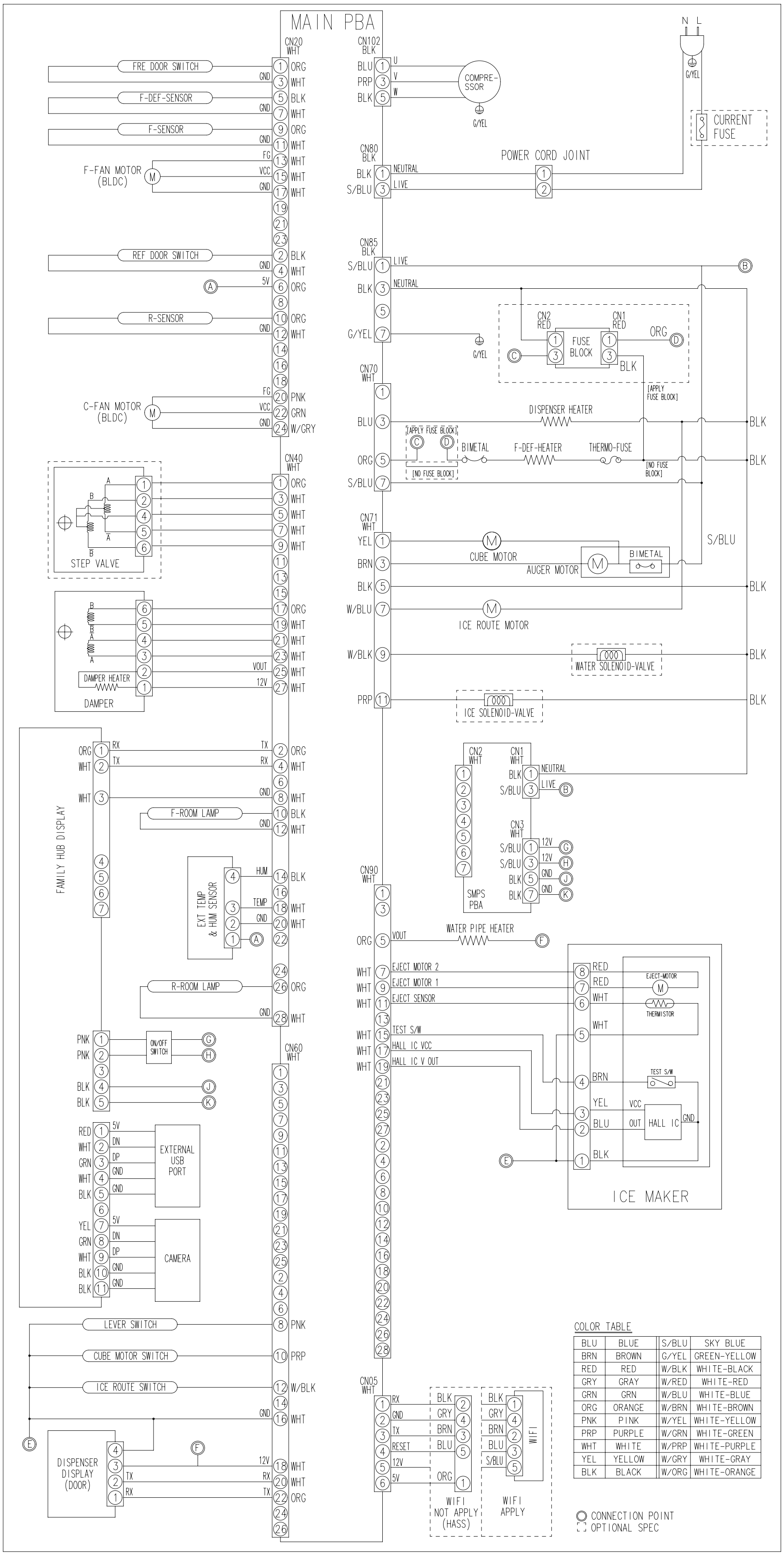
PCB DIAGRAM

WIFI
CN1
1. WIFI_NRESET
2. GND
3. WIFI_WOL
4. GND
5. GND
6. WIFI_DP
7. GND
8. WIFI_DN
9. WIFI_PHY
10. GND

BT
CN2
1. WIFI/BT 5V
2. BT WELCOM
3. GND
4. GND
5. BT_DP
6. BT_NRESET
7. BT_DN
8. MODULE_WAKE
9. GND
10. BT WAKE

7. WIRING DIAGRAM

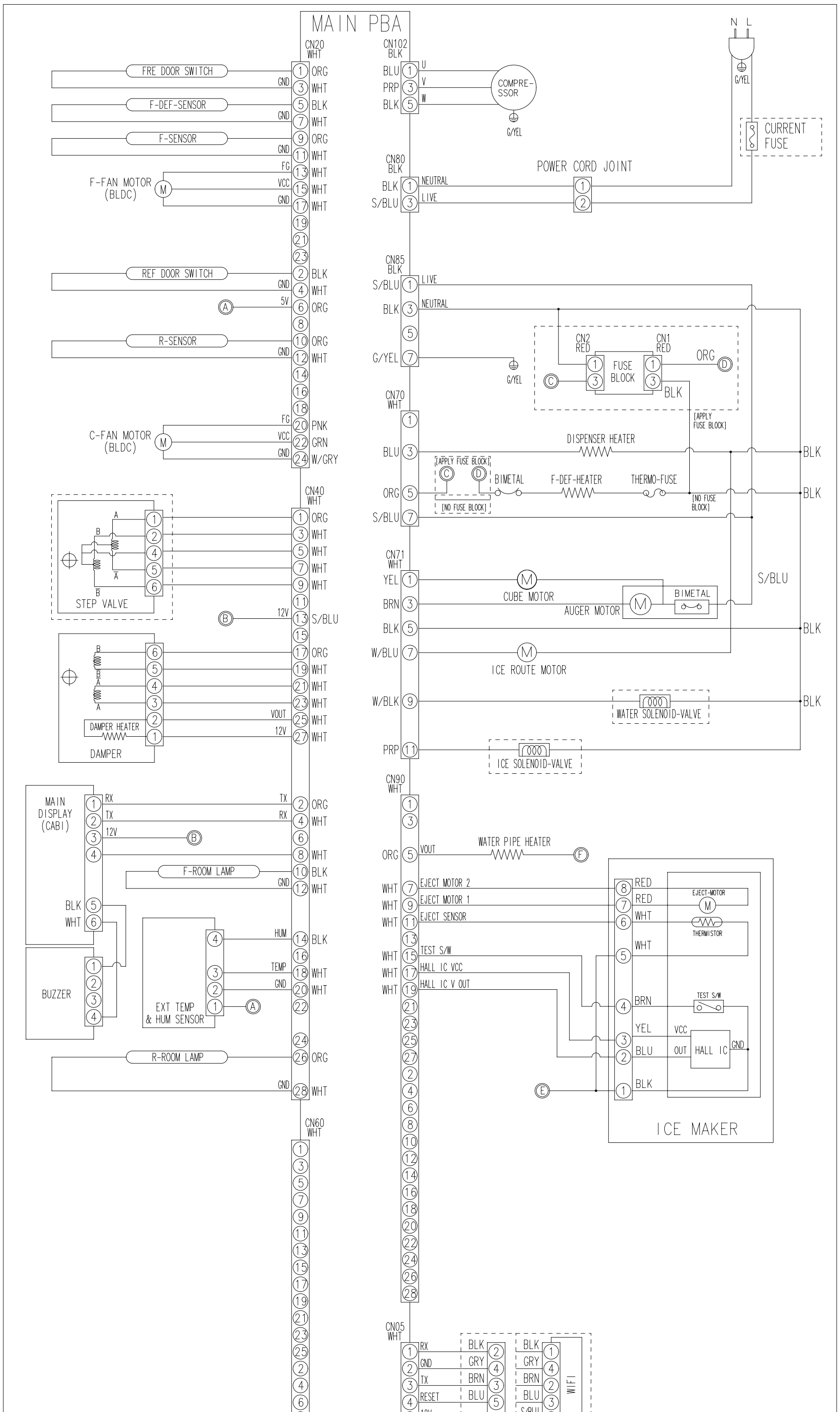
7-1. Dispenser, In-Door Ice Maker, Family Hub (RS22T5561*, RS27T5561*)



BLU	BLUE	S/BLU	SKY BLUE
BRN	BROWN	G/YEL	GREEN-YELLOW
RED	RED	W/BLK	WHITE-BLACK
GRY	GRAY	W/RED	WHITE-RED
GRN	GRN	W/BLU	WHITE-BLUE
ORG	ORANGE	W/BRN	WHITE-BROWN
PNK	PINK	W/YEL	WHITE-YELLOW
PRP	PURPLE	W/GRN	WHITE-GREEN
WHT	WHITE	W/PRP	WHITE-PURPLE
YEL	YELLOW	W/GRY	WHITE-GRAY
BLK	BLACK	W/ORG	WHITE-ORANGE

WIRING DIAGRAM

7-2. Dispenser, In-Door Ice Maker (RS22T520*, RS27T520*)

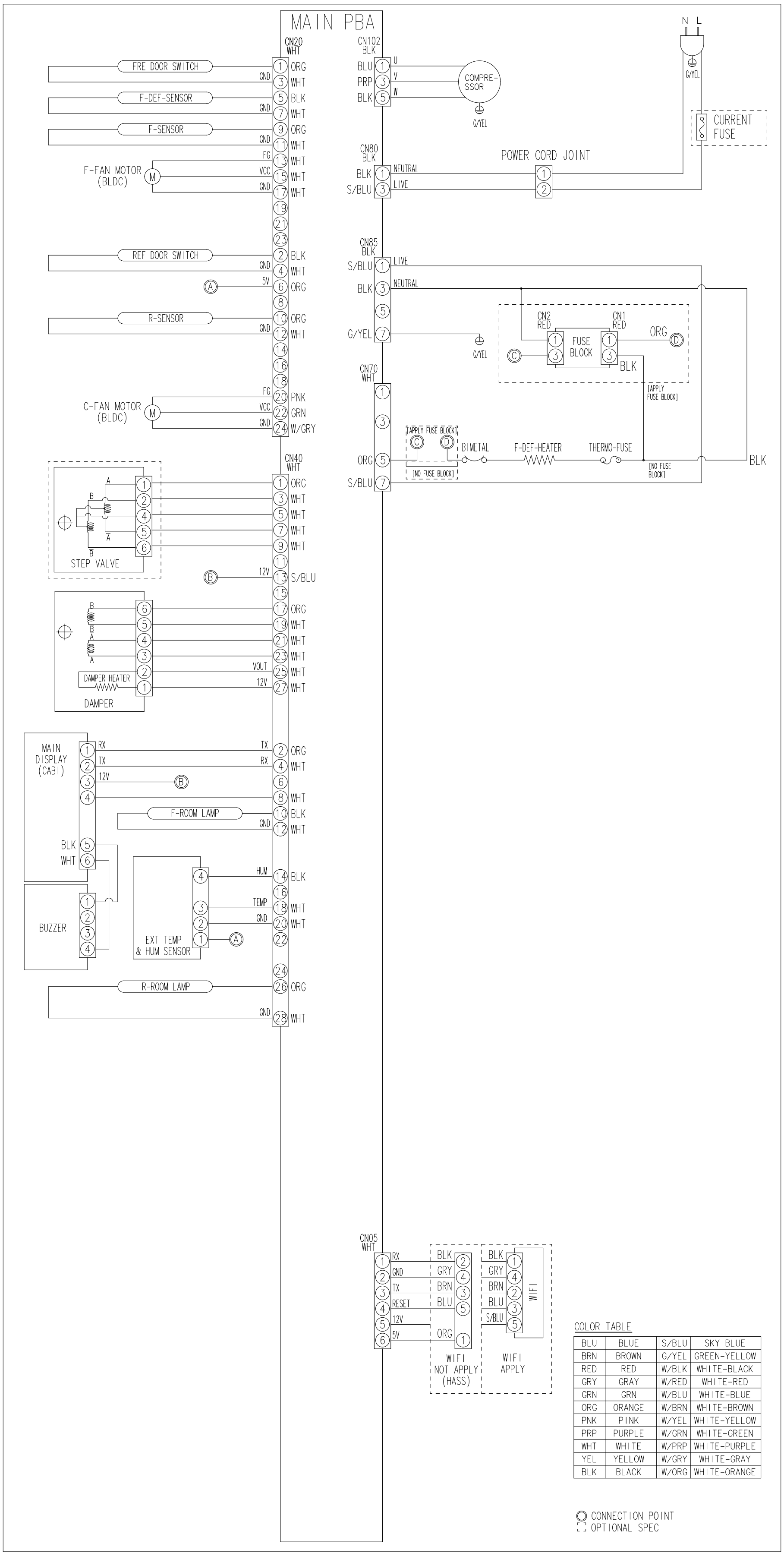


COLOR TABLE

BLU	BLUE	S/BLU	SKY BLUE
BRN	BROWN	G/YEL	GREEN-YELLOW
RED	RED	W/BLK	WHITE-BLACK
GRY	GRAY	W/RED	WHITE-RED
GRN	GRN	W/BLU	WHITE-BLUE
ORG	ORANGE	W/BRN	WHITE-BROWN
PNK	PINK	W/YEL	WHITE-YELLOW
PRP	PURPLE	W/GRN	WHITE-GREEN
WHT	WHITE	W/PRP	WHITE-PURPLE
YEL	YELLOW	W/GRY	WHITE-GRAY
BLK	BLACK	W/ORG	WHITE-ORANGE

WIRING DIAGRAM

7-3. Basic (RS28T5B0*)

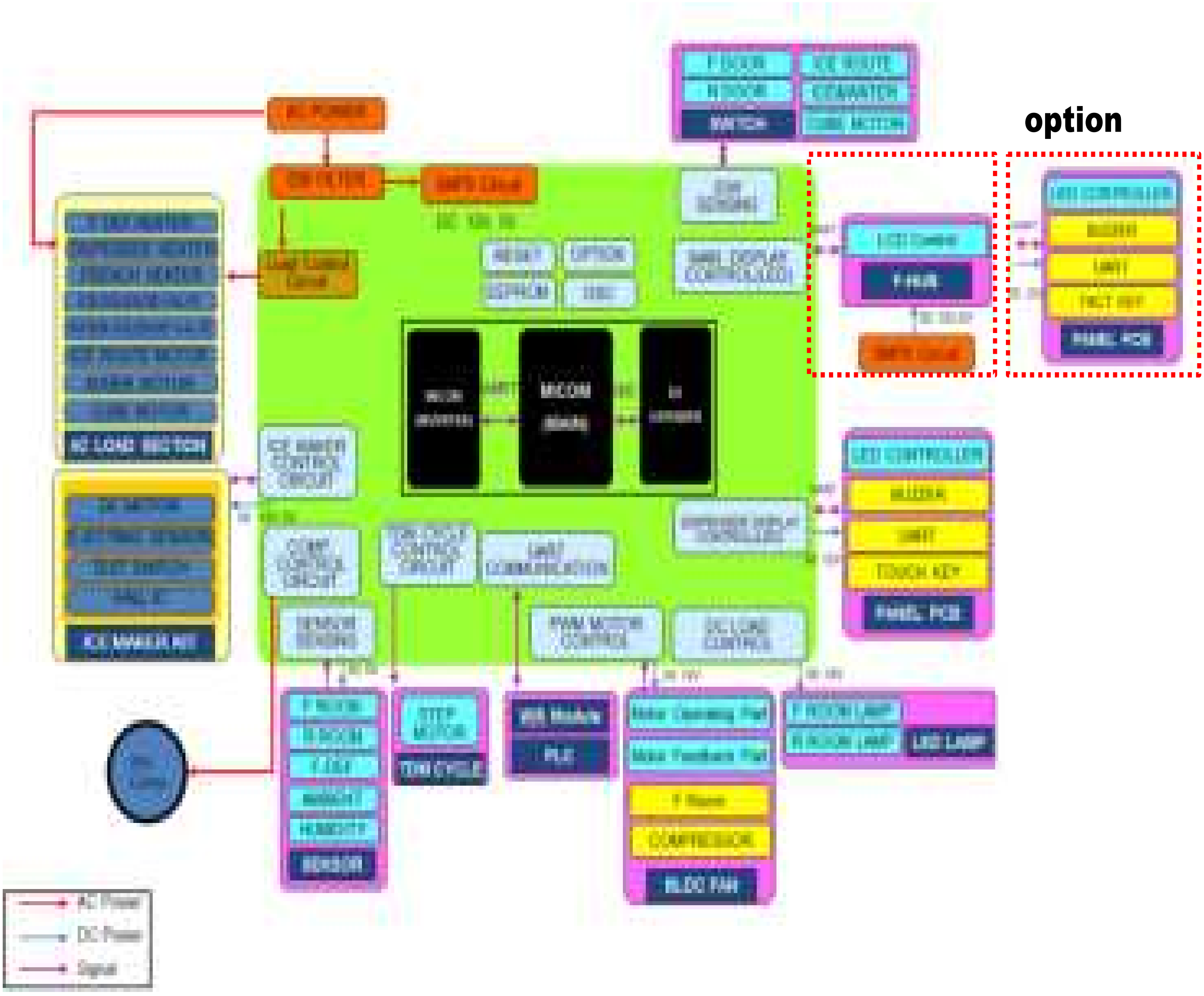


COLOR TABLE

BLU	BLUE	S/BLU	SKY BLUE
BRN	BROWN	G/YEL	GREEN-YELLOW
RED	RED	W/BLK	WHITE-BLACK
GRY	GRAY	W/RED	WHITE-RED
GRN	GRN	W/BLU	WHITE-BLUE
ORG	ORANGE	W/BRN	WHITE-BROWN
PNK	PINK	W/YEL	WHITE-YELLOW
PRP	PURPLE	W/GRN	WHITE-GREEN
WHT	WHITE	W/PRP	WHITE-PURPLE
YEL	YELLOW	W/GRY	WHITE-GRAY
BLK	BLACK	W/ORG	WHITE-ORANGE

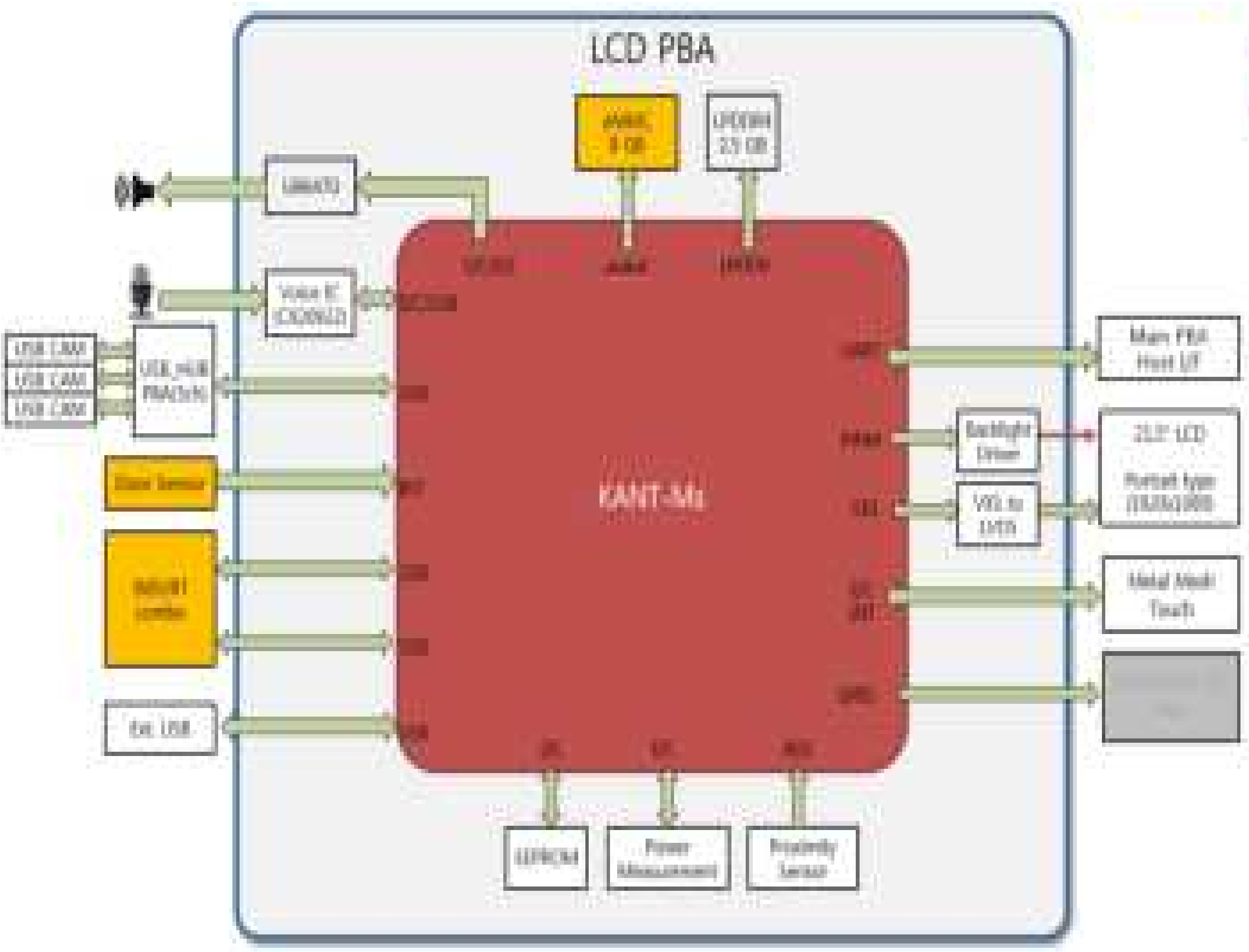
8. BLOCK DIAGRAM

8-1. Block Diagram



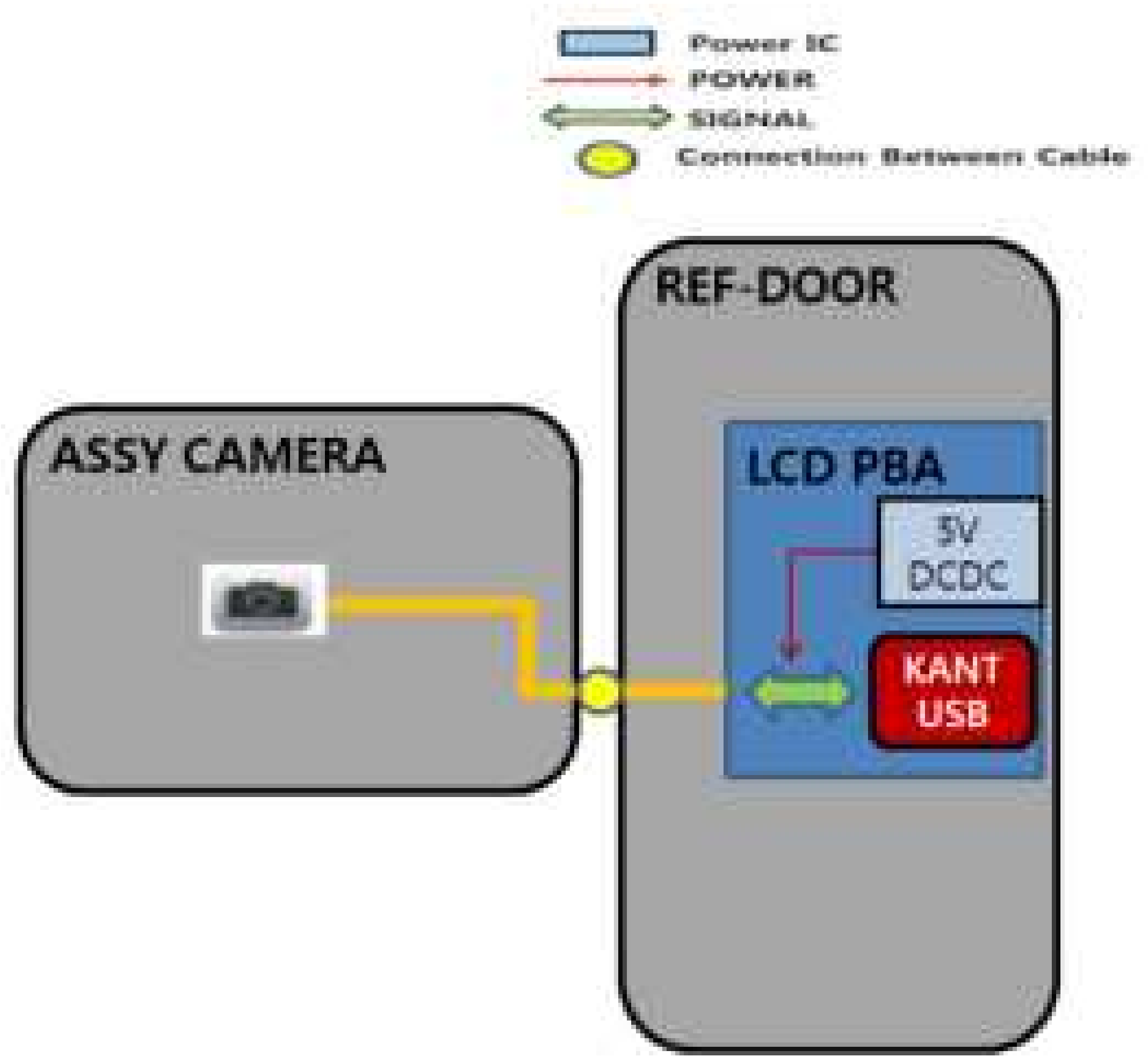
BLOCK DIAGRAM

8-2. LCD PBA



BLOCK DIAGRAM

8-3. Camera



9. REFERENCE INFORMATION

9-1. Model Code Table

Type		Capa	Series 1	Year	Series 2	Exterior Option	Interior Option	Energy	Color		/		
R	S	2	7	T	5	5	6	1	S	G	/	A	A

Capa		Series 1		Year		Series 2		Exterior Option		Interior Option		Energy		Color	
2 (20 cf↑)	7	I&W Disp.	Freestanding	T	2020	5	RS5300T RS5300TC	5	Family Hub + I&W Disp. (In-door I/M)	6	Internal Camera, F-2 Drawer	1	Energy Star + Wi-Fi Embedded	SR	Real Stainless
	2		Counter Depth					2	I&W Disp. (In-door I/M)	0	F-1 Drawer	0	DOE	SG	Black Caviar
	8	Non-Dispenser	Freestanding					B	Non-Dispenser (twist i/m)					WW	Snow White
														S9	Matt Doi metal
														B1	Matt Doi Black

9-2. Trouble Shooting

Problem	Solution
The Refrigerator does not work at all or it does not chill sufficiently.	• Check that the power plug is properly connected.
	• Check the set temperature on the digital display is warmer than freezer or fridge inner temperature.
	• Try setting it to a lower temperature.
	• Is the Refrigerator in direct sunlight or located near a heat source?
	• Is the back of the Refrigerator too close to the wall and therefore keeping air from circulating?
The food in the Refrigerator is frozen.	• Check the set temperature on the digital display is too low.
	• Try setting it to a warmer temperature.
	• Is the temperature in the room too low? Try setting it to a warmer temperature.
	• Did you store the food which is juicy in the coldest part of the Refrigerator? Try moving those items on the other shelves in fridge instead of keeping them in the Cool Select Pantry™.
You hear unusual noise or sounds.	• Check that the Refrigerator is level and stable.
	• Is the back of the Refrigerator too close to the wall and therefore keeping air from circulating?
	• Try locate the refrigerator keep away from the wall over 2 inches.
	• Was anything dropped behind or under the Refrigerator?
	• A "ticking" sound is heard from inside the Refrigerator. It is normal and occurs because various accessories are contracting or expanding according to the temperature of the Refrigerator interior.
The cabinet-door sealing area of the appliance is hot and condensation is occurring.	• Some heat is normal as anti-condensators are installed in the vertical hinged section of the refrigerator to prevent condensation.
	• Is the refrigerator door ajar? Condensation can occur when you leave the door open for a long time.
You can hear water bubbling in the Refrigerator.	• This is normal. The bubbling comes from the Refrigerator coolant liquid circulating through the Refrigerator.
There is a bad smell in the Refrigerator.	• Check for spoiled food.
	• Foods with strong odors(for example, fish) should be tightly covered.
	• Clean out your Freezer periodically and throw away any spoiled or suspicious food.

Problem	Solution
Frost forms on the walls of the Freezer	• Is the air vent blocked? Remove any obstructions so air can circulate freely.
	• Allow sufficient space between the foods stored for efficient air circulation.
	• Is the Freezer drawer closed properly?
Condensation or water drops forms on both sides of fridge door's central part.	• Condensation or water drops may form if the humidity is too high while the refrigerator is operating.
	• If condensation or water drops appears on both sides of fridge door's central part, turn the Energy Saver mode off after removing the condensation.(See the explanation of the Energy Saver button in "Using the Control Panel" on page 21.)
What Do I Need to Use the E-Smart and Smart Grid functions?	To use the Smart Grid (Demand Response) and E-Smart function on your Refrigerator, you need the following devices and apps: - Devices: 1) Wireless Router, 2) Samsung E-Smart Refrigerator, 3) Smartphone (Recommended: Galaxy S4,Galaxy Note3/ Android OS Jelly bean) - Applications: 1) "Samsung E-Smart App" from the Google Play Store or Samsung Apps. You also need to: Connect the Refrigerator and the Smartphone to the same Wi-Fi network in your home. • Install and run Samsung E-Smart App on your Smartphone (to use the Samsung E-Smart function). • In addition, to use the Smart Grid (Demand Response) function, you must Register for the service with your electric company.The company must have an EMS (Energy Management System) that supports SEP (Smart Energy profile).
Why isn't the E-Smart function working normally?	• Confirm that the router in your home and the Internet are working properly. - Connect a Smartphone to the router (AP, Access Point), and then confirm that you can browse the Internet on the phone. • Confirm that refrigerator is connected to the AP. - Check the E-Smart icon on the PANEL of refrigerator.If the refrigerator is connected, the icon will be on. • Confirm that the refrigerator and Smart Phone are connected to the same router.
Why isn't the Delay Defrost Capability working normally?	• Confirm that the router in your home and the Internet are working properly. - Connect a Smartphone to the router (AP, Access Point), and then confirm that you can browse the Internet on the phone. • Confirm that the area where you live is properly entered into the Smart phone App. • Confirm that the Smart Grid function works correctly.(Check the PANEL. L3 andL4 should not be displayed on the PANEL.)

REFERENCE INFORMATION

9-3. Q&A

Descriptions of symptoms	Check Points	Corrective Measures
▸ Noise (resonance) problems keep on even though the noise generating BLDC motors for both of the compartments are replaced several times. What does generate the resonance and how can it be settled down?	When the BLDC fan motor rotates in low RPM, The friction with air is quite high and it generates "grinding" noises.	If you replace the ambient thermistor with a 2.7K resistance (detecting 43°C), the BLDC fan motor rotates in high RPM, which reduces friction with air resulting in reduction of the "grinding" resonance.
▸ What causes the "knocking" noises? How to solve it?	It makes "knocking" or "branch breaking" noises when the liner and the shelves hit each other due to the fluctuation of the inside air pressure upon door open/close. Also, these noises occur when the liners and the shelves hit each other as the liners expand and contract due to the temperature change in both of the compartments.	Check the clearance between the selves and the liners. ① Freezer Shelves: Remove the trim shelves already attached and replace them with those supplied for service. ② Fridge Shelves: Because noise-preventing trim selves are not attached, it needs attaching.
▸ What is the solution if the same problem occurs even though the counter action in the service bulletin has been implemented already for the "knocking" noises?	Check if the selves wobble. If they do, have shelves sit firmly on their places.	
▸ What causes the liquid passing noises from the back of the refrigerator?	Refrigerant goes into the evaporator via the capillary tube in which the refrigerant expands as it circulates the cooling cycle. At this time, the refrigerant is in its liquid state and it starts evaporating as it reaches at the inlet of the evaporator with a bigger diameter, which causes the refrigerant noises. And, it gets worse when the refrigerant does not flow freely.	
▸ What is the solution for the compressor noises?	For new refrigerators.	
	Check if the refrigerator is leveled.	Check if the refrigerator wobbles by shaking with hands.
	Check if there is enough clearance at the back of the refrigerator for the ventilation of the machine compartment.	If there is not enough clearance or it is blocked by things such as newspaper, there could be resonance noises.
	Required clearance around the product.	More than 5cm from the back, 30cm from the top and 10cm from its sides.
▸ What is the solution for the compressor noises?	For old refrigerators	
	Dust could get built-up in the machine compartment. Then, its ventilation would get restricted which makes the refrigerator overheated resulting in the increase of the noise level.	Explain it to customers and let them clean dust or any other foreign substances in the machine compartment.
	As the vibration proof rubbers get hardened, noises generate during the comp operation. (The noise level is quite high.) → Replace the vibration proof rubbers.	The compressor is dislocated due to impact during its transportation such as moving-in. → Check if it's dislocated when it is more noisy after moving-in.

<table>
<tr><th>Descriptions of symptoms</th><th>Check Points</th><th>Corrective Measures</th></tr>
<tr><td>▸ During the comp start-up, iron friction noises occur. What causes them?</td><td colspan="2">The reciprocation piston could get worn out or inner components could get dislocated.</td></tr>
<tr><td>▸ What can be checked when the unit sends out noises?</td><td colspan="2">1. Check its symptoms and patterns.
2. Check if the unit is installed on a firm and leveled floor.
3. Check if the unit is installed close to the customer's living area.
4. Check if the panel on the machine compartment hits on the rear wall and the unit has enough clearance with the rear wall.
5. Check if the refrigerant pipes are shaped as normal.</td></tr>
<tr><td>▸ Why is the fridge compartment not cool? (Not a defect)</td><td colspan="2">Advise customers to adjust its temperature level to one or two step higher. For example, when the ambient temperature is low such as in winter (especially, when you use it in the morning with the door not being opened or closed during the night), the compartment temperature could get increased by 1~2°C. So, advise customers to shift its temperature level and explain to them that it does not affect its power consumption that much when its temperature setting is adjusted to one or two step lower.</td></tr>
<tr><td>▸ Why is the food melt even though the display of the freezer compartment shows -20°C?</td><td>Check the compartment temperature with a thermometer.</td><td>If it is considered to be low cooling, When the BLDC motor fan does not rotate because its restriction is not picked up. When the evaporator is frozen-up (defrost it) Temp detection error according to the characteristic change of the thermistor (set the compartment temperature or replace the thermistor)</td></tr>
<tr><td>▸ What is the reason that vegetables get frozen even though the fridge compartment is set to MIN?</td><td colspan="2">Replace the fridge thermistor because it could be faulty.
1st: Check if the thermistor works after referring to the self-diagnosis checklist on the MAIN-PCB cover. If the over-cooling keeps on even though there is no problem with the above, replace it.</td></tr>
<tr><td>▸ What can be done when frozen food gets melt in the freezer compartment or it does not cool down?</td><td colspan="2">Defrost it by using hot water and check the defrost system for any fault. And then, eliminate the root causes so as to prevent it from reoccurring.</td></tr>
<tr><td>▸ Why doesn't the compressor operate upon power supply?</td><td colspan="2">Upon the initial power on, the compressor starts operating after a five minute delay to protect the compressor. So, please wait until it starts operating.</td></tr>
<tr><td rowspan="2">▸ Why does it send out "Ding Dong" or alarm sounds with the doors closed?</td><td colspan="2">Check if the food sticks out preventing the doors from closing properly. If it send out the above sounds with the door closed well, the door switch may not have been pressed down completely. So, make sure for the door switches to be pressed completely. Still, if it does not stop going off, check the wiring connections because the door switch signals may not be inputted into the PCB. And, when the door switch is faulty or it is not pressed down completely, the fan does not rotate and it causes low-cooling advancing to a defrost problem.</td></tr>
<tr><td colspan="2">NOTE
When it comes to automatic models, the fan motor does not operate right on with the door closed after being opened. The fan motor for dual-evaporator refrigerators starts after a 10- second delay and when the ambient temperature is higher than 33°C, it starts after a minute.</td></tr>
</table>

REFERENCE INFORMATION

<table>
<tr><th>Descriptions of symptoms</th><th>Check Points</th><th>Corrective Measures</th></tr>
<tr><td>▸ What can be done when it sends out much smell in the fridge and the freezer compartments after 2 months?</td><td colspan="2">When the stored food sends out much smell.
• Check if there is any food sending out sustaining smells.
Dried squid, dried laver : Hold on them
Pounded garlic : Put it in an airtight container
Medical herbs : Make sure they are packed airtight. Replace the old packing or wrap with a new one.
Others : Check if the container is sealed or the food is packed airtight.
• Check if the compartment temperature is normal and the food is contaminated.
• Check if there are any overflow of side dishes on the shelves or the bottom of the compartments.
• Put the food sending out much smell in an airtight bag or container.
• Open the door and ventilate it. Also, clean liners, shelves, containers and door bins.</td></tr>
<tr><td rowspan="2">▸ What is the cause and its counter action for chemical smells with new products?</td><td colspan="2">During its delivery to customers, chemical smells from various components could build up inside of the compartments. So, please let the doors open for some time to use the unit.</td></tr>
<tr><td colspan="2">Precautions : Smells tend to get soaked into the liners or other components. If food generating much smell is stored inside, it would stick onto the liners and other components and it is so difficult to remove the smell. Especially, customers should take care in storing smelly food properly with its sealing being tight during the early period of the product’s use.</td></tr>
<tr><td>▸ What can be done when the smell keeps on even thought the deodorizers are cleaned?</td><td colspan="2">1) Turn off the refrigerator (unplug the unit) and remain the door opened.
2) Take out the food stored in the refrigerator. And then, take out all the shelves, door bins and containers, and put them in warm water. After cleaning them by using dish detergents and drying them, put them back to their locations.
3) Remove the deodorizer and soak it in warm water more than 4 hours. After drying it in sunlight, put it back to its location.
4) Throw away the smell-soaked plastic bags and put the food in new ones.</td></tr>
<tr><td>▸ What is the cause and its counter action for chemical smells with new products?</td><td>When it tastes and smells funny
• It tastes funny even though it does not smell funny.</td><td>It could happen when remnants of the water filter or organics have been built up in the water tank. So, replace the water filter and the water tank together. If there is no replacement part and the water tank need cleaning, use dish detergents and make sure to clean the inside without any detergents remaining inside.</td></tr>
</table>

SAMSUNG

GSPN (GLOBAL SERVICE PARTNER NETWORK)

Area	Web Site
Europe, CIS, Mideast & Africa	gspn1.samsungcsportal.com
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